Projectile Tracking Chronograph (PTC)

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1 Introduction

[To be completed - Introduction content goes here]

2 Project Description

2.1 Background and Motivation

The ability to accurately measure projectile velocity has long been fundamental to the study of ballistics and the advancement of shooting sports. Early methods relied on manual observation and empirical estimation, which were often inconsistent and dependent on environmental conditions. The development of mechanical ballistic chronographs in the late nineteenth century introduced a scientific approach by measuring velocity from distance over time, laying the groundwork for modern ballistic instrumentation.

In contemporary applications, ballistic chronographs have become essential tools for competitive shooters, hunters, and militaries as they provide quantitative data about projectile performance. Even small variations in muzzle velocity can significantly impact a projectile's trajectory and bullet drop, specifically at long ranges. Competitive shooters depend on accurate velocity measurements to refine ballistics and ensure the target is consistently hit. Other measurements of extreme spread and standard deviation are of interest, where lower variability in these metrics directly correlates with tighter shot grouping and improved accuracy.

Modern commercial chronographs further enhance usability by performing real-time calculations of muzzle energy, allowing users to simply gather projectile data rather than manual processing. However, many advanced chronograph systems rely on technology that can be expensive for hobbyists. Additionally, the commercially available systems focus primarily on velocity measurement and provide limited insight about the projectile's trajectory without the use of specialized or high-cost radar technology.

The PTC project is motivated by the need for an accessible and transparent ballistic measurement system that goes beyond the basic velocity calculation. By leveraging optical sensing, precise timing, and embedded computation, PTC aims to provide accurate projectile velocity measurements while enabling trajectory estimation through angular and spatial analysis. Beyond its technical objectives, PTC serves as an interdisciplinary senior design experience; this effort brings together optical engineering expertise focused on laser optics and beam deflection, electrical engineering contributions centered on analog sensing and signal processing, and computer engineering responsibilities of embedded firmware, timing analysis, and user interface implementation. Through collaboration, the team aims to develop a ballistic measurement system while fostering teamwork and problem-solving skills essential for professional engineering practice.

2.2 Current Technologies and Relevant Projects

2.2.1 The First Patented Ballistic Chronograph

Early projectile velocity measurement can be traced back to the late nineteenth century through purely mechanical systems such as the Le Boulenge ballistic chronograph, which was patented in 1874. This device relied on electromagnetic release mechanisms to drop two metal rods at different times as a projectile passed through two spaced wire screens. Knowing the acceleration due to gravity and the fixed spacing between the screens allowed the projectile's velocity and flight time to be calculated. Although entirely mechanical, this approach established the fundamental concept of velocity measurement through precise timing, a principle that remains central to modern electronic and optical chronographs [1].

2.2.2 Optical Chronograph

These devices operate by detecting the passage of a projectile across two optical sensors spaced at an identified distance. As the projectile passes over each sensor, the light received through the sensor is interrupted, generating electrical signals that are time-stamped to compute the projectile's velocity [2].

While optical chronographs are relatively affordable and straightforward to operate, they are sensitive to environmental conditions. Measurement errors are introduced when there are changes in sunlight intensity, shadows, or improper projectile positioning. These commercial chronographs only display the velocity data. PTC builds upon the fundamental principle of measurement used by optical chronographs, but seeks to mitigate environmental sensitivity by designing laser-based detection and precise electronic timing, enabling improved consistency and expanded trajectory analysis beyond simple velocity measurement.

2.2.3 Magnetic Chronographs

Magnetic chronographs measure projectile velocity by detecting disturbances in a magnetic field caused by a projectile passing near a sensor array. These systems are typically mounted directly to the firearm barrel and do not rely on external lighting conditions [2].

Magnetic chronographs introduce certain limitations despite their reliability. The attachment of the device to the barrel can slightly alter the harmonics of the barrel, affecting where the projectile impacts. Magnetic chronographs are not suited for spatial position movement since their purpose is to measure velocity close to the muzzle. In contrast, PTC is designed as a non-contact measurement system, allowing projectile velocity and angular trajectory to be determined without physical interaction with the firing mechanism, preserving ballistic integrity.

2.2.4 Doppler Radar Chronographs

Doppler radar chronographs represent the most advanced of the commercially available projectile measurement systems. These devices transmit radar signals that reflect off the projectile during flight, allowing the system to continuously track velocity based on Doppler frequency shifts [2].

While radar-based chronographs offer excellent accuracy and ease of use, they are costly and rely on complex signal processing hardware, limiting user access to customization of raw data. PTC aims to explore an alternative approach by leveraging optical scanning techniques and real-time signal processing to achieve comparable spatial resolution at a lower cost. By using laser interruption timing and angular position mapping, PTC seeks to bridge the gap between basic optical chronographs and high–end radar systems.

2.3 Goals and Objectives

The PTC project is designed to extend traditional ballistic chronograph functionality by combining optical sensing, embedded computation, and user-friendly data presentation into a single system. The goals define the overall vision of the system and its intended capabilities, while the objectives describe measurable outcomes that guide implementation. The stretch goals are identified to encourage innovation beyond the minimum functional scope of the project.

2.3.1 Goals

Basic Goals:

- Develop an optical-based system capable of detecting a projectile as it passes through a defined measurement region.
- Accurately measure projectile velocity using precise timing and optical sensing techniques.
- Ensure the system operates reliably in indoor environments without dependence on ambient lighting conditions.
- Provide calculated results, including projectile velocity, to the user within 10 seconds of projectile detection.
- Present projectile-related information in an intuitive manner suitable for novice users.

Advanced Goals:

- Extend the system's measurement capabilities beyond velocity to include trajectory-related information, such as the direction of projectile travel.
- Display processed data on an integrated screen, enabling standalone operation without the need for external devices.
- Reconstruct and visualize a two-dimensional representation of the projectile's trajectory.

Stretch Goals:

- Determine whether a projectile's trajectory would intersect a target with a one-meter diameter at a distance of ten meters.
- Integrate environmental sensing (such as temperature and humidity) to support improved ballistic analysis in outdoor environments.
- Incorporate atmospheric effects such as wind, temperature, humidity, and pressure into trajectory and offset calculations based on user-input target distance.
- Support a wide range of projectile velocities, from as low as 5 m/s to as high as 100 m/s.
- Determine and report the compass direction of projectile travel using onboard orientation and navigation sensors.

2.3.2 Objectives

Basic Objectives:

- Implement an optical detection system that does not require precise alignment by the end user.
- Capture projectile event data using optical sensors, cameras, or laser-based detection methods.
- Present results to the user via an integrated display with minimal user interaction required.
- Utilize a microcontroller such as a Raspberry Pi-based platform to coordinate sensing, timing, computation, and user interface tasks.

Advanced Objectives:

- Develop a calibration procedure that is intuitive, such as using known reference objects with the camera's field of view.
- Synchronize optical detection events with strobe lighting and camera/image acquisition to capture multiple images of the projectile in flight.
- Implement image processing and mathematical analysis to extract spatial information from captured images.

Stretch Objectives:

- Integrate a magnetometer to determine the compass orientation of the device and infer the compass direction of the projectile motion.
- Use an accelerometer to determine the vertical orientation of the system and calculate the projection elevation angle.
- Combine orientation, environmental, and optical data to improve trajectory estimation accuracy.

2.3.3 Projectile Tracking Chronograph: Features/Functionalities

The PTC is designed to provide a comprehensive set of features that extend beyond the traditional ballistic chronograph by incorporating various subsystems such as optical sensing, embedded computation, and spatial analysis. PTC implements an optical light detection system (light gate) that identifies a projectile as it enters the sensing region. By interrupting a modulated beam from a light source, the system generates high-speed timing events that are captured using hardware interrupts on the microcontroller. These events can be used to enable accurate time-of-flight measurements across a known distance, allowing for precise velocity calculations. The use of a modulated carrier frequency along with photodetector(s) that have a sufficient bandwidth will allow the system to operate in indoor and ambient lighting conditions. Additionally, PTC uses an optical chamber with synchronized strobe light illumination and camera imaging to capture multiple instances of the projectile as it is moving through the chamber. Once the initial velocity is obtained from the light gate, the microcontroller activates the strobe flashes and camera exposures. This in turn produces a single image containing multiple shadows of the projectile, each corresponding to a known interval of time. Image processing algorithms can then be executed on a Raspberry Pi to extract the spatial position data from the shadows of these images, allowing for the direction of the projectile and its angular trajectory to be approximated.

This system also supports embedded ballistic and trajectory computation, combining the previous timing data and spatial measurements to calculate the projectile velocity, angular deviation, and estimated trajectory path through the measurement region. In our advanced and stretch goals, we would like to further enhance these calculations by incorporating other environmental data from other types of sensors. Integrating information from temperature, humidity, and pressure sensors can be used to further refine the ballistic calculations to increase the overall accuracy of the system by accounting for atmospheric effects. Furthermore, combining other sensors such as an accelerometer and magnetometer would allow the system to determine its own orientation. This capability would support other advanced features, like determining the compass heading of the projectile's travel path.

To enhance the usability of PTC, a user interface consisting of simple input controls and a display has been incorporated into the system. Processed ballistic results are to be presented to the user within seconds of the detection of the projectile. This interface is designed to be user-friendly while still providing meaningful data to the user. Collectively, these features make the PTC a powerful tool for projectile measurement and analysis.

2.4 Required Specifications

2.4.1 System Specifications

General System Specifications:

TABLE I: Specification of System Components

Parameter	Description	Specification
Maximum speed error	The maximum error of the speed detected of the projectile	$< 20\%$
Maximum system response time	The maximum time from a projectile traveling through the system until information is displayed to the user	< 5 s
Detectable projectile size	Maximum diameter of a spherical projectile that can be detected and characterized	Between 15 mm and 50 mm
Detectable speeds	The speed of a projectile that can be detected and characterized	Between 10 m/s and 50 m/s
Angular Resolution Error	The maximum error of the angle detected of the path of the projectile	$<$ 1/5 radian
Power Requirements	The maximum power the system will draw on average over the course of 10 seconds	< 300 W
Sensing Area	The area of the system that will be able to sense projectiles through it	$>$ 10 cm x 10 cm
System Size	Total size of the system	$<$ 3 m x 3 m x 3 m
System Weight	Total weight of the system	< 50 pounds
Ambient Light Tolerance	Maximum ambient light the system can tolerate	$<$ 20 lumens per square foot

2.4.2 Component Specifications

Specific Component Specifications:

TABLE II: Specification of Project Components

Component(s)	**Description**	**Specification**
Camera(s)	The sensor area, the quantum efficiency of detecting the photons in the wavelength of the strobe, and the resolution of the sensor	$\geq$ 6 mm sensor width; 600-700 nm spectral sensitivity; 50-70% quantum efficiency @ 650 nm; 1600 x 1200 - 1920 x 1200 pixel target resolution; 200-500 fps target frame rate; $\geq$ 60 dB dynamic range
Photodetector(s)	The responsivity of the photodiode in the wavelength of the emitting laser or LED wavelength, the noise floor of the photodiode, and the bandwidth of the photodiode	$>$ 100 kHz bandwidth; 0.3-0.6 A/W @ 650 nm; 5-10 mm width; $\geq$ 20 dB SNR during beam interruption
Gate light source(s)	The light source for the initial light gate for detecting the projectile entering into the system	635-670 nm; $\leq$ 1 μs rise/fall time; 10-100 kHz carrier frequency; $\geq$ 80% modulation depth; $\geq$ 100 mm line length
Environmental Sensor(s)	Accuracy of the humidity, temperature, and pressure sensors	$<$ $\pm$10 hPa accuracy at 1000 hPa; $<$ $\pm$5% humidity at 30%-80% humidity; $<$ $\pm$5°C at 10°C-50°C
Power Supply	Provides a regulated voltage and current to power the electronic devices	Minimum 50 A continuous current; 12-48 V output voltage range; $\leq$ 2% ripple under load; $\leq$ 5 ms for transient response recovery
Strobe Light Mechanism	Capabilities of the subsystem that creates the strobe flashes	$<$ 200 μs flash time; $<$ 100 μs turn on time; $<$ 100 μs turn off time; $>$ 1000 lumen output
Physical System Structure	Physical layout of the system	$>$ 10 cm width and height sensing area; $<$ 2 m sensing area length; $<$ 3 m overall height; $<$ 2 m width; $<$ 3 m overall length
User Interface	User interface specifications	$>$ 2 cm width and height of screen; $<$ 1 m width and height of screen
Microcontrollers/ Computers	Microcontroller and computer specifications	$>$ 1 MHz MCU clock speed; Image processing capability for computer

2.5 Optical Hardware

The system will consist of a light gate (or sensor gates) and an area for strobe photography of the projectile (optical chamber). The initial light gate will get a projectile speed that will determine the timing of the strobe flashes. The camera(s) will take one photo that contains multiple shadows of the projectile at the points in time where the strobe is triggered.

The cameras, lenses, and strobe LEDs depend on each other. The LEDs must be bright enough to drown out the ambient light leaking into the box in order to provide a good signal to the cameras. The lens system must be designed to capture enough light to provide a sufficient signal to show the projectile on the cameras. The LEDs will need to flash for a short enough time to avoid too much blurring of the projectile in the chamber; however, they should flash for a long enough time to provide sufficient light to activate the camera sensor enough to get an acceptable picture. A rough estimate of a camera that operates at 30 fps can comfortably film at typical bedroom levels of illumination, which is about 10 lumens/square foot. Given an approximate strobe area of 2 square feet, and about 20% illumination efficiency, that would be 100 lumens for the 1/30 s or 33 ms that a camera frame lasts. With a 100 microsecond strobe time, that's 330 times the brightness needed for an equivalent amount of light to hit the camera sensor during the 100 microsecond strobe, which gives at least 3000 lumens required from the strobe for a simple camera.

The initial light gate will be used to detect the projectile entering the system and its approximate speed. The photodiode must be of sufficient bandwidth to detect a fast-moving projectile (a paintball is about 17mm and travels at 90 m/s, which gives 180 microseconds of signal time) would require at least 5 kHz of bandwidth. A projectile that is 1/4 inch long and traveling at 1000 feet/s would require a 50 kHz bandwidth sensor to detect the projectile. In order to filter ambient light, a carrier wave of sufficiently higher bandwidth would be required, so 100 kHz would be the minimum for detecting a bullet.

In the following, the diagram elaborates on the design of the sensor gates and optical chamber responsible for detecting a projectile and taking a picture of the projectile as it moves through the enclosure. By interrupting the line of sight of the light source to its corresponding photo-sensor, a projectile can be detected for initial calculations to be made. As the projectile moves through the optical chamber, multiple cameras can be used to make further calculations that estimate position and trajectory.

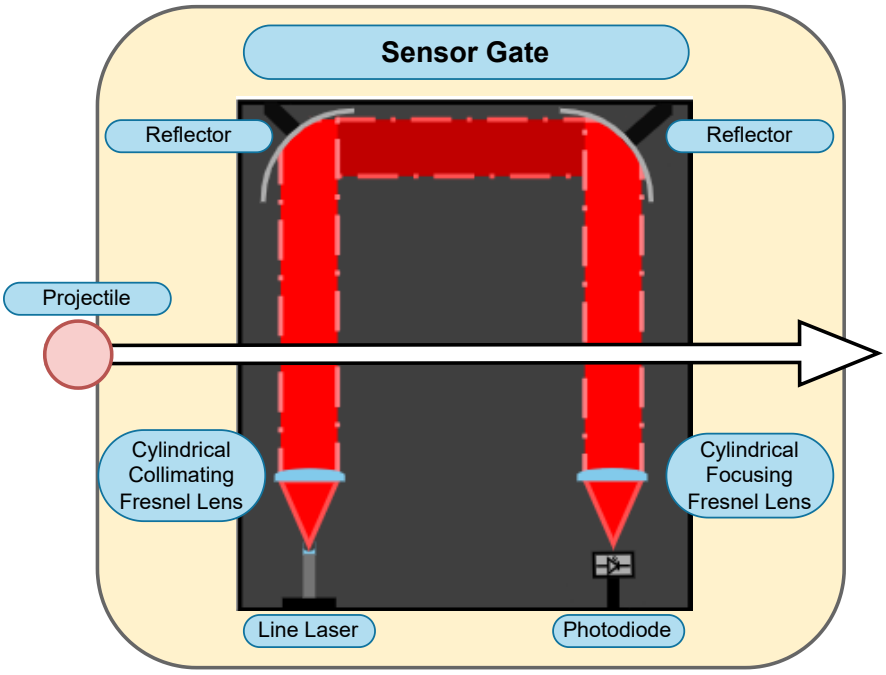


Fig. 1: Sensor Gate Design

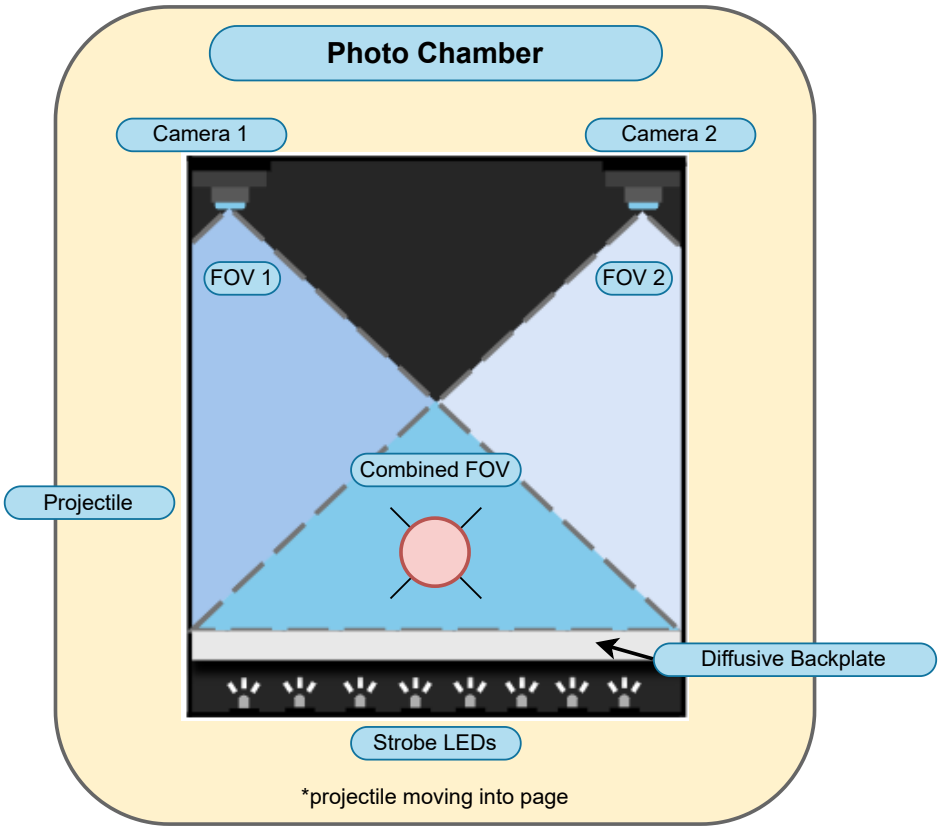


Fig. 2: Figures 1 and 2 elaborating on the sensor gates and optical chamber of the PTC system, illustrating the interaction between external projectile input and the optical system in place.

2.6 Electronics and Software

The hardware block diagram illustrates the necessary hardware and connections to facilitate our goals and objectives, as well as their current status at the time of this writing. Each block in the diagram is color-coded to indicate which team member will be primarily responsible for its research, design, and operation. The RP2350 MCU will be primarily responsible for handling sensor information and hardware-level interactions, whereas the Raspberry PI computer will be used for image processing of the projectile and to control the user interface. An optical sensor and chamber will be used to detect a projectile and take pictures, and these signals will be used by the RP2350 and the Raspberry PI to make calculations.

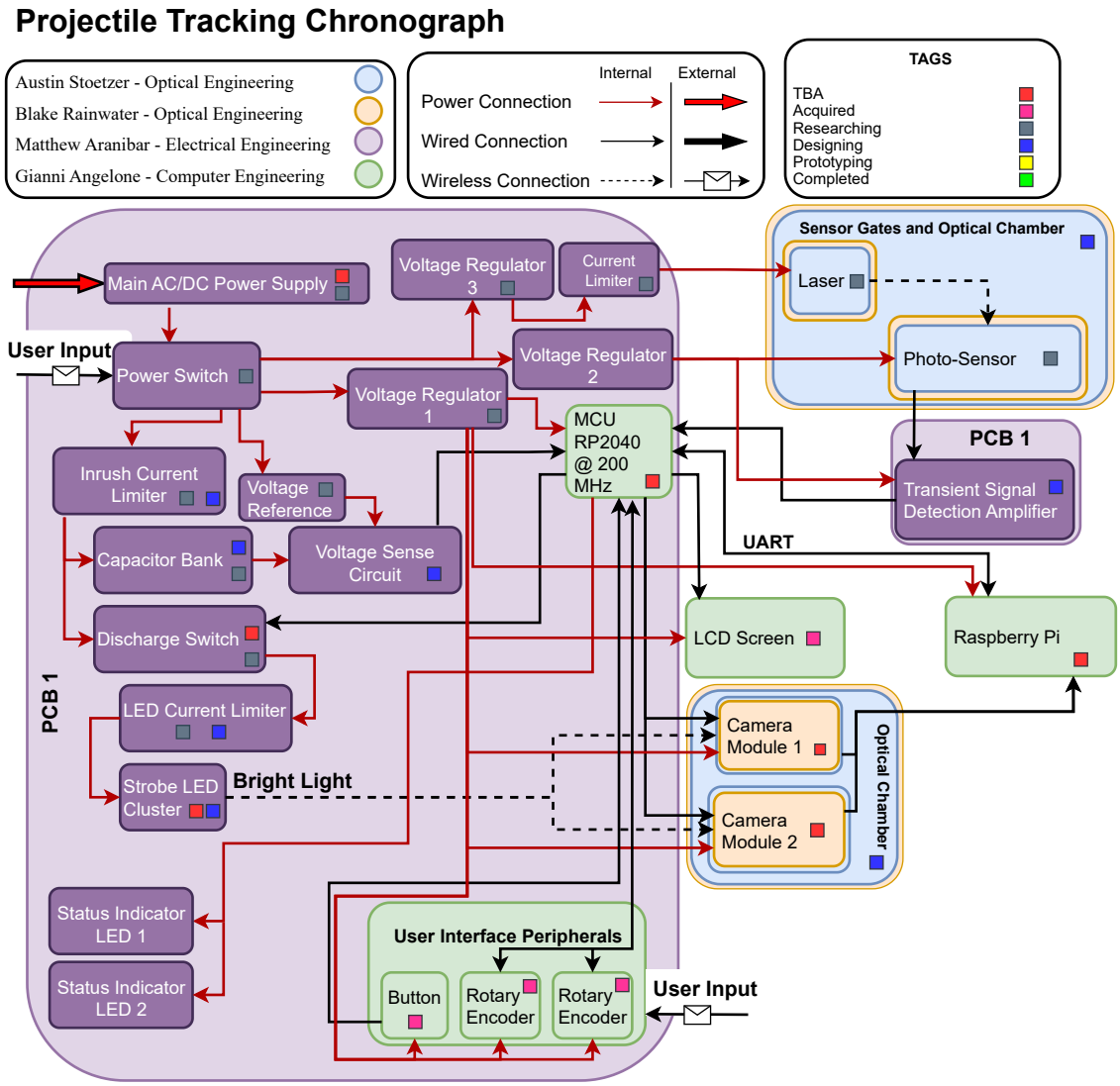


Fig. 3: Hardware Diagram of the PTC system that illustrates component connections, status, and division of workload.

The software flowchart begins with the system power-on and initialization of the Raspberry Pi subsystems. After loading the configuration settings and performing the necessary calibrations, the system enters a charging state to prepare the capacitors for the strobe LED cluster. Once the capacitors are fully charged, the software transitions into an armed ”idle” mode that continuously monitors for a projectile detection signal from the sensor gate. Once a projectile is detected, the system will trigger synchronized strobe pulses and camera captures, followed by image processing to extract positional data from the projectile. The ballistic and trajectory calculations are then performed, with the results being displayed through the user interface.

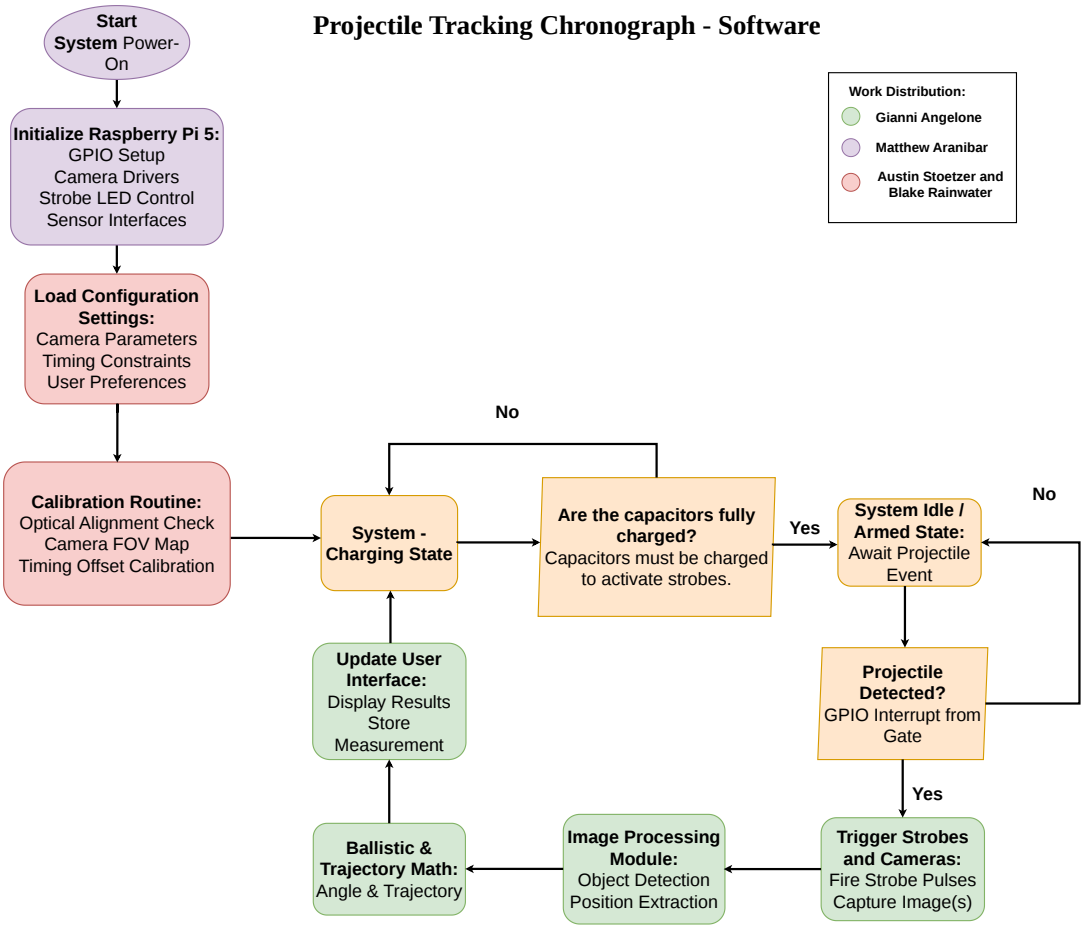


Fig. 4: illustrates the software flow of the Projectile Tracking Chronograph, detailing system initialization, projectile detection, data acquisition, processing, and reset to an idle state.

The data flowchart of the PTC system, details the flow of data from initial projectile detection to final system output. The diagram outlines how optical interruptions at the gates trigger hardware interrupts that prompts the MCU to synchronize the cameras and LED strobes, enabling the capture of multiple images of the projectile as it moves through the system. These images are then processed using computer vision techniques to extract positional data, allowing for the velocity and trajectory to be calculated in the ballistic and trajectory engine. The resulting data from these subsystems is then communicated to the user interface, where the results are displayed.

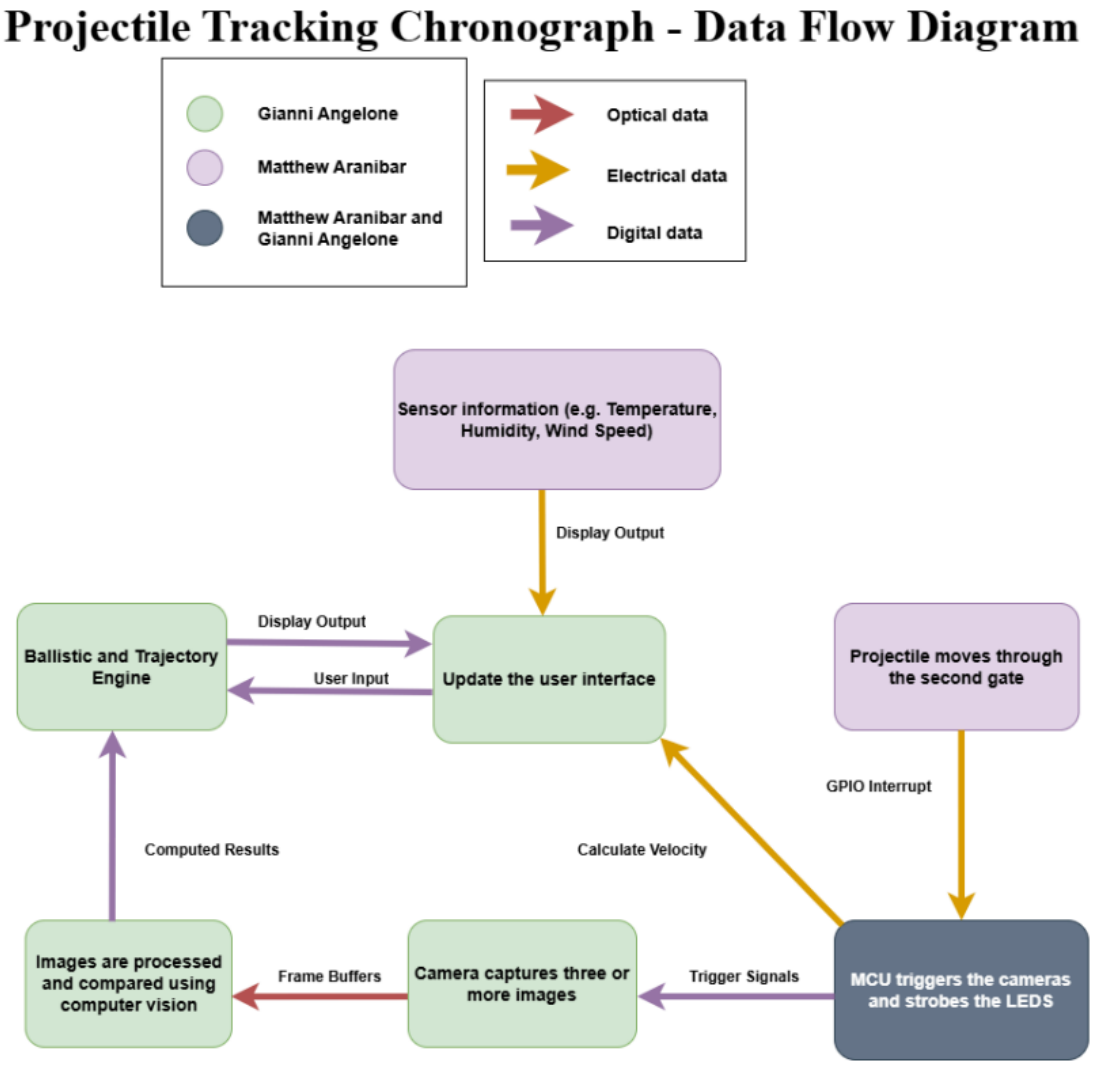


Fig. 5: Flowchart of the PTC system that illustrates the end-to-end data flow from optical event detection and hardware interfacing through synchronized strobe and cameras, image processing, and ballistic computation, outputting to the user via the user interface.

The illustrations below present the current prototype configurations developed for the PTC. These figures provide visual context for the mechanical layout of the various system components described in the previous sections. These sketches are intended to provide a clear framework for system functionality during development and testing.

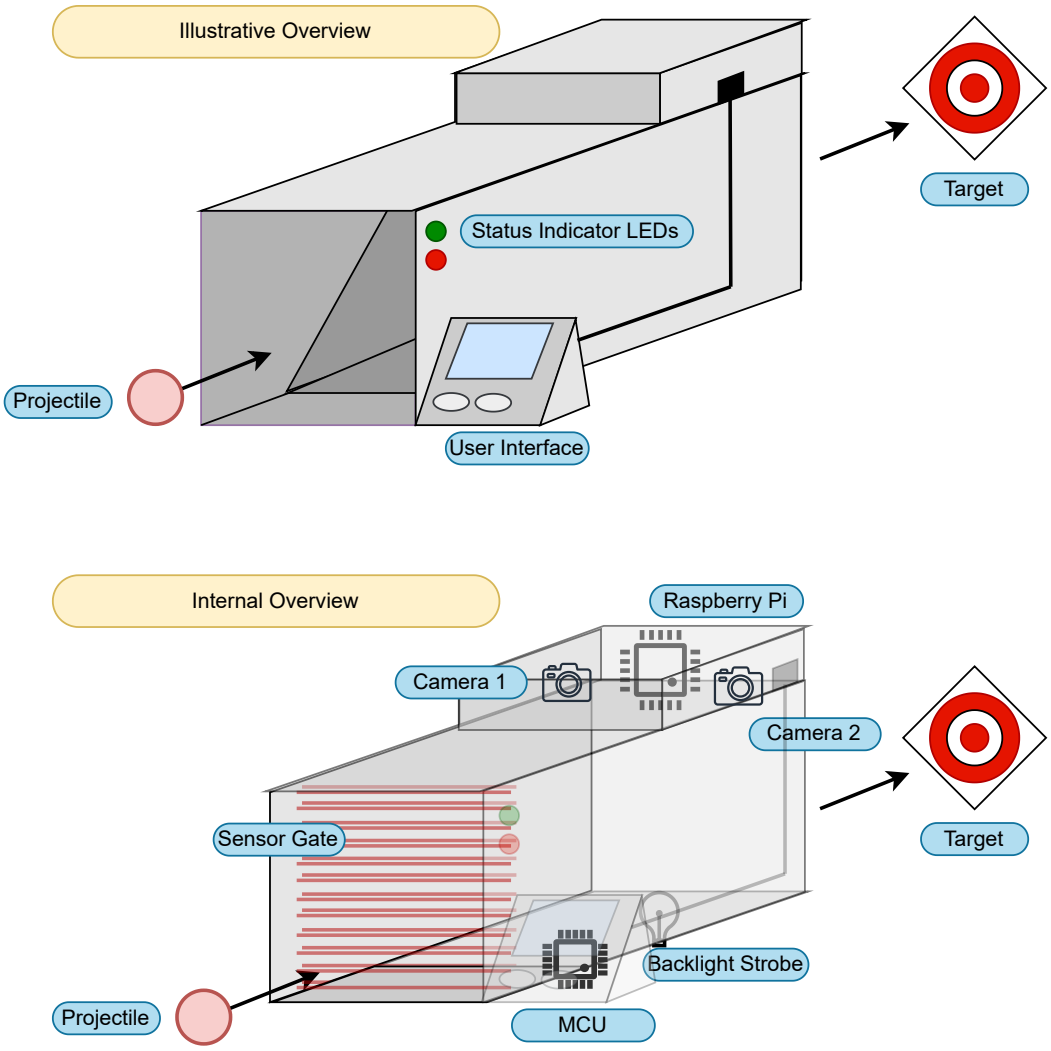


Fig. 6: Prototype illustrations detailing the various subsystems from different viewing points.

The last diagram in this section is a House of Quality showing the relationships (main body) between various customer and engineering requirements, as well as the correlations between those engineering specifications (roof). The relationships in the main body between customer and engineer have different weights depending on how closely one affects the other, with the strongest relationship having a weight of 7 and the weakest having a weight of 1. Each requirement also has a polarity, denoted by an up or down arrow, that displays the desirability said requirement. For example, ”Affordability” would have a positive polarity because a cheaper product is a more desirable trait. In the roof of the House, the correlations between engineering tradeoffs can be seen. A positive correlation means that improving one engineering requirement also benefits another, while a negative correlation means that improving one engineering requirement comes at the detriment of another. A neutral correlation means there is no tradeoff between two requirements if one is improved.

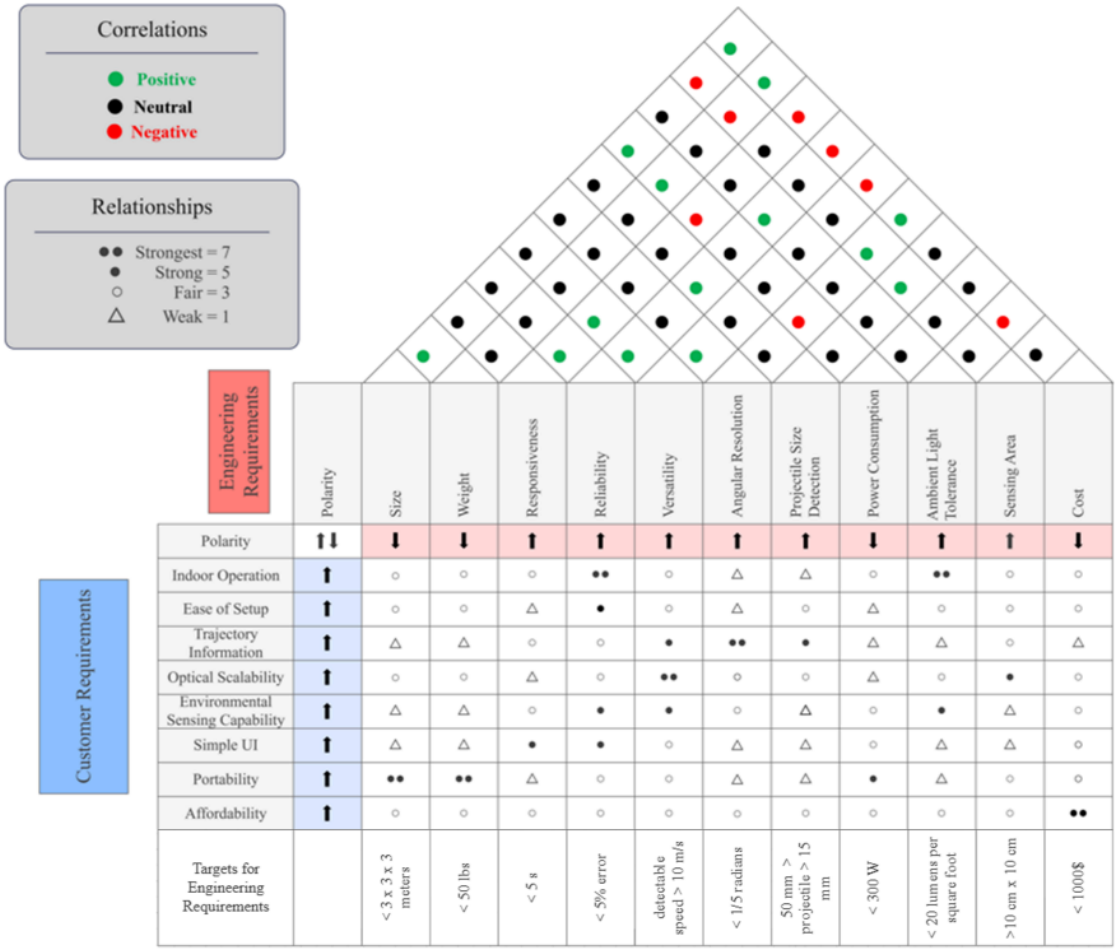


Fig. 7: House of Quality diagram that shows the tradeoffs between the engineering and customer requirements of the project. The customer requirements (left-most column) depict the features that are more important to the user, while the engineering requirements (top row) show the features that are considered for various engineering aspects of the project.

3 Research and Investigation

3.1 Camera Technology

3.1.1 Camera Outcome

The cameras are primarily responsible for capturing an image of the projectile in flight within the strobe imaging chamber. Due to the potential speed of the projectile, these images need to be taken quickly, which is possible with a larger amount of incoming light.

3.1.2 Color vs Monochrome

Monochrome cameras are simpler, and are capable of higher resolutions than color cameras. Monochrome cameras do not provide color information, and therefore do not provide as beautiful of images compared to color cameras. For the goals of this project, a monochrome camera is advantageous for a number of reasons.

First, the camera lens design is significantly simpler due to the elimination of chromatic dispersion, so elements such as doublets and triplets are unnecessary to correct for this. Chromatic dispersion is the difference in effective refractive index of a material for different wavelengths (or colors) of light. This difference in refractive index causes a change in the focal length of simple lenses for different colors of light, so a lens system set up perfectly for yellow light images will blur the blue and red light, causing chromatic aberration. Chromatic aberration can be compensated for by using doublets or triplets, which are lenses with different materials, which have different chromatic dispersions, and are arranged in such a way as to have nearly identical focal lengths for multiple colors of light across the visible spectrum.

Second, monochrome imaging would allow for a bandpass color filter which would greatly reduce the effect of ambient light on the system. With a narrow bandwidth strobe illumination, a bandpass color filter would not significantly reduce the light into the cameras from the strobe system, but would reduce light from outside, such as sunlight reflecting into the strobe imaging chamber during outdoor usage. Sunlight reflecting into the chamber would cause the imaging of the projectile to include times other than when the strobe is illuminated, which would cause a blurry streak that could interfere with the processing of the image.

Third, monochrome imaging allows for higher resolution. Color cameras use a grid color filter in front of a monochrome 2d sensor array in order to get color information. A process of demosaicing is used to recover the color information for the full resolution of the camera sensor; however, information or resolution is lost in the process. Monochrome imaging does not require a color filter grid for retrieving color information, which allows a higher resolution image to be produced. The higher resolution of a monochrome camera would allow more precise projectile tracking, giving higher accuracy on the position,

angle, and speed of the projectile from the strobe photography stage of the system. [Cite TechNexion color vs monochrome]

While color imaging would allow us to produce more impactful strobe photographic images to include with our presentation, the advantages of monochromatic imaging outweigh those of color imaging.

TABLE III: Camera Color Technology Comparison

Feature	**Color**	**Monochrome**
Lens Design Complexity	High	Low
Ambient Light Sensitivity	High	Low
Resolution	Lower	Higher
Aesthetics	Better	Worse

3.1.2.1 Removing the Color Bayer Filter

Removal of the color Bayer filter on a camera can convert a color camera into a monochrome camera. This process is detailed in the youtube video https://youtu.be/y39UKU7niRE [Cite bayer filter removal video] Also discussed in the forum https://forums.raspberrypi.com/viewtopic.ph [Cite raspberry pi forum hq and global shutter monochrome]

3.1.3 NoIR vs Standard

Camera sensors are sensitive to IR wavelengths that are invisible to humans, and therefore many cameras have a filter that blocks those IR wavelengths in order to better replicate human vision. Sometimes it is valuable to be able to detect those IR wavelengths, such as when debugging a TV remote it can be advantageous to point your cell phone camera at the remote to see if it is transmitting. Additionally, since we are using white LEDs for lighting, there will be little IR light that would distort the image. The selection of NoIR vs standard is not of great importance, except for the showcase high-speed strobe photographic images.

TABLE IV: Camera NoIR vs Standard

Feature	**NoIR**	**Standard**
IR Sensitive	Yes	No
Color Fidelity	Worse	Better

3.1.4 Global Shutter vs Rolling Shutter

Global Shutter cameras have a shutter that changes from transparent to opaque or from opaque to transparent all at the same time, and samples all of the pixels at the same time. Rolling Shutter cameras use a technique that either samples the lines of pixels or blocks the light of the lines of pixels one line at a time, which causes a slight time delay between the top of the image being taken and the bottom of the image. Rolling

Shutter is a less expensive technique, it allows for higher resolutions at a lower price, but there are significant issues for high speed photography.

TABLE V: Camera Shutter Technology Comparison

Feature	**Global Shutter**	**Rolling Shutter**
High Speed Photography	Better	Worse
Cost	Higher	Lower
Resolution	Lower	Higher

3.2 Camera Modules

3.2.1 Raspberry Pi Camera Module 2

A low cost option, the camera module 2 is the camera we will use for integration testing while the proper cameras are designed and created. There may be problems with the rolling shutter in the high speed photography, which can be seen for example in the bend of moving propellers when imaging with a cell phone camera.

3.2.2 Raspberry Pi High Quality Camera

The high quality camera has a higher resolution than the global shutter camera, but has rolling shutter distortion that would interfere with proper capture of the high speed projectiles in the system.

3.2.3 Raspberry Pi Global Shutter Camera

The global shutter camera is specialized for high-speed photography. While it has a lower resolution than the High Quality camera, the global shutter camera is capable of taking photographs over 500 times per second (at low resolution), or a single photograph as short as 30 μs using an external trigger. The advantages and rather low cost of the Raspberry Pi Global Shutter camera an ideal choice for the system.

[cite Wilbrandt website]

3.2.4 e-CAM21_CURB - 2MP OV2311

[To be completed]

3.2.5 ArduCam OV9281 Global Shutter NoIR Mono Camera

[To be completed]

Unfortunately, the lens is not interchangeable, so it would not work unless we can use the built-in lens.

TABLE VI: Camera Selection Comparison

Name	Price	Resolution	Shutter	External Trigger	Color
Raspberry Pi Global Shutter Camera	$50	1456 × 1088	Global	Yes	Color
Raspberry Pi Hi Quality Camera	$55	4056 × 3040	Rolling	Yes	Color
Raspberry Pi Camera Module 2	$15	3280 × 2464	Rolling	No	Color
e-CAM21_CURB	$90	1600 × 1300	Global	Yes	Monochrome
ArduCam OV9281 Global Shutter NoIR Mono Camera	$42	1280×800	Global	No	Monochrome

3.3 Illuminative Technology - Strobe Imaging

3.3.1 LED vs Gas Discharge

Light Emitting Diodes (LEDs) as a light source for strobe photography is a cheap and inexpensive light source. The major challenge for LEDs is to provide a large current in a short period of time.

Discharge flash lamps are a good way to create short flashes of light, however they are broadband, expensive, and require high voltages. Different gasses can be used for different wavelengths of light, e.g. Xenon often produces a white light. There are details about the high voltage electric spark discharge triggering that make for a complex use case. Gas discharge lamps are often used to pump lasers. [Cite Perkin Elmer Flashlamp catalog]

Due to the low cost and simplicity of using LEDs, we have selected to use LED lights.

TABLE VII: Strobe Illumination Technology

Feature	LED	Gas Discharge
Wavelength selectivity	High	Low
Cost	High	Low
Voltage Requirements	High ($> 1\,\text{kV}$)	Low (2–20 V)
Intensity	High	Lower

3.3.2 Wavelength

There are advantages and disadvantages to using different wavelengths, or colors, of light.

3.3.2.1 UV

UV leds are typically more expensive due to the exotic high-bandgap materials required. They require a higher forward voltage than other LEDs due to the higher energy of UV photons. The UV light will breakdown materials, causing lower operating lifespans for UV LEDs. However, UV LEDs have the shortest wavelength, which allows for higher resolution imaging than other wavelengths, but the sizes and distances we are working with do not get anywhere near the diffraction limits imposed by the wavelength of light. Cameras are less sensitive to UV light due to a few effects, including the higher energy per photon creating a lower response per unit power, and the absorption of UV light from the optics.

3.3.2.2 Blue

Blue LEDs are inexpensive due to their common use in white LEDs. White LEDs are typically made with a blue LED coated with a phosphor that converts the blue light into a wide range of wavelengths. Cameras are moderately sensitive to Blue light.

3.3.2.3 Red

Red LEDs are ubiquitous, but are often low powered. There are very few high-power red LEDs available, and those that are are quite expensive. Cameras are often very sensitive to red light.

3.3.2.4 IR

IR LEDs in the 850 nm - 960 nm range are some of the cheapest LEDs due to the fact the bandgap of Silicon is in the energy range of IR photons. One of the advantages of IR is the invisible nature, so that it wouldn't interfere with vision nor would it alert enemies in military applications. Cameras are moderately sensitive to IR light in the 700 nm - 800 nm range, however they have a low sensitivity to the light in the 850 nm and up range, with about 0.4 responsivity to 850 nm light.

TABLE VIII: Strobe Illumination Wavelength

Feature	**UV**	**Blue**	**Red**	**IR**
Cost	Moderate	Moderate	High	Low
Longevity	Low	Moderate	High	High
Diffraction-limited Resolution	High	Moderate	Low	Low
High Power Common	Yes	Yes	No	Yes
Visible	No	Yes	Yes	No
IMX296 Responsivity	Low (< 0.5)	Moderate (0.5–0.8)	High (0.9–1)	Moderate (0.5–0.9)

[Cite FSM-IMX296 Datasheet]

3.3.3 LED Field of View

The field of view (FOV) of an LED is the angle at which the light comes out of the LED. The FOV of the LED is one point of comparison, since a wider FOV will provide more even lighting within the sensing area of the strobe imaging chamber, while a narrower FOV will provide more light on target, and therefore be more efficient. A good baseline is to have the LED FOV be at least as big as the FOV of the camera, so that it can approximately illuminate the entirety of the sensing area within the FOV of the camera from between the cameras in the same plane as the cameras. Given that the FOV of the cameras will likely be less than $90°$, an FOV of the LEDs of between $90°$ to $120°$ would be ideal.

3.3.4 Strobe LED component comparison

The top two LEDs are both from Solidlite, the AG3535C-YRR-1000mA-H1 and the AC3535A-YRR-1000MA, the main difference being the viewing angle. For a more uniform illumination, the AG3535C with it's $120°$ viewing angle would be good, but for requiring fewer LEDs due to more concentrated light, the AC3535A with a $90°$ viewing angle would be better. One of the advantages is that the IV curve flattens out around 3.5V, which allows for a simpler driving circuit.

TABLE IX: Comparison of IR LEDs

Parameter	**AG3535C-YRR-1000mA-H1**	**AC3535B-YRR-1000mA-H1**	**AC3535L-YRR-1000MA / AC3535A-YRR-1000MA**
Manufacturer	Solidlite	Solidlite	Solidlite / Solidlite
Price ($)	1.1	1	1 / 1
Current - DC Forward (If) (Max)	1 A	1.5 A	1.5 A / 1.5 A
Radiant Intensity (Min @ If)	430 mW/sr @ 1 A	900 mW/sr @ 1.5 A	175 mW/sr @ 1 A / 550 mW/sr @ 1 A
Viewing Angle	$120°$	$60°$	$150°$ / $90°$
mW/$	1228.08	757.61	814.97 / 1012.17
mW/$ (wall-plug)	2454.55	4650	4650 / 4950

3.4 Computational Technology

3.4.1 Micro-controller

Sensing fast moving projectiles and measuring their physical characteristics is inherently a challenge that absolutely requires high speed processing. Therefore, it is essential that the processor we select is capable of receiving inputs and running calculations much

faster than the projectile is moving. Our stretch goals list the PTC as possibly being capable of tracking projectiles moving as fast as 100 m/s. That is 0.01 seconds per meter and 0.1 milliseconds per millimeter, which corresponds to a frequency of 10 kilohertz. In an ideal world, if a processor is running with a clock speed of 10 kHz, the projectile sensors are 1 mm apart, and the projectile is moving at 100 m/s, the processor would have only a single clock cycle to detect the projectile at both sensors and calculate speed. This is clearly insufficient and eliminates any slower running chips. In order to detect the projectile and to minimize error in doing so, our processor must be running as fast as possible while remaining reasonable in price, accessibility, and complexity. Even if we decide to use an external component for the detection of projectiles, these criteria still hold true as we would want to minimize the amount of time spent running calculations, potentially throwing off the accuracy of triggered flashes and image captures. Luckily, there are a wide variety of micro-processors available to both industry and consumers that operate at significantly higher speeds than 10 kHz while remaining extremely cheap, particularly in the hobbyist space with brands such as Arduino, Raspberry Pi, Espressif, and Adafruit. We prefer microcontroller architectures in the hobbyist space because they are guaranteed to have development boards that will be easy to develop and debug our programs on.

3.4.1.1 Arduino

Arduino is a brand of development boards with a wide range of products available to consumers, at widely varying price points. The most expensive Arduino product at 77.90 euros (about $89.69) is the Arduino GIGA R1, with Wifi capability and a dual core architecture, running at 480 and 240 MHz with the STM32H747XI MCU [Arduino Hardware]. This is quite expensive, and does not include the cost of a separate MCU by itself. The only option in stock for the STM32H747XI MCU costs $19.56. Among the cheapest of Arduino products is the Arduino Nano Every at 14.50 euros (about $16.69), with an ATMega4809 MCU [Arduino Hardware]. The ATMega4809 has a maximum frequency of 20 MHz [ATMega4809]. In the middle of the pack lies the Arduino UNO family, with the UNO R4 Minima sporting a maximum clock frequency of 48 MHz [RA4M1] at a price of 22.00 euros ($25.33) [Arduino Hardware]. At the time of writing, the cheapest option in stock for the ATMEGA4809 itself costs $1.74 on DigiKey, and the cheapest option in stock of the RA4M1 MCU costs $5.36 [DigiKey].

3.4.1.2 Raspberry Pi Pico

Raspberry Pi mostly specializes in Single Board Computers, however, they do have a series of development boards with their Pico line, the Pico 1 at $4.00 and the Pico 2 at $5.00 [Raspberry Pi]. The Pico family consists of 2 MCU architectures, the Pico 1's RP2040, and the Pico 2's RP2350 family [Raspberry Pi]. The RP2040 can run up to 133 MHz [RP2040 Datasheet], and the RP2350 series can run up to 150 MHz [RP2350/RP2354 Datasheet]. The RP2040 and RP2350 chips themselves can each be found on Digikey at prices of $0.70 and $1.00, respectively. However, the RP2040 is not in stock, but the RP235X MCUs are.

3.4.1.3 Espressif

Espressif, like Arduino, hosts a wide variety of products, many of which include features like Wifi and Bluetooth. But with a far wider variety of products than Arduino, Espressif's website hosts a 'Product Selector' that can help narrow down choices based on a customer's needs. Since Espressif's product list still includes products that are Not Recommended for New Designs (NRND) or are only available as sample products [Espressif], it is important to filter for only products under a Mass Produced status, meaning they are available and not too complex or specialized for new designs. Doing this reveals‘ that nearly all Espressif microcontrollers available have processing speeds exceeding 100 MHz and that about 90 have processor clock speeds of 240 MHz, but it does not narrow down options all that much with 162 results. Additionally, most products are System on Chip (SoC) or modules that include Wife and Bluetooth capability that we will not need for the PTC. Coupling this large variety of over 160 different products with features that are unnecessary for this project, and the higher price point as a result, makes choosing an ESP32 microcontroller difficult. Looking at DigiKey and searching up Espressif development board products, most viable options (options that are actually development boards and not particular modules, as DigiKey's search tool is unable to make the distinction) are upwards of $8.00. These cheaper $8.00 options actually are general purpose development boards. The ESP32-C6-DEVKITM is a general purpose development board that costs $8.00 and is built around the ESP32-C6-MINI-1 MCU, which itself costs $4.40 [DigiKey], and has a clock speed of 160 MHz [Espressif Product Selector]. Another option is the ESP32-DEVKITM-1, also $8.00, that uses the ESP32-MINI-1 MCU ($4.07) with a clock frequency of 240 MHz.

3.4.1.4 Technology Comparison

TABLE X: Microcontroller Technology Comparison

Name	**Max Clock Speed (MHz)**	**Development Board Cost (USD)**	**Standalone MCU Cost (USD)**	**Total Cost (USD)**	**In Stock (Mouser, DigiKey)**
Arduino GIGA R1	480	$89.69	$19.56	$109.25	Yes
Arduino UNO R4	48	$25.33	$5.36	$30.69	Yes
Arduino Nano Every	20	$16.69	$1.74	$18.43	Yes
Raspberry Pi Pico 1	133	$4.00	$0.70	$4.70	No
Raspberry Pi Pico 2	150	$5.00	&1.00	$6.00	Yes
ESP32-C6-MINI-1	160	$8.00	$4.40	$12.40	Yes

ESP32-MINI-1	240	$8.00	$4.07	$12.07	Yes

The Arduino GIGA R1 will not be chose. Not only does it include features we do not currently have a use for, it and it's MCU are far and away the most expensive options, which would not be friendly to our budget if we order more for redundancy or because of malfunction. The Raspberry Pi Pico 1 and 2 meet our needs quite well, being among the fastest options for chips and development boards, designed for simplicity, and also being extremely affordable. The total costs outlined in the table are also not final, as it would be prudent to order more individual MCUs in case of component failure or other issues. The Pico series retain a high clock frequency and still remains the cheapest option when ordering more individual IC's, as the cost of the Espressif MCU's is significantly higher. However, it would be worth noting that the ESP-32-MINI-1 is a great secondary option if the Raspberry Pi Picos do not work out.

The Pico architecture is split into two families, the RP2040 used by the Raspberry Pi Pico 1, and the RP2350 series used by the Raspberry Pi Pico 2.

3.4.1.5 RP2040

The RP2040 is a microprocessor that was developed by the Raspberry Pi team for use in their Raspberry Pi Pico development board, and is available to industry and the general public as a standalone IC. According to the datasheet, it runs at 133 MHz, but can be safely overclocked to 200 MHz with some setting changes specified by Raspberry Pi. This is a point in the RP2040's favor, as this overclocking ability is specifically pointed out by the designer and will not negatively affect the processor's architecture.

3.4.1.6 RP2350

The RP2350 family of processors shares many similarities with the RP2040, but offers a variety of enhancements as well. Not only does the RP2350 have a dual core architecture, it operates by default at a clock speed of 150 MHz. This would allow us a higher base clock speed without the extra effort of overclocking, as well as provide us extra processing power to potentially facilitate concurrent calculations at a high speed. The RP2350 however, does not have internal flash storage for writing programs.

3.4.1.7 RP2354

The RP2354 variant of the RP2350 series is exactly the same as the RP2350, but has 2 MB of internal flash storage, reducing the number of external components we would need to incorporate in our PCB in order to get a functioning chip.

3.4.1.8 RP235X A / RP235X B

Of the RP2350 and RP2354, there are also A and B variants of each type. The A type is the base, and has 30 GPIO pins, 4 of which are capable of receiving analog inputs. The

B variant expands upon this with a larger form factor and more GPIO pins; 48 GPIO pins total with 8 of them being capable of analog inputs. While extra GPIO pins may be useful, they may not be necessary for our project.

3.4.1.9 Part Comparison

3.4.1.10 Development Board

Instead of trying to program and constantly debug a chip soldered to a board, it makes sense to also acquire a development board so that software can be developed alongside hardware designs, and both can be tested together.

While there are many of the same options as above in earlier sections for a development board, familiarity and software compatibility are of the utmost importance. And so, the discussion of which development board to choose is a short one. It makes sense to develop functionality with hardware similar to what we will actually be using, so the Raspberry Pi Pico 2 development board was chosen to accompany our choice of RP2354A MCU, a chip in the same series as those used in Raspberry Pi Pico 2's.

3.4.2 Image Processor

One objective of the PTC project is to capture multiple images of the projectile in flight and extract spatial information from these images. Due to the complexity of digital image processing, a normal micro-controller does not have sufficient processing power to handle the procedural analysis of photographs. (drop proof) To solve this issue yet still comply with project constraints, a more powerful computer will be used to handle image processing. The image processor will only be used for image processing, and all other signals and data must be handled by our RP2354A micro-processor. Much like our options for the micro-processor, there are a few small computers that are capable of image processing, such as the Raspberry Pi, Orange Pi, and Odroid.

3.4.2.1 Raspberry Pi

3.4.2.2 Orange Pi

3.4.2.3 Odroid

3.4.2.4 Technology Comparison

Insert Table Below:

Of these, we chose Raspberry Pi...

3.4.2.5 Raspberry Pi Module 4

3.4.2.6 Raspberry Pi Module 5

3.4.2.7 Raspberry Pi Compute Module 5

3.4.2.8 Parts Comparison

Insert Table Here

We chose Raspberry Pi 5 because it had 2 camera ports, cheaper than Compute Module 5...

3.4.3 Interfacing Technology

3.4.3.1 Buttons

Buttons are some of the most commonly seen components of user interfaces. Buttons work by either closing or opening a circuit, and this change in voltage can be detected by the MCU to perform an action. Each button will occupy a single GPIO pin of the MCU, and software will be able to control its function, as well as filter out bouncing, a phenomenon that occurs when the physical contacts of the button bounce on the conductors, sending false signals to the MCU. De-bouncing, as the process is called, can also be achieved with other methods.

3.4.3.2 Switches

Switches are also very commonly seen components of user interfaces, and work similarly to buttons, either closing or opening a circuit, or switching between circuits, but for an indefinite amount of time rather than a short moment like a button press. A switch could occupy one or more GPIO pins of the MCU, and software could be used to detect which state the switch is in. Since a switch remains in its position until an action is performed by the user, the software may be less fluid than if the component was a button.

3.4.3.3 Potentiometers

Potentiometers use a wiper on a resistive pad to create variable resistance, adjustable by the user. With this variable resistance, variable voltages can be sensed by the MCU on a single GPIO pin, and such values can be processed to finely tune or adjust some setting. Potentiometers are, however, limited in the range that they can spin and the voltage range and resolution they can provide, all dependent on the type of potentiometer and the voltage source in use. Additionally, since this is a resistive circuit component, power will be consumed and let off as heat.

3.4.3.4 Rotary Encoders

Rotary Encoders are a more complex component of user interfaces, but are commonly seen on devices that require a large range of input values, like volume and frequency

controls on a radio. While such uses cases can be serviced with a potentiometer, Rotary Encoders are more conductive than resistive, and do not consume much power like a potentiometer would. Additionally, they do not have a limited range of turns, and can be rotated indefinitely. Consequently, they do need more GPIO pins to use, requiring two for encoders without a built in button, and three for encoders with a built in button. Rotary Encoders function by alternating two voltage signals as they spin, with an inherent phase shift between the two signals. The MCU is then able to determine which signal alternates first, essentially conveying whether the rotary encoder was rotated clockwise or counter-clockwise.

3.4.3.5 Touch Pads

Touch pads are commonly seen on laptops, and are used to control cursor position on screen. This will use up many GPIO pins, and since our device's functions should be simple to the user, will likely unnecessarily complicate and obfuscate our design by forcing the user to click and drag to perform actions instead of just pressing a button.

3.4.3.6 Touch Screens

Touch screens are popular in advanced technology, and serve as a great way to display information and also have the user interact with this information at the same time, in the same localized space. Touchscreens, however, are more expensive than other components, and will no doubt take up a significant amount of GPIO pins.

3.4.3.7 Technology Comparison

Insert Table Here...

3.4.3.8 Part Comparison

3.5 Illuminative Technology - Light Gate

3.5.1 Light Sources

3.5.1.1 LEDs

LEDs are semiconductor devices that are capable of emitting light through a process called electroluminescence. This happens when charge carriers (electrons and holes) recombine within a semiconductor junction and produce photons. In an LED, this type of emission is mostly driven by spontaneous emission, which means that the emitted photons have a random direction, phase, and polarization upon emission. These types of light emitter lack an optical cavity as well, leading to an incoherent light source that has a broad spectral distribution. [utmel, unitopled, libretexts]

Due to their broad emission angle, LEDs are useful for many applications that require wide or diffuse illumination. Examples of this include general lighting, displays, and indicators. LEDs generally operate with low to moderate efficiency and can achieve

long operating lifetimes as well. These light sources typically have lower optical power density and broader spectral output compared to laser diodes, making them less suited for high-precision optical measurements. [utmel, unitopled]

3.5.1.2 Laser Diodes

Laser diodes, like LEDs, are semiconductor light emitters that make use of carrier recombination to produce photons. Unlike LEDs, however, laser diodes utilize stimulated emission within an optical resonant cavity to produce monochromatic light that is coherent and highly directional. In stimulated emission, an incoming photon causes the release of an identical photon in phase and direction. Since the optical cavities operate as a positive feedback loop, these photon interactions get amplified, leading to a narrow beam of light that is concentrated at a single wavelength. [utmel, lumimetric]

Due to these properties, laser diodes can generate significantly higher optical power, maintain a narrow spectral linewidth, and support fast modulation at timescales much shorter than LEDs. These characteristics make laser diodes ideal for applications such as communications, sensing, and high-speed measurement systems. Drawbacks of these components are that they typically tend to be much more complex, have higher costs, and require better thermal management techniques. [utmel, lumimetric]

3.5.1.3 Technology Comparison -

For the PTC project, we want modulated illumination across the sensing region, enabling reliable projectile detection when the gate is disturbed by a passing object. This light source must maintain sufficient intensity and uniformity to ensure consistent triggering while also being resistant to variations in ambient lighting. Due to these conditions, the light source chosen for this subsystem was the laser diode.

The coherent and greatly directional output of laser diodes enables the design to illuminate specified detection zones with minimal spreading and high intensity. Additionally, the advantages of laser diodes will improve timing accuracy and reduce the susceptibility of the sensor gate to ambient lighting variations. In contrast, LEDs produce broad light that would require large optics or complex lens systems to achieve a similar power density, increasing overall system complexity without delivering equivalent precision to the laser diode.

TABLE XI: Illuminative Technology Comparison

Feature	**LED**	**Laser Diode**
Emission Type	Spontaneous	Stimulated
Coherence	Incoherent	Coherent
Spectral Width	Broad	Narrow
Beam Directionality	Broad	Very Directional
Optical Power Density	Lower	High
Modulation Capability	Lower	Higher
Thermal Sensitivity	Low	Higher

Cost	Low	Higher

3.5.2 Wavelengths

3.5.2.1 Ultraviolet

Ultraviolet (UV) light occupies the portion of the electromagnetic spectrum with wavelengths between 100 and 400 nm. Due to its relatively short wavelength, UV light carriers higher photon energy compared to visible and infrared radiation. This higher energy enables strong interactions with biological compounds and photosensitive materials, making it very useful in applications for fluorescence analysis, photolithography, and sterilization. UV light can be divided into three main sub-categories: UV-A (315-400 nm), UV-B (280-315 nm), and UV-C (200-280 nm). [reddotled, iluxred]

Although UV light is very useful, there are several challenges associated with using this light for optical sensing systems. For instance, many common optical materials and protective coatings absorb UV radiation, reducing the amount of light that can be sensed on photodetectors. Additionally, common silicon photodiodes have low sensitivity in the UV region because much of the incident light is absorbed by the protective window materials or the surface of the detector itself before photoelectric conversion can take place. These limitations can significantly increase the difficulty of trying to use ultraviolet detection for optical sensing without the use of specialized optical components. [hamamatsu, izakscientific]

3.5.2.2 Infrared

Infrared (IR) light consists of wavelengths longer than visible light, ranging from 700 nm all the way to 1 mm. In optical sensing systems, the near-IR region (700-1000 nm) is most commonly used because it can be easily generated by semiconductor emitters and efficiently detected by silicon photodiodes. Infrared emitters are used for a wide range of applications, such as remote control, night-vision imaging, and proximity sensors. [lumex]

One advantage of IR illumination is its ability to penetrate materials and scattering media more effectively than shorter wavelengths. Silicon photodiodes are able to maintain high responsivity throughout much of the near-IR region, often reaching peak sensitivity between 800 nm and 900 nm. This makes infrared illumination particularly viable for optical sensing systems that use photodiode detection. However, one drawback of IR light is that it is invisible to the human eye, making beam alignment and system testing particularly difficult. [izakscientific]

3.5.2.3 Visible (Red)

Visible light spans wavelengths from approximately 400 to 700 nm, corresponding to colors that are perceptible to the human eye. The red portion of the spectrum (620-700 nm) is particularly useful for optical sensing systems due to its compatibility with silicon

photodetectors. Red light sources are frequently used in optical sensors, alignment tools, and various measurement systems because they provide a good balance between practical usability and detector sensitivity. [reddotled]

Silicon photo-detectors exhibit strong responsivity across the visible spectrum, especially the red region, allowing efficient conversion of photons into electrical signals. Furthermore, visible red light offers practical advantages during system development and operation, since the light can be easily observed by users. This visibility simplifies beam alignment and calibration, making troubleshooting processes much simpler compared to infrared systems. For these reasons, red wavelengths are commonly selected for optical sensing applications that require reliable detection and simpler system setups. However, one limitation of visible light is its sensitivity to ambient light. Since many common indoor lighting sources also emit strongly within the visible spectrum, background light can introduce noise and reduce the signal-to-noise ratio of optical detectors. [izakscientific]

3.5.2.4 Technology Comparison - Wavelengths

When selecting an illumination wavelength for an optical sensing system, several factors are to be considered, including detector sensitivity, ambient light susceptibility, alignment convenience, and overall system complexity. UV light offers high photon energy and strong material interactions, but often requires specialized optical components due to factors such as reduced detector sensitivity and material absorption. IR light provides strong compatibility with many photo-detectors and is widely used in sensing systems, but its invisibility to the human eye can make system alignment more difficult. [hamamatsu, lumex]

Visible red light provides many of the advantages of infrared illumination and adds more as well. Silicon photodiodes maintain high responsivity in this region, and the emitted light is visible, allowing simple system alignment and calibration. Due to these factors, the visible red wavelength was chosen as the primary wavelength for the sensor gate subsystem.

TABLE XII: Wavelength Technology Comparison

Feature	**UV**	**Infrared**	**Visible (Red)**
Typical Wavelength Range	100–400 nm	700 nm – 1 mm	620–700 nm
Photon Energy	High	Low	Moderate
Detector Compatibility	Lower for standard Si detectors	High for Si detectors	High for Si detectors
Visibility to Human Eye	No	No	Yes
Common Applications	Sterilization, fluorescence, photolithography	Remote sensing, night vision, proximity sensors	Optical sensing, alignment systems
Alignment Convenience	Difficult	Difficult	Easy

System Complexity	Higher	Moderate	Lower

3.5.3 Types of Lasers

3.5.3.1 Point Laser

A point laser is a light source that emits a highly focused, collimated beam concentrated into a small spot. This type of laser typically produces a Gaussian beam profile, where the intensity is highest at the center of beam and decreases radially outward. Due to its high optical power density and narrow beam divergence, a point laser is useful in applications requiring precise alignment, localized illumination, or high-accuracy targeting. Examples of this can be found in systems such as interferometers, laser cutting tools, and laser-targeting technologies.

The high coherence nature of a point laser allows for extremely accurate spatial resolution, making it particularly useful in systems where only a single point in space being disturbed is desired. However, it is this same property that limits the point laser in applications that require coverage over a large sensing region. For applications like this, additional optical components will be required to expand the effective detection area, increasing overall system complexity and costs.

3.5.3.2 Line Laser

A line laser is a laser source that has been modified to project the laser beam in a line rather than a point. This is typically achieved through the use of optical components, such as cylindrical lenses or other diffractive elements. These optics spread the beam in one dimension while maintaining collimation in the perpendicular direction. Due to this modification, the emitted light forms a continuous line across a surface, providing a much larger area of illumination than that of a point laser. [edmundoptics]

These types of laser are commonly used in applications such as alignment systems, dimensional measurement systems, and machine vision technologies. Line lasers are particularly beneficial in systems where you want to illuminate an entire cross-sectional region simultaneously. However, since the optical power is distributed over a larger area, the intensity at any given point across the line is lower than that of a point laser, which can increase the need for more robust sensing in some systems. [edmundoptics]

3.5.3.3 Technology Comparison - Types of Lasers

When selecting between point and line lasers, the primary consideration for the PTC project is the tradeoff between spatial coverage and optical intensity. The sensing system benefits from illumination across a defined detection region rather than a single point. A line laser enables the creation of an optical ”gate” that can be interrupted by a passing projectile. While the reduced intensity of a line laser compared to a point laser is a consideration, the benefits provided by the increased coverage of a line laser outweigh the drawbacks.

TABLE XIII: Laser Type Comparison

Feature	**Point Laser**	**Line Laser**
Beam Shape	Focused point	Expanded line
Optical Power Density	High	Lower (distributed)
Coverage Area	Small	Large (1D spread)
Alignment Sensitivity	High	Lower
Need for Scanning	Often required for coverage	Not required
System Complexity	Higher (for large coverage)	Lower (built-in coverage)

3.5.4 Component Selection - Laser Diodes

3.5.4.1 Adafruit 1057 Laser Diode

The Adafruit 1057 laser diode module is a compact red line laser designed for low-power optical applications. This device operates at a wavelength of approximately 650 nm and provides an optical output power of up to 5 mW. It is designed to operate over an input voltage range of 2.8-5.2 V, with a maximum current draw of 25 mA. This is a Class IIIR laser that is made to have direct compatibility with low-voltage embedded systems without the need for complex driver circuitry. This module incorporates an integrated cylindrical lens that converts the emitted beam into a line pattern with a wide fan angle of approximately 120°, allowing for broad area coverage. The physical dimensions of this device are 10 mm in diameter and 32.8 mm in length, with wire length of 100 mm. This makes the module very easy to mount and integrate into compact systems. While this module provides low power consumption and overall simplicity, it's performance is primarily suited for general-purpose sensing tasks. The datasheet for this particular device is limited in optical specifications, indicating that it is not optimized for high-precision measurement applications. [1057-Web]

3.5.4.2 Quarton Inc. VLM-650-23 LPO

The VLM-650-23 LPO is a miniaturized red line laser diode designed for precision alignment and sensing applications. This device operates in a wavelength range of about 640-655 nm and produces a line-shaped beam using a cylindrical lens system. It is designed to operate at an input voltage of 3-6 V with a maximum operating current of 50 mA. The module produces a total optical power of less than 5 mW, while the accessible laser output is limited to less than 1 mW, classifying this diode as a Class 1 laser device. The optical system consists of a collimating glass lens and a cylindrical quartz line lens, producing a line beam with a fan angle of $50° \pm 5°$. The resulting line has a width of less than 1.5 mm at a distance of 2.5 meters and a line accuracy of approximately 4/1000, which is a deviation of less than 1.6 mm at 400 mm. Mechanically, this device has a cylindrical diameter of 5 mm and a length of 22.8 mm, with an approximate mass of 6 g. It operates in a temperature range of -20°C to $+65$°C and includes various safety features such as surge protection and reverse polarity features. This device has suitable

characteristics for sensing applications, but can be a little restricted due to its smaller fan angle. [Quarton-diode-1]

3.5.4.3 Quarton Inc. VLM-650-27 LPT-10

The VLM-650-27 LPT-10 is an industrial red line laser module designed for applications requiring higher optical output and a large coverage area. This device operates in a wavelength range of 645-655 nm and requires an input voltage of 2.6-6 V. The VLM-650-27 LPT-10 is able to support two output regimens, with the first being a low-power configuration with an optical output of less than 2 mW and the second being a higher-power configuration producing 12-15 mW. These two modes have operating currents of below 35 mA and below 100 mA respectively. For this project, the low-power configuration was considered as this is a safer laser being categorized as a Class 1 laser device. The optical system consists of an ashperical plastic collimating lens and a quartz cylindrical lens, producing a line-shaped beam with a fan angle of $90°$. The projected line has a width of 3 ± 0.5 mm at 5 m with an accuracy of approximately ± 1 mm at 5 m. Physically, this device has a cylindrical brass housing with a diameter of 12.5 mm and a length of 30 mm and operates in a temperature range of $+10°$ to $+40°$C. This device is intended for general industrial alignment applications, with a recommended working distance of more than 2 meters on average. This device has a fairly moderate fan angle, making it better suited for more compact systems. [Quarton-diode-2]

3.5.4.4 Quarton Inc. VLM-635-55 LPO

The VLM-635-55 LPO is an industrial-grade red line laser module that is optimized for precision alignment and high visibility. This device operates in a wavelength range of 630-645 nm and requires an input voltage of 3-6 V, with a maximum operating current of less than 80 mA. This module produces a total optical output power of 4-5 mW, while the accessible laser output is limited to less than 0.39 mW, making this a Class 1M laser device. The optical system uses both a glass collimating lens and a glass line lens to produce a uniform, line-shaped beam. Similar to the VLM-650-27 LPT-10, the VLM-635-55 LPO generates a line that has a width of 3 ± 0.5 mm at 5 m with an accuracy of approximately ± 1 mm at 5 m. This laser has a fan angle of $60°$, which is less than that of the VLM-650-27 PLT-10. This laser module features an aluminum housing with a diameter of 10 mm and a length of 38 mm, with a mass of about 6 g. It operates in a wide range of temperatures, from $-20°$C to $+65°$C and can be stored in temperatures up to $+85°$C. This laser was designed for precision alignment applications as well, but with a bigger working distance of up to 20 meters. Like the VLM-650-23 LPO, the fan angle is a limiting factor due to its smaller beam spread over short distances. [Quarton-diode-3]

3.5.4.5 Component Comparison - Laser Diodes

When comparing these laser diode modules, several key tradeoffs are evident, including optical power, beam coverage, physical size, and system integration complexity, to name a few. The Adafruit 1057 module emphasizes simplicity, low power consumption, and a very wide fan angle, but offers limited optical precision characterization compared to the industrial-grade alternatives. The VLM-650-23 LPO provides a balanced combination

of compactness, low power operation, and a well-defined line being generated, making it well-suited for applications requiring greater optical control. The VLM-650-27 LPT-10, on the other hand, delivers higher optical power and wider coverage, improving performance in more compact systems and in environments with greater amounts of ambient light. One drawback of these improvements though is the increased cost and and power demand required by this laser. The VLM-635-55 LPO offers enhanced visibility due to its shorter operating wavelength and higher brightness, but also has the same disadvantages as the VLM-650-27.

Cost further differentiates these different lasers. The Adafruit 1057 is the cheapest, being prices at $8.95, whereas the Quarton Inc. VLM-650-23 LPO, VLM-650-27 PLT-10, and VLM-635-55 LPO are significantly more expensive at $79.40, $75.16, and $83.22, respectively. Although the Quarton Inc. modules provide improved optical performance, their increased cost is not justifiable for a system focused on reliable projectile detection rather than high-precision measurements. For the PTC project, the primary objective of the sensor gate is to achieve consistent detection across a defined sensing region while maintaining a compact, low-power design while also being cost-effective. The Adafruit 1057 meets our requirements by providing a very wide fan angle ($120°$) for sufficient coverage in a compact system, while also having adequate optical output power. [digikey-laser-diodes]

TABLE XIV: Laser Diode Component Comparison

Characteristic	**Adafruit 1057**	**VLM-650-23 LPO**	**VLM-650-27 LPT-10**	**VLM-635-55 LPO**
Wavelength (nm)	∼650	640–655	645–655	630–645
Optical Output Power (mW)	≤ 5	< 5 (total), < 1 (accessible)	< 2 (low power), 12–15 (high power)	4–5 (total), < 0.39 (accessible)
Laser Class	Class IIIR	Class I	Class I (low power mode)	Class 1M
Operating Voltage (V)	2.8–5.2	3–6	2.6–6	3–6
Operating Current (mA)	≤ 25	≤ 50	< 35 (low) < 100 (high)	< 80
Beam Shape	Line	Line	Line	Line
Fan Angle (°)	~ 120	50 ± 5	~ 90	60
Line Width	N/A	< 1.5 mm @ 2.5 m	3 ± 0.5 mm @ 5 m	3 ± 0.5 mm @ 5 m
Line Accuracy	N/A	$\sim 4/1000$ (< 1.6 mm @ 400 mm)	± 1 mm @ 5 m	± 1 mm @ 5 m
Collimating Lens	Integrated (unspecified)	Glass	Aspherical plastic	Glass
Line Lens	Cylindrical (integrated)	Quartz cylindrical	Quartz cylindrical	Glass
Housing Material	N/A	N/A	Brass	Aluminum
Dimensions (mm)	$\phi 10 \times 32.8$	$\phi 5 \times 22.8$	$\phi 12.5 \times 30$	$\phi 10 \times 38$
Mass (g)	N/A	~ 6	N/A	~ 6
Operating Temperature (°C)	N/A	-20 to $+65$	$+10$ to $+40$	-20 to $+65$
ESD Protection	N/A	Yes	30 kV	30 kV
Recommended Distance	N/A	N/A	> 2 m	1–20 m
Cost ($)	8.95	79.40	75.16	83.22

3.6 Sensor Technology

3.6.1 Photodetectors

3.6.1.1 Photodiodes

Photodiodes are semiconductor devices that convert incident light into an electrical current through the generation of electron-hole pairs within a p-n (or p-i-n) junction. Photodiodes

are commonly operated in a reverse-bias mode, meaning that the diode provides a high resistive path to the flow of current. When photons are absorbed in the depletion region, charge carriers are generated and flow across the junction under a reverse-bias, thus producing a photocurrent proportional to the incident optical power. Operation in this mode allows for improved response speed and reduced junction capacitance, making them suitable for high-speed detection tasks.

Silicon photodiodes have strong responsivity across the visble red and near-IR spectrum, with peak responsivities typically ocurring between 600-900 nm. At wavelengths near our desired wavelength of 650 nm, responsivity values typically range from 0.4 to 0.6 A/W. These devices are very desirable to precision timing applications due to their very fast response times, often in the nano- to microsecond range. They are able to respond this quickly due to their low capacitance and lack of internal gain mechanisms, however, this also results in a naturally small photocurrent being generated from the device. This necessitates the inclusion of transimpedance amplifiers (TIA) or other gain amplification circuits in order to generate a usable signal. [hamamatsu, thorlabs]

3.6.1.2 Phototransistors

Phototransistors are another light-sensitive semiconductor device like the photodiode. However, these photodetectors operate more similarly to a bipolar junction transistor, with the incident light acting as the base current. When photons are absorbed in the base-collector junction, a small photocurrent is generated and then amplified by the current gain of the transistor. This internal gain results in significantly higher output current compared to photodiodes, often reaching micro- to milliampere ranges under similar lighting conditions. [onsemi]

The increased sensitivity of phototransistors allows for the detection of much lower light levels without the need of additional amplification circuitry. One drawback of this, however, is that the internal amplification often leads to slower response times due to the charge storage effects within the device. Additionally, phototransistors tend to exhibit reduced linearity, greater sensitivity to thermal fluctuations, and have increased noise compared to photodiodes. As a result, phototransistors are commonly used in applications where sensitivity in low-light situations are more important than speed and precision. [onsemi, thorlabs]

3.6.1.3 Technology Comparison - Photodetectors

When comparing photodiodes and phototransistors, several key tradeoffs are evident, including response speed, sensitivity, and circuit complexity. Photodiodes provide a much higher response speed and low noise, with the drawback being the necessity for additonal amplification circuitry. The phototransistor offers much larger output signals compared to photodiodes, at the cost of increased noise and lower response speeds. For the PTC project, the primary requirement for the photodetector subsystem is to detect rapid interruptions of the optical signal as a projectile passes through the sensing region. Therefore, we believe that photodiodes are the more suitable choice for this system as they provide the necessary speed and precision required to meet our goals for projectile detection.

TABLE XV: Photodetector Technology Comparison

Characteristic	**Photodiode**	**Phototransistor**
Operating Principle	Photon-generated carriers in reverse-biased junction	Photon-generated base current amplified by transistor gain
Internal Gain	None	High ($\beta \approx 50$–300)
Output Current Range	nA to μA	μA to mA
Responsivity @ 650 nm	~ 0.4–0.6 A/W	Gain-dependent (not directly specified)
Response Time	ns to μs	μs to ms
Bandwidth	High (MHz range)	Low (kHz range)
Linearity	High (proportional to incident light intensity)	Nonlinear (gain and saturation effects)
Noise	Low	Higher (due to internal gain)
Sensitivity	Moderate	High
Saturation Behavior	Minimal	Susceptible at high illumination
Temperature Sensitivity	Lower	Higher
Circuit Complexity	Requires amplifier (TIA)	Simple (resistor load sufficient)
Power Consumption	Low	Moderate

3.6.2 Component Selection - Photodiodes

3.6.2.1 [Insert photo-detector 1]

3.6.2.2 [Insert photo-detector 2]

3.6.2.3 [Insert photo-detector 3]

3.6.2.4 [Insert photo-detector 4]

3.6.2.5 Component Comparison - Photodiodes

3.7 Reflector Technology

3.8 Transimpedance Amplification Technology

Since it has been established that the PTC will shoot lasers into photo-diodes and wait for the interruption of the beam in order to detect projectiles, a system must be put in place to make the output of the photo-diodes legible to the MCU. The solution is a Transimpedance Amplifier that will convert the current output of the photodiodes into a

voltage, and then impart massive gain on that voltage to get it high enough to be read by another device.

3.8.0.1 Dedicated IC

Dedicated ICs exist for transimpedance amplifiers, however many options are quite expensive.

3.8.0.2 General Purpose Design - Op-Amps

Using general purpose op-amps, a transimpedance amplifier can be designed and assembled that meets our requirements. However, such designs may be more susceptible to excessive noise and may have a lower bandwidth, limiting response speed.

3.8.0.3 General Purpose Design - BJTs

Using general purpose BJTs, a transimpedance amplifier can be designed and assembled that meets our requirements. However, such designs may require more components to impart a similar gain and may be more susceptible to excessive noise.

3.8.0.4 Technology Comparison

Insert Table Here...

3.8.0.5 General Purpose Design - Op-Amps

3.8.0.6 TL084

3.8.0.7 Something with higher bandwidth

3.8.0.8 Part Comparison

Insert Table Here...

3.9 Transimpedance Amp Output Sensing Technology

When the laser light beam is interrupted by a projectile, the reduction of incident light on our photo-diode sensors will cause a drop in output voltage. The projectile will interrupt the beam twice, and so both of these transient signals must be detected and interpreted so that the MCU can use the time difference between these two signals to calculate the projectile's speed and estimate when the strobe signal should be activated for image capture. Below are a small variety of methods we can use to accomplish this.

3.9.0.1 Time of Flight Sensing

With a time of flight sensor, at the added cost of more components and circuitry, it would be possible to send two separate triggers (the signals from the projectile interrupting the

beam) and receive a coded message that would detail how much time passed between these two triggers, possibly on the nanosecond scale. This has the potential benefit of reducing latency and also providing more accurate timing results without relying purely on the processing speed of the MCU. However, this likely still would require some clever circuitry to make the output of the transimpedance amplifier suitable for the ToF sensor.

3.9.0.2 Level Detection

In an ideal environment, the output of the photodiodes would be constant until the projectile arrives, and therefore the output of the transimpedance amplifiers would be constant. Theoretically then, it is conceivable that coupling this output with capacitors would prevent a voltage from reaching the end of the signal chain, and when the projectile does arrive, a transient would then be generated, propagating past the capacitor and into a GPIO pin of our MCU to be detected and timed. Of course, there is much potential for our environment to be non-ideal, and this method may prove highly unreliable.

3.9.0.3 Standalone MCU Measurement

This is the brute force method, and involes constantly polling the output values of our transimpedance amplifier to determine an average value and threshold values above and below it. Whenever the transient signal arrives, it should be possible for the MCU to detect it and begin counting clock cycles to determine the time between transient signals. This relies purely on MCU clock speed, and may be susceptible to noise and place hard limitations on how fast a projectile can be to be accurately measured.

3.10 Power Technology

3.10.1 Voltage Regulation

To ensure components are powered properly, and that their requirements and specifications are met and not exceeded, voltage regulation is necessary. This can be achieved with a small variety of technologies.

3.10.1.1 Switching Voltage Regulator

Employs high frequency switches to interrupt the source voltage and produce a waveform with a lower average voltage. A filter is then used to turn this resultant square wave into a smooth DC output. Due to the alternating nature of the output, a small ripple will still appear on the output, varying the output voltage by a small amount. Suitable for large voltage differences from input to output.

3.10.1.2 Linear Voltage Regulator

Output voltage is controlled linearly by an Op-Amp and transistor switch, causing excess power to be emitted as heat. Typically unsuitable for large voltage differences from input

to output. Exhibits dropout behavior, where if the input voltage falls to close to the output voltage, the output voltage can no longer be correctly maintained, and falls as a result.

3.10.1.3 Low Dropout Voltage Regulator

Just like a standard linear voltage regulator, but with optimized circuitry inside to minimize the effects of dropout behavior, greatly reducing the margin necessary between input and output voltage to maintain a proper output voltage. This allows for greater headroom of the input voltage level so that any unexpected dips likely will not interfere with the output.

3.10.1.4 Zener Diode

A Zener diode is a type of diode that has a controlled avalanche breakdown, and is meant to operate in such a state. The breakdown voltage can be manufactured, and this allows a us to create a rudimentary voltage regulation system by placing a zener diode in reverse bias from the power rail to ground. This accounts for over-voltage conditions by causing a short when the power rail voltage exceeds the breakdown voltage of the Zener diode, which may not be ideal if such a condition persists, and alone does not account for under-voltage conditions.

3.10.1.5 Technology Comparison

Insert Table Here...

The LEDs in use for the strobe, the JR5050 6V Q-Class, typically have a forward voltage of 6V while maintaining a current draw of 500mA [White LED J Series Datasheet]. This means that in order to power them in series, a large voltage would be necessary. This is a driving factor in our choice for power supply, as many LEDs are necessary to create a bright enough environment for image capture, and we would like to reduce the amount of parallel circuits drawing large currents. For example, 24V would allow us to place 3-4 LEDs in series.

For the actual Strobe circuits, a large capacitor will be charged to the supply voltage, and a Low Dropout Regulator can be used to maintain a suitable voltage for the LEDs upon discharge, while the capacitor supplies the necessary current.

The MCU requires an input voltage of 3.3V, and so a switching regulator would be most suited for powering the MCU, as it would be able to handle a higher voltage and do so efficiently.

The Line laser in use also requires a lower voltage at a different level than the MCU, so another switching regulator would also be suitable here.

3.10.1.6 AP5100

The AP5100 is suitable for our needs, being able to handle a rated input voltage anywhere from 4.75V to 24V, while supplying 1.2A [AP5100 Datasheet]. The Raspberry Pi RP2350

Hardware Design guide suggests a 1A supply current to be suitable for computationally intensive projects, so the AP5100's current capabilities are suitable. The AP5100 is adjustable, and its datasheet also supplies an example application circuit that perfectly suits our needs, a 3.3V regulator [AP5100 Datasheet].

3.10.1.7 R78K5.0-0.5

Like other switching regulators, the R78K5.0-0.5 can efficiently regulate voltages, and this one would also be able to handle a 24V source. The added advantage of this regulator is that it is completely housed within a single package, requiring no external components. As is, this regulator can be directly connected to a circuit.

3.10.1.8 Other Switching Regulator

3.10.1.9 LD1086DTTRY

The LD1086 is a reasonably priced option that will allow us to deliver large currents of up to 1.5A while maintaining a constant voltage. Additionally, the output is adjustable with a simple formula that can be calculated by hand [LD1086DTTRY Datasheet].

3.10.1.10

3.10.1.11 Other LDO

3.10.1.12 Parts Comparison

Insert Table Here...

We went with the AP5100 and the LD1086 because...

3.10.2 Current Regulation

- Normal Resistor, NTC, IC current limiter, current source

3.10.2.1 Soft Start Circuit

Using a MOSFET, it is possible to implement a soft start circuit that slowly reduces resistance as the inrush current settles, allowing inrush current to be effectively limited without reducing the voltage on the power rail by a large amount like a resistive element would. This requires more and more expensive components than a simple resistor or thermistor, but would reduce power loss and keep the power rail at a consistent voltage that will not change much even if the load draws much more current.

3.10.2.2 Thermistor

A thermistor changes resistance based on its temperature. With a Negative Thermal Coefficient thermistor, it would be possible to limit inrush current in a simple manner. However, because this component is still resistive, it would still have a voltage drop

across itself, reducing the available voltage on the power rail. Thermistors also often have long 'reset' times, on the order of severl seconds to tens of seconds, to return to their base state and resistance. This means that inrush currents may not be effectively dealt with if the system is shut off then immediately turned on again, as the thermistor may still be at a temperature that correlates with low resistance and therefore minimal inrush limiting.

3.10.2.3 Typical Resistor

While simple, a typical resistor must be able to handle the power it absorbs, and this could mean that this resistor ends up being larger than others in the circuit, and depending on how much power is expended, could necessitate a heat sink or other method of dispersing heat due to lost power. If used in the main power rail, a resistor would also limit the maximum voltage available on the power rail due to the voltage drop across the resistor, and higher load currents would increase this drop.

3.10.2.4 Technology Comparison

Insert Table Here...

Since we intend to rely on a large capacitor bank to locally provide current to the strobe circuit, and since the LEDs of the strobe have such large forward voltages, managing inrush current with minimal voltage drop on the main supply voltage is important. Because of this, it makes sense to use a gate-driven soft-start circuit. This will also prevent damage to the PTC if it is turned off and on again, whereas an NTC thermistor may not be able to protect the circuitry in such circumstances.

The strobe circuit itself will no doubt be discharging a significant amount of power. To prevent damage to the LEDs, current limiting is necessary. Since each LED in series will cause a large voltage drop, the current limiter itself will not have a massive voltage drop across it while handling half an amp of current. Still, the current limiter must be able to withstand at least 1 whole Watt. Nothing special is necessary for this use case, so to save on costs, we chose to use typical resistors with appropriate power ratings.

3.10.2.5 Parts Comparison

Insert Table Here...

3.10.3 Transistor Switches

3.10.3.1 BJT

Bipolar Junction Transistors are current controlled semiconductor devices known for their fast switching and high current carrying capabilities. The downside is that currents from the source and load can flow into and out of the base, leading to current draw on the MCU, or potentially backflow, if voltages or currents exceed the transistor's specifications.

3.10.3.2 JFET

Junction Field Effect Transistors share an insulated Gate with other transistor technologies, and are able to be used with AC signals without extra considerations due to the absence of a 'body diode', but are relatively uncommon and expensive.

3.10.3.3 MOSFET

Metal-Oxide Semiconductor Field Effect Transistors don't often use a metal oxide anymore, but the name stuck around. These transistors have an insulated gate, preventing any current draw from or backflow into the MCU. MOSFETs often require driver circuits to ensure they are properly powered and switch fast enough for their applications due to gate capacitances, but these capacitances also allow them to be used in soft-start circuits. The orientation of a MOSFET must be taken into consideration, as the wrong orientation can cause forward-biasing of an internal 'body-diode' and cause unwanted conducting, invalidating the purpose of the switch.

3.10.3.4 Enhancement Mode / Depletion Mode

Field Effect Transistors come in either Enhancement Mode or Depletion Mode. Depletion mode transistors are far less common than enhancement mode transistors, and therefore more expensive. This project likely would not require the unique properties of a depletion mode transistor.

3.10.3.5 NPN/PNP/N-Channel/P-Channel

BJTs come in either NPN or PNP variants, and FETs come in either N-channel or P-channel variants as well. NPN and N-channel transistors turn 'on' with a positive voltage applied to their base or gate relative to the emitter or source pins, and PNP and P-channel transistors turn 'on' with a negative voltage applied to their base or gate relative to the emitter or source pins.

3.10.3.6 Technology Comparison

Insert Table Here...

3.10.3.7 Part Comparison

Insert Table Here...

3.10.4 Power Source

3.10.4.1 Battery

Batteries would allow our device to work in any suitable environment without the need for a power receptacle nearby. In order to use a battery, however, we must conform to project requirements. Batteries can also become unreliable at colder temperatures, or dangerous at higher temperatures, necessitating strict safety standards.

3.10.4.2 Switched-Mode Power Supply

Switched-mode power supplies, much like switched converters, use a switching action to modulate the input voltage then filter it to create an average voltage at a lower magnitude with minimal power loss. While more expensive than a linear power supply, the switched-mode is capable of handling far higher input and output voltages.

3.10.4.3 Linear Regulated Power Supply

A linearly regulated power supply uses a mixture of passive and active components to control the output voltage level, but is not very efficient. These kinds of power supplies also aren't often able to support high current outputs, and those that do are more expensive.

3.10.4.4 Technology Comparison

3.10.4.5 Part Comparison

3.11 Communication Technologies

Dedicated sensors and external ICs can accomplish various tasks, and perhaps to a greater degree of effectiveness than the MCU, without using up much of the MCU's processing power. Although, in order to use these external sensors and IC's, they must be able to communicate with the MCU and trade information. This process can be facilitated through a variety of communication protocols that are common in industry. Additionally, our MCU and Image Processor must be able to communicate with each other, and this may necessitate using another protocol.

3.11.1 Communication Protocols

3.11.1.1 I2C

I2C, which stands for Inter-Integrated Circuit, is a synchronous communication protocol that utilizes two wires from the MCU as a bus to send and receive information from multiple devices. I2C is able to facilitate communication between many devices with a total of 2 wires: a clock wire driven by the master chip, and a data wire that is used intermittently based on which device the master chip wishes to receive information from. This allows us to control when information is queried from sensors, partially allows us to control how fast this process happens, and saves space on our board and conserves our GPIO pins by limiting the amount of wires in use.

3.11.1.2 SPI

SPI stands for Serial Peripheral Interface, and is another communication protocol that is commonly used, especially for small displays. Unlike I2C however, it is not thoroughly standardized, and there can be small differences in how different pins used for SPI may be named. SPI is also a synchronous protocol, but in order to use multiple devices with

the SPI protocol, they must be daisy-chained. In a daisy-chain, data that sent to a device further down the chain must first pass through all previous devices, which can complicate software and slow down the transfer of information. SPI also typically involves 4 data pins, more than I2C or UART.

3.11.1.3 UART

UART is an often used asynchronous protocol, and stands for Universal Asynchronous Receiver-Transmitter. UART is simple and especially useful for communications between devices that operate at different clock speeds, and can be configured as such. UART typically requires two data lines and a ground connection. The PTC will likely require an asynchronous connection between the MCU and the Image Processor, since the Image processor has a higher clock speed. This is facilitated by using an agreed upon Baud rate, or rate of transmission. By doing so, both devices can send and receive data at the same speed, while their processors operate at different speeds.

3.11.2 Technology Comparison

The PTC project requires a user-interface and communication between two computational devices. The user-interface will incorporate a screen display, and this will likely be driven by the SPI protocol. As for communication between the MCU and Image Processor, it makes sense to use UART, since speed of computation and data transmission is of less importance after the superimposed images of the projectile have been taken and processed. If extra sensors are added to improve calculations and extrapolation of results, these will likely use I2C. Therefore, all of these technologies are extremely useful and will be beneficial to the implementation of the PTC.

3.12 Software Technologies

3.12.1 Integrated Development Environments

3.12.2 Available Languages

3.12.3 Image Processing Software

3.12.3.1

4 Standards and Design Constraints

The PTC system was designed according to established engineering standards to ensure safe operation, reliable measurements, and maintainable system development. Since the project integrates lasers, optical components, embedded electronics, and real-time software processing, multiple categories of standards were considered during both the design and implementation phases. [ieee.org]

4.1 Engineering Standards

While a variety of standards are in use, there are a few in particular that we absolutely must understand and adhere to in order to successfully implement our design.

4.1.1 Communication Standards

In order to program certain devices, and for the end user to receive trajectory information from the PTC, certain devices must be able to communicate with each other and transfer data. To power the entire project, our MCU and Image Processor must first be programmed with one communication standard. These two devices must also communicate with each other during operation with another, and, additionally, the display screen must be driven with yet another standard.

USB The USB, or Universal Standard Bus, is an industry standard maintained by the USB-IF (USB - Implementers Forum), a non-profit corporation founded by the companies that originally worked together to create the USB protocol, companies such as Apple, Microsoft, and Texas Instruments [USB.org]. The USB-IF is thus solely capable of defining what criteria a device must meet in order to be considered a proper USB device. It is our responsibility as engineers to ensure the products we use conform to the USB-IF's specifications like the USB 2.0 Specification, released in the year 2000 and continually updated to optimize performance across a variety of technologies. Other specifications exist for newer USB versions. In order to do this, we can simply check that the manufacturer of a so-called USB device is in fact recognized by the USB-IF as a Standard Adopter, meaning that the manufacturer has proven that their devices meet the USB-IF's specifications. The USB 2.0 specification itself and the list of standard adopters can both be found on the USB-IF's website, USB.org.

According to the USB 2.0 standard, in section 4.1, a USB system is made up of 3 categories: the USB interconnect, the USB devices, and the USB host. Every USB system only has one host in a tiered system, and, according to the USB 2.0 standard, the host can only maintain a connection with a maximum of six lower tiers, with a hub at the center of all tiers except the last one [USB 2.0 Standard]. This means that should the PTC project require it, we could build a USB connection several tiers deep (From host to hub, hub to hub, etc.) without violating the standard, ensuring proper functionality. However, our use case is limited to just programming our devices.

4.1.1.1 UART

4.1.1.2 SPI

4.1.1.3 CAN

4.1.2 Electrical and Hardware Safety Standards

Electrical safety standards guided the hardware architecture to prevent shock hazards, unsafe operating conditions, and damage to components. The system design aligns with

principles described in IEC 61010, which governs safety requirements for laboratory equipment and measurements. As the Projectile Tracking Chronograph functions as a precision measurement device, these guidelines influence the decisions of grounding or power distribution. [iec.ch]

In addition to grounding and power distribution considerations, protection against transient voltage spikes and improper GPIO usage was also evaluated. Since the Raspberry PI 5 interfaces directly with external sensors and triggering circuitry, improper voltage levels could permanently damage the processor. As a result, voltage regulation and level shifting techniques were considered where necessary to ensure compatibility between components. Furthermore, current-limiting resistors and protective circuit elements were incorporated to reduce the risk of short circuits and component failure during testing and operation. [raspberrypi.org]

Another important consideration was electromagnetic interference due to the presence of high-speed switching elements such as strobe lighting and digital signals. Proper wire routing and grounding references were used to minimize noise that could affect sensor readings or image acquisition. These practices will help maintain signal integrity and ensure accurate projectile detection under real operating conditions. [ieee.org]

4.1.3 Data and User Interface Standards

User interaction design emphasizes clarity and consistency to reduce operational errors. Measurement results are displayed using standardized formatting so users can easily interpret velocity outputs. The interface provides consistent feedback during operation, including system readiness, detection events, and processing status. The real-time display expectations guided the interface responsiveness to ensure the users receive a successful measurement.

In addition to clarity and consistency, the interface design follows principles inspired by human-machine interaction standards, emphasizing minimal cognitive load and clear visual hierarchy. Important system states such as “charging,” “armed,” and “detection complete” are visually distinct to prevent user confusion during operation.

Color coding and text indicators are used together to ensure accessibility and reduce ambiguity in interpretation. For example, warning or non-ready states may be highlighted differently than operational states to clearly communicate system readiness. These design considerations are critical in systems involving high-speed measurements, where user misinterpretation could lead to improper operation or missed data capture.

Furthermore, response latency in the user interface is minimized to ensure that feedback appears instantaneous to the user. This aligns with real-time system expectations, where delays in feedback could negatively impact usability and perceived system reliability. [nngroup.com]

4.1.4 Software and Coding Standards

Software development for the PTC system follows structured and modular programming practices to ensure maintainability, scalability, and reliability. The system is implemented in Python due to its flexibility, readability, and strong support for image processing libraries such as OpenCV. Code organization is divided into functional modules, including image acquisition, projectile detection, trajectory computation, and user interface control. This modular design allows individual components to be development, tested, and debugged independently. [python.org]

To maintain consistency across the development team, coding practices follow general guidelines inspired by the Python PEP8 standard. These include consistent naming convention, indentation rules, and code formatting to improve readability and collaboration. [peps.python.org]

Additionally, inline comments and documentation are used to explain complex logic, particularly in image processing routines and mathematical computations.

Error handling is also a critical component of the software design. The system must handle potential failures such as camera disconnection, improper image capture, or unexpected input values. Instead of allowing the system to crash, an exception handling mechanism will be implemented to ensure recovery and continued operations. These software standards contribute to a stable and maintainable codebase that supports real-time projectile tracking and future system improvements. [softwareengineering.stackexchange.com]

4.1.5 Laser Safety Standards

Laser safety is the discipline of identifying, evaluating, and controlling the hazards associated with the usage of lasers. The primary concern with these devices is the tightly concentrated beam of optical power that the laser produces. A 1 mW laser operating the visible spectrum produces an irradiance at the retina approximately 100,000 times greater than looking directly at the sun. This is because the laser is a coherent source with a low-divergence, which can cause great damage to the eye of a person. Higher-power lasers that are used in fields such as medicine, research, and manufacturing cause injuries that are often permanent and severe. [abridgedoptics]

Laser hazards are often categorized into two broad categories: beam hazards and non-beam hazards. Beam hazards are associated with the emitted radiation from the laser itself either through direct, reflected, or scattered contact with the eye or skin. Non-beam hazards come from interactions that are typically caused by laser usage, such as electrical shock from power supplies, fire ignition by high-powered lasers, or toxic fumes generated by certain laser-material interactions. [abridgedoptics]

4.1.5.1 Regulatory Framework

There are three principle documents for laser safety that govern the manufacturing, sale, and workplace use of lasers:

- **FDA/CDRH Regulations (21 CFR 1040):** The FDA's Center for Devices and Radiological Health (CDRH) regulates the manufacturing and sale of laser products under the Federal Laser Product Performance Standard (FLPPS) in the United States. This standard establishes requirements for laser product classification, labeling, and other safety features.

- **ANSI Z136.1:** The American National Standards Institute (ANSI) publishes the ANSI Z136.1, which is documentation on how lasers should be used in the workplace. This standard provides many details on training requirements, control measures, program administration, and other workplace-related protocols.

- **IEC 60825-1:** The International Electrotechnical Comission (IEC) publishes the IEC 60825-1 document, which defines safety requirements and hazard classifications internationally. This standard shares many similarities with the U.S. standards, but there are differences in labeling requirements and certain numerical limits.

Commercial lasers that are purchased with a hazard classification provided by the manufacturer must conform to the classification requirements set by the FLPPS. These standards all share a common technical foundation, but differ in qualities such as administrative detail and certain requirements/limits. [slac.stanford]

4.1.5.2 Laser Hazard Classification System

The laser hazard classification is a system that ranks every type of laser product based on its potential to cause biological harm. The classification is determined by comparing the laser's accessible emission to the Accessible Emission Limit (AEL) for each class, with characteristics such as maximum power output, wavelength being determining factors for this classification. The different classes of lasers are as follows:

- **Class 1 Lasers:** These lasers emit radiation that may be visible or invisible, but are incapable of producing radiation levels that would be damaging under normal operating conditions. These lasers may have an accessible beam with very low intensities or even potentially fully enclosed. Class 1 laser are not required to follow personal protective equipment (PPE) or control measure requirements due to their low-level of potential harm. [slac.standford]

- **Class 1M Lasers:** These types of lasers are safe for viewing to the naked eye, but could potentially cause harm if viewed through optical instruments such as magnifying lenses. These lasers are still low-power, but could cause harm, so they should have special consideration when optical components are present in the same environment. [slac.stanford]

- **Class 2 Lasers:** Class 2 lasers are categorized by their emission in the visible spectrum (400-700 nm) and their higher intensity beams. These lasers trigger the eye's natural aversion response, but are generally safe under normal circumstances. [slac.standford]

- **Class 3R Lasers:** These lasers are still considered ”low-risk” for causing injury to the eye, but exceed the limits of Class 2 lasers. Direct beam viewing could be dangerous, especially for prolonged exposure, so these lasers require basic administrative controls. [slac.standford]

- **Class 3B Lasers:** Class 3B lasers are more hazardous to the eye from direct beam viewing as well as reflections. These lasers can also present a minor hazard to biological tissue under prolonged exposure conditions. These lasers require administrative and engineer controls, and necessitate protective eyewear to be worn from exposed individuals. [slac.stanford]

- **Class 4 Lasers:** These are the lasers that present the highest hazard classification. They are capable of causing significant eye injury from direct, reflected, or scattered radiation. They also present burning hazards to skin or fire hazards to combustible materials. These laser require the most strict administrative controls and procedures as well as mandatory PPE for personnel. [slac.standford]

4.2 Design Constraints

Engineering constraints significantly influenced the PTC system architecture by limiting available design choices and forcing tradeoff decisions among safety, performance, cost, and usability considerations.

4.2.1 Safety Constraints

Safety constraints are the most influential factors in system development. Laser exposure risk requires strict beam control and enclosure design to maintain eye-safe operation. Since the projectiles we will measure travel at a higher velocity, projectile containment and controlled firing direction are necessary to prevent unintended impacts. Indoor operation requirements further restricts the power levels for lasers and demands careful management of reflections. Electrical isolation between subsystems will be required to prevent shock hazards as well as protecting sensitive electronics.

In addition to laser and projectile hazards, mechanical stability and enclosure design play a critical role in ensuring safe operation. The system housing must securely contain all optical and electrical components to prevent unintended exposure or movement during operation. Loose components or misalignment could result in inaccurate measurements or increased safety risks.

Another key consideration is fail-safe system behavior. In the event of power loss, sensor failure, or unexpected software errors, the system must default to a safe state. This includes disabling laser outputs and preventing further triggering until the issue is resolved. These safety measures reduce the likelihood of accidental exposure or incorrect system operation.

User training and clear operational procedures are also considered part of the safety constraints. Even with engineered protections, improper use can still introduce risk. Therefore, the system design emphasizes intuitive operation and clear instructions to minimize user error. [iec.ch]

4.2.2 Performance Constraints

Performance limitations were defined by measurable system requirements such as camera frame rate and processing latency. Accurate velocity measurement requires precise synchronization between projectile detection, strobe activation, image capture, and computational analysis. The system must detect a projectile, trigger a strobe event, capture sequential images, and compute velocity within strict timing constraints. Camera hardware limits the frame acquisition speed while the software processing time introduces latency that must remain low enough to maintain measurement accuracy.

Another critical performance consideration is system synchronization accuracy. The alignment between the laser detection subsystem, strobe illumination, and camera capture must occur within extremely small time windows to ensure valid measurements. Even minor timing mismatches can introduce significant errors in trajectory estimation.

Additionally, computational performance must be balanced with energy consumption and thermal limitations of the Raspberry Pi 5. Prolonged high CPU usage can lead to thermal throttling, reducing processing speed and affecting real-time performance. As a result, efficient algorithms and optimized processing pipelines are necessary to maintain consistent system operation.

The tradeoff between image resolution and processing speed is also a key factor. Higher resolution images provide more accurate positional data but increase computational load. Therefore, the system must carefully select an optimal resolution that balances accuracy with real-time processing capability. [opencv.org]

4.2.3 Economic Constraints

Economic limitations strongly influenced the component selection. This project was not sponsored, therefore all the funding came from the team itself. After researching the current technologies and what is commercially available, a budget of five-hundred was set. Optical sensing was chosen over radar-based measurement due to lower cost and easier integration. A Raspberry Pi platform was selected instead of an FPGA to reduce development complexity and expense while still meeting the processing requirements. These decisions illustrate the engineering tradeoffs between performance, complexity, and affordability. [ieee.org]

4.2.4 Environmental Constraints

Environmental conditions such as indoor lighting variation, temperature changes or vibrations can affect the optical sensing accuracy. The system must maintain reliable operation despite the fluctuations in light and other mechanical disturbances, requiring further

controlled lighting and a stable housing structure.

Lighting variability is one of the most significant environmental challenges affecting system performance. Changes in ambient lighting conditions can impact the effectiveness of image differencing and contour detection algorithms. To mitigate this, the system relies on controlled strobe illumination to provide consistent lighting during image capture.

Vibrations and mechanical disturbances can also introduce alignment errors in both the optical and camera systems. Even small deviations in alignment can affect trajectory accuracy. Therefore, rigid mounting structures and vibration-resistant designs are necessary to maintain system stability.

Temperature variations may also impact electronic components and sensor performance. Prolonged operation in elevated temperatures can affect processing efficiency and potentially reduce the lifespan of components. These environmental constraints necessitate careful system design to ensure reliable operation under varying conditions. [ieee.org]

4.2.5 Manufacturing Constraints

Manufacturing considerations include the limitations of available fabrication methods, particularly with respect to 3D printing and mechanical assembly. Components must be designed with tolerances that can be reliably achieved using accessible manufacturing techniques. Misalignment in optical components, such as lasers and mirrors, can significantly impact system accuracy, making precision in mounting structures a critical requirement.

Additionally, the availability of standard hardware components influenced the design process. The system was designed to utilize parts 'off the shelf' whenever possible to reduce cost and simplify assembly. This includes standard mounts and enclosure components. These constraints required careful consideration of part compatibility and assembly procedures to ensure that the final system could be constructed reliably without specialized manufacturing equipment.

4.2.6 Maintainability Constraints

The system is to be designed with modular subsystems to allow easy replacement of sensors and simplified troubleshooting. The use of the Raspberry Pi 5 will allow software updates without hardware modification, granting long-term usability.

To further enhance maintainability, the system incorporates clear separation between hardware and software components. This allows updates or replacements to occur without requiring full system redesign. For example, camera modules or sensors can be upgraded independently of the core processing software.

Documentation is also an important aspect of maintainability. Clear code documentation, wiring diagrams, and system architecture descriptions ensure that future developers or users can understand and modify the system effectively. Without proper documentation,

even a well-designed system can become difficult to maintain over time.

In addition, modular software design allows for easier troubleshooting by isolating faults within specific subsystems. This reduces debugging time and improves overall system reliability during long-term use. [softwareengineering.stackexchange.com]

4.2.7 Usability Constraints

Usability requirements focus on minimizing calibration efforts and allowing novice users to operate. A simplified touchscreen interface and an automated measurement workflow will reduce the complexity of setup while maintaining accurate performance.

4.2.8 Programming Constraints

Programming constraints are largely influenced by the limitations of the Raspberry Pi platform and the use of Python as the primary development language. While Python provides ease of development and strong library support, it is not optimized for high speed processing compared to lower-level languages such as C or C++. This introduces challenges when processing high-resolution images or performing computations within strict timing constraints.

Additionally, the Linux-based operating system used by the Raspberry Pi introduces non-deterministic scheduling, meaning the precise timing control is more difficult to guarantee. This is particularly important when synchronizing camera capture with strobe lighting and projectile detection events. To mitigate these limitations, efficient coding practices and lightweight processing techniques are used wherever possible.

OpenCV operations must also be carefully optimized to reduce processing overhead. Techniques such as reducing image resolution, limiting region-of-interest processing, and minimizing unnecessary computations are used to maintain acceptable performance. These programming constraints directly influence system design decisions and require tradeoffs between computation complexity and real-time responsiveness.

4.2.9 Time Constraints

Project scheduling constraints between Senior Design 1 and 2 required a prototype first development strategy. Early functional demonstrations are prioritized over optimization to help ensure completion deadlines are met.

Time constraints also influenced the prioritization of system features. Core functionality, such as projectile detection and velocity measurement, was prioritized over advanced features like enhanced visualization or extended analytics. This ensured that a working prototype could be completed within the academic timeline.

Iterative development was adopted as a strategy to manage time effectively. Instead of attempting to implement a fully optimized system from the beginning, the team focused on incremental improvements, validating functionality at each stage before progressing

further. This approach reduces risk and ensures steady progress toward project completion.

Testing and validation time was also limited, requiring efficient debugging and verification methods. As a result, simulation and controlled testing environments were used to accelerate development while still ensuring system accuracy and reliability. [pmbook.org]

5 Application of ChatGPT and Other Similar Platforms

[To be completed - Application of ChatGPT content goes here]
-We each need to do one case study

5.1 Case Study 1 - Image Processing Design Using Large Language Models

Please refer to Appendix C, [1a],[1b],[1c],[1d] for the Case Study 1 prompt and outcomes generated by ChatGPT, Google AI, and DeepSeek respectively.

This case study was conducted to evaluate how effectively large language models (LLMs) assist in the design of a real-time image processing pipeline for the PTC projectile tracking chronograph. Unlike previous prompts focused on hardware comparisons, this study emphasized an engineering problem directly related to the project implementation. The prompt requested a complete workflow capable of detecting a high-speed projectile using synchronized camera images captured through a strobe illumination system, followed by velocity and trajectory computation using OpenCV.

The objective of this comparison was not to determine which model produced the most information, but rather to evaluate the practicality and engineering usefulness of each response when applied to an embedded vision system operating under real hardware constraints.

ChatGPT [1b] provided the most structured response in terms of overall system architecture. The model organized the solution into clear processing stages including image acquisition, motion detection, and trajectory calculation. The suggested techniques such as frame differencing and centroid tracking aligned well with standard OpenCV workflows and were directly applicable to the PTC design. One notable strength was the explanation of why each processing step was necessary, allowing the response to function as both documentation guidance and conceptual design support. However, the response assumes ideal imaging conditions and did not account for limitations such as Raspberry Pi processing latency or lighting variability. Though the architecture was useful, engineering judgment is still required to adapt the recommendations.

Google AI [1c] approached the same prompt differently by emphasizing algorithm descriptions and referencing commonly used computer vision methods with stronger ties to external documentation. The information was technically accurate and often included broader comparisons between detection techniques. The response did not clearly describe how individual stages connected within a real-time embedded system. While useful for research validation and terminology clarification, an engineering interpretation was necessary to translate this information into an executable workflow.

DeepSeek [1d] demonstrated strengths in proposing solutions for implementation suggesting pseudo code structures and optimization considerations. The model discussed alternative tracking strategies and offered suggestions for improving robustness under noisy image conditions. In several cases, DeepSeek proposed computationally intensive solutions that may be better suited for desktop GPUs rather than embedded processors. This highlighted a recurring limitation observed across models: a lack of awareness of hardware constraints unless explicitly specified within the prompt.

Across all responses, a consistent observation emerged that large language models are highly effective during early designs but require careful validation. ChatGPT proved most beneficial for establishing system architecture and documentation structure, Google AI supported verification through a broader context, and DeepSeek contributed implementation focused ideas that expanded possible choices. None of the models independently produced a fully deployable solution, but proved to benefit in accelerating different stages of the engineering design process.

From an engineering perspective, this case study demonstrates that large language models function best as a collaborative effort rather than a single technical source. The results indicate that AI-assisted development can significantly reduce initial research time while still requiring traditional engineering analysis to ensure accuracy, feasibility, and system reliability.

5.2 Case Study 2 - Document Formatting and Table Generation Efficiency

Please refer to Appendix C, [2a.1],[2b],[2c], for the Case Study 2 prompt and outcomes generated by ChatGPT and Copilot respectively. An additional prompt, [2a.2], was used for the case study.

In this case study, the effectiveness of using an LLM to help create a standardized table format was tested. Specifically, two LLMs were leveraged to organize and refine the tables generated by the different authors of this report into one unified structure. The investigation focused on assessing the model's ability to reduce manual formatting effort and improve uniformity. Additionally, the limitations of this approach were studies, including the need for human oversight to verify the accuracy of the translated tables and to ensure compatibility with the pre-established formatting structure of the document.

Through the evaluation of this study, insight was gained into the benefits and trade-offs of using LLMs for documentation editing purposes.

ChatGPT [2b] was structurally consistent with its response when converting tables to the defined ”Standard Table Format” (STF). The model maintained the row separation, caption placement, and usage of the longtable command. In particular, this model strictly maintained the column widths. In the original query, the reference table used had only four columns, with the follow-up query containing a table with 5 columns to be converted to the STF. In ChatGPT's response, it prioritized the table structure provided by the reference table over original distinctions made in the follow-up table. This caused the LLM to merge two columns together to align the column widths set by the STF. The LLM finished its query by noting the changes it made to the columns, and also provided an option for the original table's columns remain intact and dynamically update the size of the columns instead of merging them. Upon further prompting, the LLM was able to produce a table that maintained all of the original information, albeit the column sizes were not sized correctly for the document, leading to manual correction. Even though this LLM maintained most of the original information from the follow-up table, this change in table structure necessitated the need for additional human oversight to confirm if these changes were acceptable.

Copilot [2c] was asked the exact same prompt as ChatGPT and was more to the source table than ChatGPT. This response maintained all of the original columns, maintaining full data fidelity. The only correction to be made was manually adjusting column widths, as they no longer adhered to the standard size of the table set by the STF. Copilot finished its response by asking the user if to use different column widths or standardize captions/labels. This response was much closer to the desired outcome of the query, as the data fidelity is more important than strict structural compliance.

Overall, ChatGPT emphasized structural adherence to the source material over data fidelity, leading to more time spent querying to get a more suitable response, whereas Copilot was more faithful to the data in the tables in generating its response. This shows the nuance in how these two models approach a problem when it is up to them to make a decision. In ChatGPT's case, it chose alter the follow-up tables physical dimensions (rows/columns) to better replicate the exact structure of the STF, while Copilot chose to maintain the physical dimensions of the follow-up table and adapt the structure set by the STF to that table. While both LLMs were able to produce workable tables, this study showed the limitations of the transformative nature of certain AI models for tasks such as standardized table generation.

5.3 Case Study 3 - Metaprompt to use AI for ideas on how to use AI in the project

In this case study, LLMs were evaluated on the capability of determining potential applications of AI for the senior design project. Please refer to [appendix] for the prompt and responses.

Gemini clearly thought the prompt was referring to using AI within the operational phase of the project itself, not in the design or research phase as was desired. Gemini mentioned a number of potentially useful advanced uses of AI, including object detection, replacing the kinematic equations with an AI model. One of the limitations is that deep neural networks would require large amounts of training data, which would be impossible to get without either simulation or automation, either of which would be a large expenditure of effort.

Gemini mentioned using AI for sensor fusion, in which it could be useful for filtering out noise. One suggestion was to compare a Kalman filter to a Gated Recurrent Unit. Signal denoising was also mentioned for allowing lower-cost hardware and compensating for the higher noise with better filtering with AI. Unfortunately, the economics of this don't work out, since the vast majority of the cost is in development time, and BOM costs are nearly negligible. Since there is only a couple prototypes that will be made, reducing development cost and time at the expense of even $10 in parts is valuable, so spending significant time on developing a use for AI to allow a lower cost part is not beneficial.

Grok interpreted the prompt in a similar way to Gemini. It recommended a number of ways AI can be used in the operational phase of the project. First, Grok recommended using AI for object detection or localizing. Models like YOLO give a bounding box on the objects that are detected within the image.

One previous project involved using YOLO on a security camera video stream. One part of the project involved using the detection of cars to alert the user when there was a car in the driveway, indicating there was a visitor. That way when visitors inevitably knocked as if there was a baby sleeping, such that the user was unable to hear the knock on the door, the alert from the security camera was able to let the user know to go open the door. The bounding box of the detected car was useful, since the camera was pointed at the road in front of the house, so the bounding box of the car detection was used to filter out cars that were simply passing by, or parked in the neighbor's driveway. Another part of this previous project using YOLO was to use the security camera to detect where people were in the yard, and control the outdoor lights to light up only the area of the yard where the user was. This was accomplished using addressable LED strips for the outdoor lighting, so that the location used with the bounding box of the detection was used to select which LEDs lit up. Unfortunately, the positional accuracy of the bounding boxes of models like YOLO leave a lot to be desired. It is a coarse position, and is often not centered on the object. Instead, the bounding boxes of YOLO are useful for things such as determining if a car is within a driveway, like in the aforementioned project, or to get an idea if a person is on the left side, right side, or center of an area. YOLO does not have a bounding box resolution near the resolution of the image, instead it is more coarse. [Cite Ramos and Sappa 2025 YOLO] Since we will be able to take calibration pictures without the projectile, it will be significantly easier to use simple math to determine the location of the projectile rather than using AI for object detection and bounding boxes.

Grok also recommended trajectory prediction, drawing on examples such as using neural networks for trajectory estimation for mortar trajectory estimation in GPS/GNSS denied areas. It also recommended Physics Informed Neural Networks (PINNs) for 3d trajectory

estimation from 2d video. It linked to a patent on estimating 3d trajectory using 2d camera images [cite Thornbrue, Cherveny, and Haas patent].

One aspect of the image enhancement and denoising recommendation from Grok that was novel was the recommendation to compare autoencoders or diffusion models versus classical filters such as Gaussian for improving tracking accuracy. This could be useful for narrowing down the centroid of the projectile in the images for better accuracy of positioning the projectile.

5.4 Case Study 4 - AI Tools in Search Engines

It is apparent that AI has become increasingly powerful and useful, especially in the accumulation of data and recognition of patterns. As such, it has become prevalent in search engines as an automatically implemented tool to try and make research faster and deliver answers quicker. In this case study, the effectiveness of these searching tools were benchmarked with search queries of more obscure topics, engineering standards and protocols in this case.

In my research of the USB standard and protocol, I occasionally struggled to find documents in the USB-IF's archive, and this was sometimes because of a particular document's unique naming scheme, or because the document had been removed due to obsolescence of an old standard. For reference, I was trying to find the USB-IF's standard on USB micro connectors, which debuted in the early 2000's. It was unclear to me whether such connectors conformed to the USB-IF's standards, and if they did, which USB standard they conformed to. The search engine's (DuckDuckGo's) AI search tool was able to help me find the document outlining the USB 1.0 standard by providing me a link to the Wayback machine, an internet archive that preserves such documents, and a link to a Wikipedia article with suitable references. This was the inspiration for this case study, detailed below.

For this case study, the prompt and responses can be viewed in Appendix []. The search results were relevant, and led me to a blog post on the Analog devices website on how to use UART, and a Wikipedia entry. With another search, the AI tool was able to clarify that UART is not a standard like USB, but a communication protocol.

This leads me to believe this tool is useful in matters of research, but may lead me down certain avenues without clarifying my misconceptions. Thus, it is important to always ask multiple questions, and thoroughly read the sources it points me to, so that potential misconceptions don't interfere with our implementation of the PTC.

6 Hardware Design

6.1 Structural Diagram

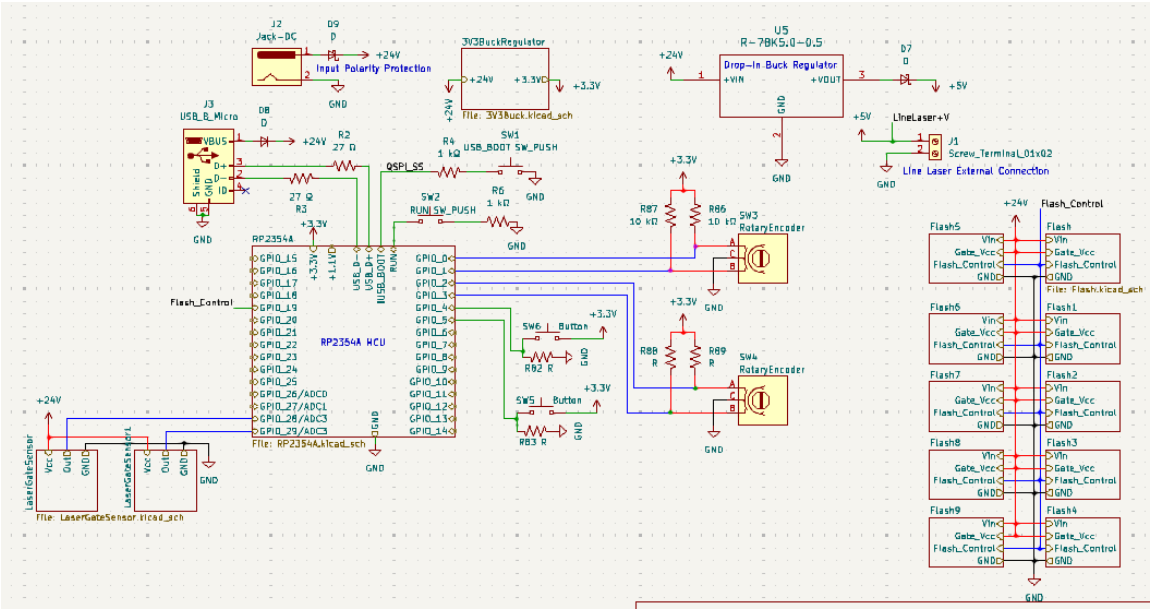
6.2 Optical System

6.3 Subsystem Block Diagram

6.4 Schematic Diagram

6.4.1 Top-Level Schematic

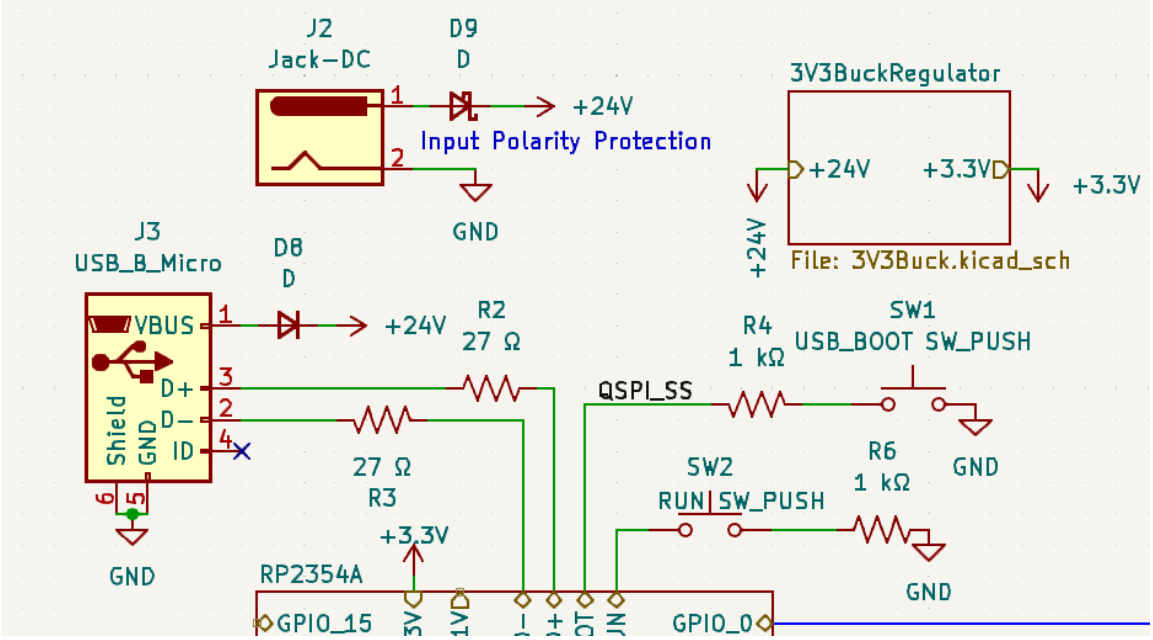
The overall schematic of the current PTC project is shown below, with essential circuits incorporated in hierarchical blocks to save space on the master schematic sheet.



The overall schematic shows power inputs, our central micro-controller, user-interface peripherals, as well as essential circuits like voltage regulators, sensors, and strobe circuits.

6.4.2 Power

We intend to use a wall outlet and 24 V DC power supply to power this project, and this device uses a Barrel Jack port. Additionally, the Raspberry Pi Pico 2 RP2354A microcontroller requires a power input at a voltage of 3.3 V. The RP2354A also is programmed via USB 2.0 protocol, and thus requires a USB port.



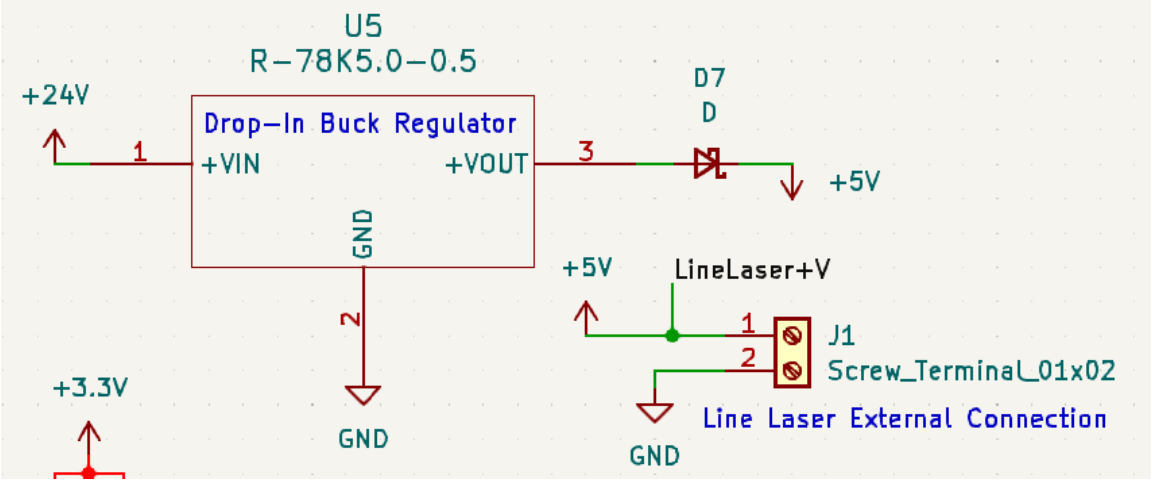
To provide 3.3V from a 24V input, the AP5100 switching buck regulator is used. This circuit will be described in detail in the schematic subsystems section. Additionally, it is in our best interests to install basic circuit protections to prevent device failure; among which are input polarity, over-voltage, and over-current protections.

To provide input polarity protection for the main power rail, a schottky diode has been placed after the output of the DC barrel jack. The schottky diode will prevent damage to components by effectively opening the circuit when a negative voltage is applied, and is particularly useful in that there will be less of a voltage drop across the diode than if a standard silicon diode was used, preserving most of the 24 V to be used in the main power rail. Other protections will be discussed in later sections.

The USB micro B port is simple in construction, and technically we only require the ground pin and the data pins. But, by incorporating a diode on the USB port's VBUS output, it will be possible to power and program the RP2354A without supplying full power to the device, as other systems will prevent excessive current draw, and the AP5100 3.3 V regulator will function properly with an input voltage ranging from 4.75 V to 24 V [AP5100 Datasheet]. Additionally, this diode serves to protect the USB port and its connections from over-voltage should the device be fully powered on, preventing the 24 V main power from reaching the port. The shield pin and ground pin of the USB port are wired together to prevent floating voltages.

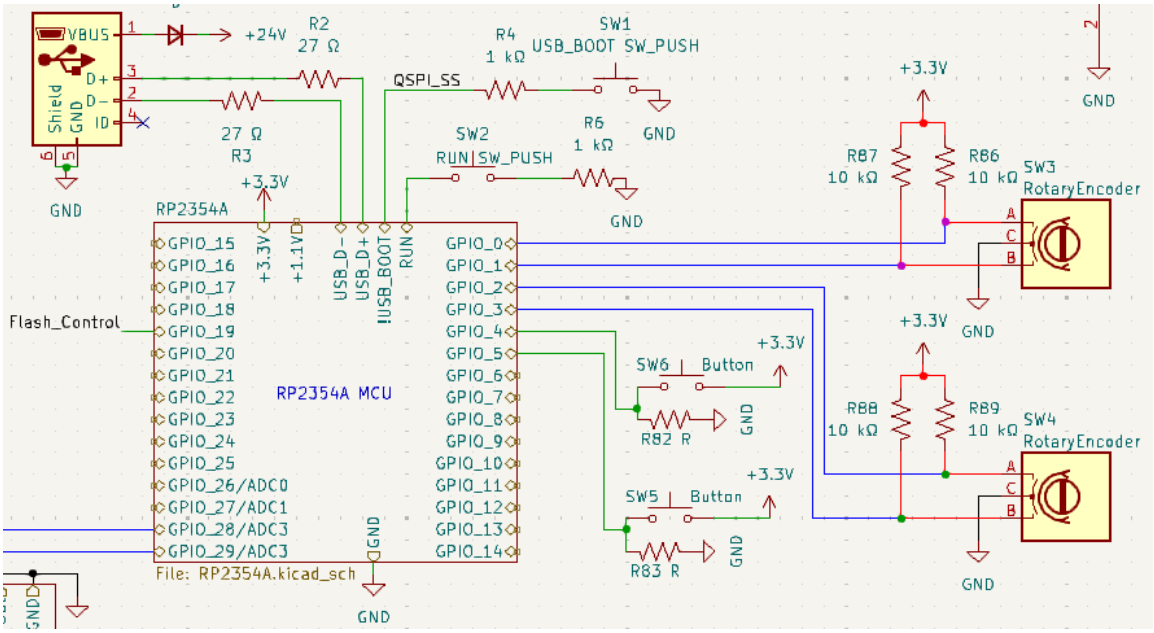
6.4.3 Peripherals

To facilitate the functions of the PTC project, a few essential peripheral components and circuits are absolutely necessary, such as the line laser for the laser gate system, and user-interface components. Below is a zoomed in capture of the circuit that powers the line laser.



The line laser is an external component, and will not be soldered to the final PCB, so a screw terminal connection is used to incorporate it. From the line laser's datasheet, it requires an input voltage within a range of 2.8 V to 5.2 V [Line Laser Datasheet]. Additionally, the laser has an internal driver and only requires an input voltage. Therefore, we have elected to use a drop-in component, a full switching buck regulator within a single confined package in a component called the R-78K5.0-0.5. This regulator is capable of taking our 24 V main power and outputting a stable 5 volts, and a schottky diode is added in series with the output for protection, as suggested by the device's manufacturer in the datasheet [R-78K5.0-0.5 Datasheet].

Of the user-interface components, we have a couple rotary encoders and a few buttons.



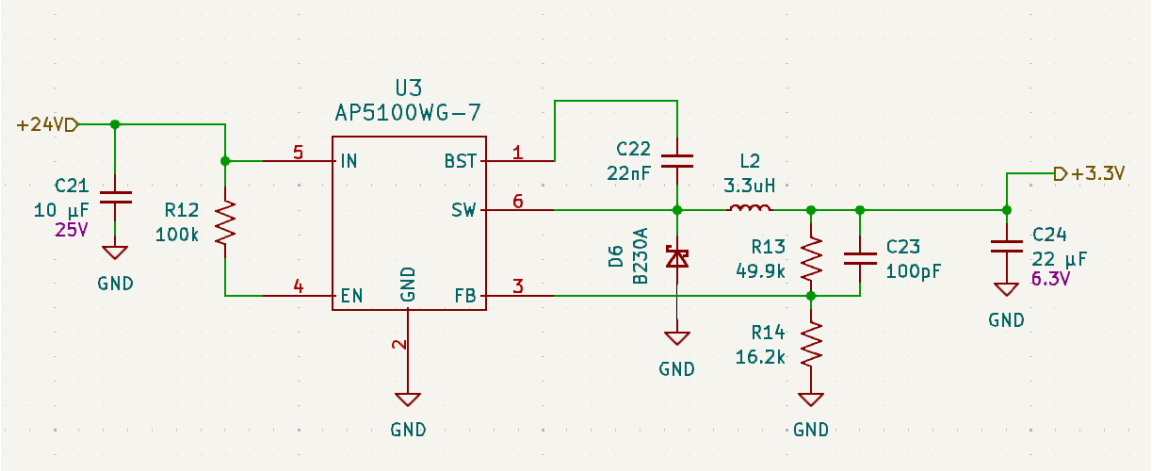
The buttons labeled USB_BOOT and RUN are necessary for a few key functions of the RP2354A, and work in an active-low principle. To enable the end-user to interact with the PTC, 2 buttons and 2 rotary encoders will be in use. This should allow enough controls

for an intuitive user-interface while still conserving GPIO pins for other potential use cases, such as a display screen and sensors. For the buttons, an active-high configuration is used by placing the micro-controller's GPIO connections between the button and pull-down resistor. The rotary encoders require signal pins A and B to be held high, so that the rotating contacts inside can ground them in an alternating fashion. The MCU will be able to determine the direction of rotation by reading each signal and comparing them with their previous states.

6.5 Subsystem Schematics

6.5.1 3.3V Regulator

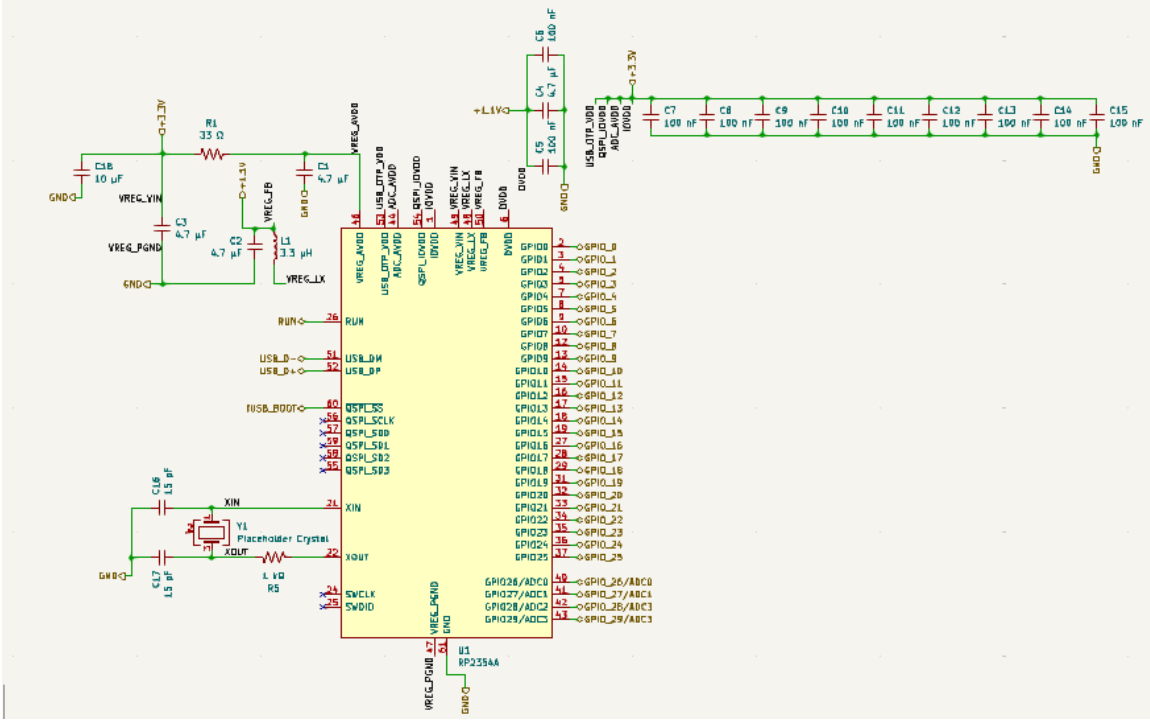
As previously mentioned, the AP5100WG7 is a switched step-down converter, and can handle input voltages from 4.75 V to 24 V at currents up to 1.2 A. This converter does have an adjustable output voltage, but this schematic was assembled with reference to the typical application circuit provided in the device's datasheet [AP5100 Datasheet], wherein it outputs 3.3 V, which is what we require to supply power to our MCU.



The values for the components are also provided in the example circuit, and we have elected to use the same values to ensure proper functionality.

6.5.2 Micro-controller

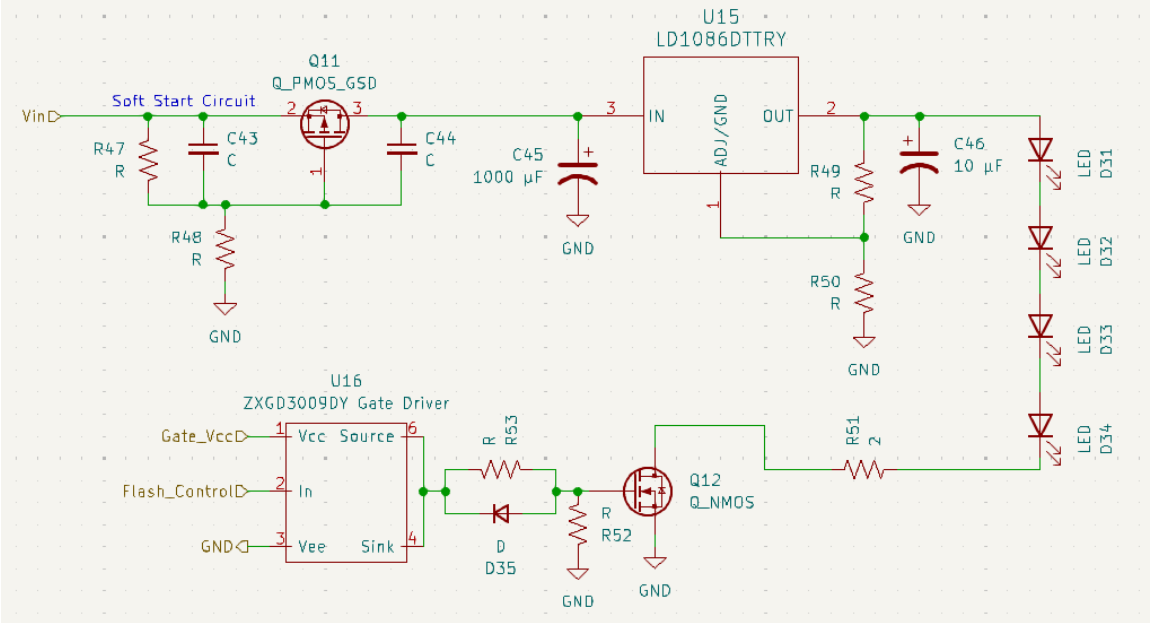
The RP2354A is a complex device, but luckily the team at Raspberry Pi has not only provided a datasheet, but also a simple user-guide for building the minimum circuitry required to run this hardware [Hardware Design with RP2350].



The RP2354A actually not only requires a 3.3 V source, but also a 1.1 V source. The RP2354A does, however, come with an internal regulator to provide this 1.1 V voltage, at the cost of more external components. At the 3.3 V input and 1.1 V output, various capacitors are used to filter the signals and provide stable DC power, while the inductor and capacitor next to it are used to configure the internal regulator. Many small capacitors are also used as coupling capacitors to filter out noise, and a crystal oscillator is used to provide a timing reference to the chip. This crystal oscillator requires a precise capacitance, which the Raspberry Pi team calculated to be 10 pF, but accounting for parasitic capacitances, chose to use two 15 pF capacitors in parallel [Hardware Design with RP2350].

6.5.3 Strobe

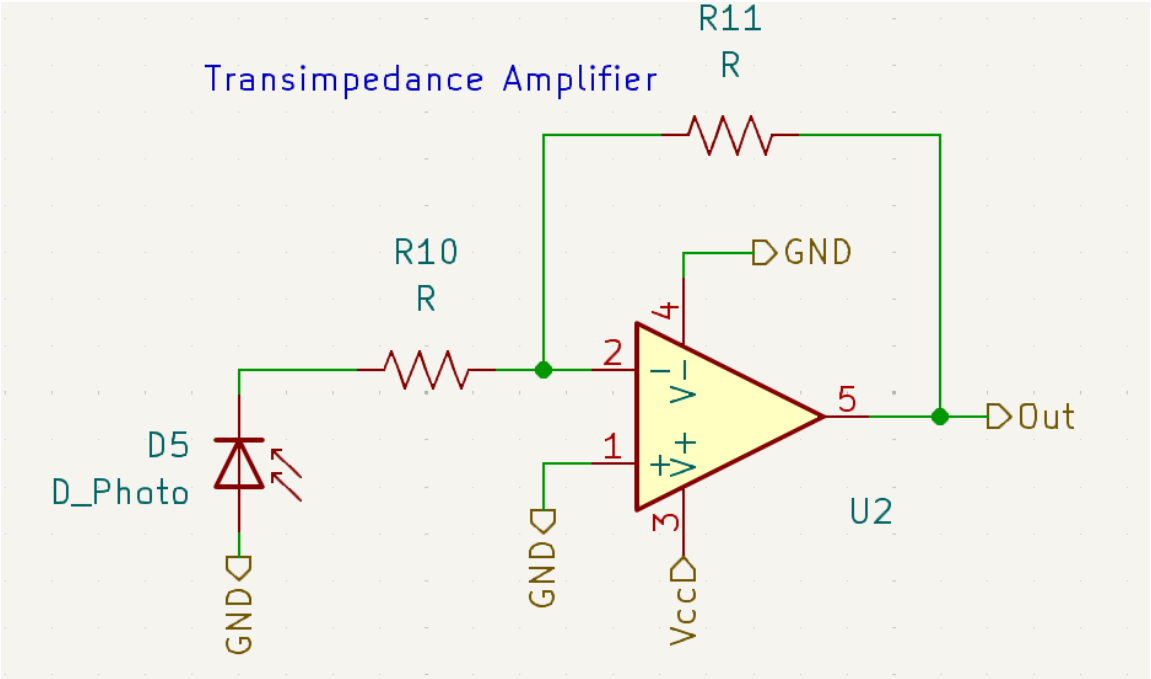
This strobe circuit has been designed by our own team to meet our requirements. We require a very bright flash for the strobe, and so elected to use high power LEDs. These LEDs not only require a larger current, but also have a large voltage drop across them, limiting the amount we can place in series before we exceed our supply voltage. In order to achieve our desired brightness then, it is necessary to build a number of identical circuits that can drive a short series of these LEDs. Additionally, since we now had identical parallel circuits that would all flash at the same time, the total current draw of all the LEDs would now exceed or take a significant portion of the current that our power source could supply, causing issues with our other components. Thus, a capacitor bank is needed. But with a large capacitor bank, now we needed to manage inrush current so that components didn't overheat when the PTC is plugged in.



In the schematic above, we implement a soft start circuit with a P-channel MOSFET that slowly turns on as the large capacitor charges. This circuit topology effectively limits inrush current while still providing a short charge time of at most a few seconds. This allows the strobe circuit to be operational long before a user activates it. The LD1086DTTY low-dropout linear regulator is used to ensure that a constant current is supplied to the strobe LEDs even as the capacitor bank discharges when the circuit is active, ensuring a stable light source with minimal variance in intensity. Since the LEDs have a high forward voltage, putting just a few of them in series creates a large voltage drop that is close to the supply voltage. Thus, a low-dropout regulator is preferred to maximize the voltage headroom the capacitor bank has before it discharges to a level that compromises the regulator's ability to maintain a stable voltage. The feedback resistors are calculated for the LD1086DTTY to supply a voltage of 20 V, allowing about 3 V of headroom. A low impedance resistor is used at the end of the LED chain to limit the maximum current and protect the LEDs. An N-channel MOSFET is placed at the low-side of the chain to activate the strobe LEDs when a signal is received through the ZXGD3009DY Gate Driver from the micro-controller. The Gate Driver is used to ensure fast switching times for the MOSFET, minimizing the delay from the flash signal to the actual flash. With this system, we should be able to reliably provide enough power and sufficient lighting for repeated flashes and image captures when the PTC is in use, with minimal delay, thus preserving the timing accuracy of the system.

6.5.4 Laser Sensor Gate

In order to capture precise images of a projectile as it travels through the PTC, it is necessary to know the speed at which it is traveling before the images are taken. With the laser gate system, this will be possible, but a sensing circuit is necessary. Since the laser gate system will work by using lasers to beam light into photo-diodes, transimpedance amplifiers can be used to turn the currents produced by the photo-diodes into readable voltage outputs that can be interpreted by the MCU.



Just before a projectile is shot through the PTC, the MCU can run a calibration to determine the current voltage produced by the transimpedance amplifiers. When the projectile is shot through, the incident light on the photo-diodes will be interrupted, briefly reducing the transimpedance amplifier's output voltage, twice, allowing the MCU to capture two distinct signals and measure the time between them to calculate the projectile's speed.

7 Software Design

[To be completed - Software Design content goes here]

7.1 Software Design Overview

The software system for the Projectile Tracking Chronograph (PTC) is responsible for integrating sensor inputs, image processing, and user interaction in a unified real-time measurement platform. The system combines data from the laser-based velocity subsystem with camera-based image processing to generate a complete representation of projectile motion. The software is designed for execution on the Raspberry PI 5, serving as the central processing unit for all computational tasks. [raspberrypi.org]

The architecture follows a modular design approach, allowing independent development of key subsystems including image acquisition, projectile detection, trajectory estimation and user interface rendering. This structure improves system maintainability and enables future optimization without requiring an entire redesign. [softwareengineering.stackexchange.com]

7.2 System Execution Flow

The overall execution flow of the PTC software system follows a sequential yet modular pipeline that ensures real-time performance and accurate data processing. The system begins in an initialization state, where all hardware components including the camera, laser subsystem, and strobe lighting are configured and verified for proper operation.

Once initialization is complete, the system transitions into a standby or “armed” state. In this mode, the software continuously monitors for trigger conditions indicating that a projectile has entered the detection region. This event-driven design ensures efficient use of computational resources by avoiding unnecessary processing when no activity is present.

Upon detection, the system immediately activates the strobe lighting and captures synchronized image frames. These frames are then passed into the image processing pipeline, where motion detection and contour extraction are performed. The centroid positions obtained from each frame are forwarded to the trajectory computation module.

Simultaneously, the laser subsystem provides an independent velocity measurement, which is integrated with the positional data. The combined dataset is then used to estimate the projectile’s trajectory.

Finally, the computed results are transmitted to the user interface, where both numerical and graphical outputs are displayed. After completing the measurement cycle, the system resets to the armed state, ready for the next detection event.

This structured execution flow ensures deterministic behavior and efficient coordination between hardware and software components. [raspberrypi.org]

7.3 Image Processing Pipeline

The image processing pipeline is responsible for detecting the projectile within captured frames and extracting positional data required for trajectory estimation. The system will use OpenCV as the primary computer vision library due to its extensive functionality and compatibility with Python. [opencv.org]

Image acquisition is performed using a calibrated camera system synchronized with a strobe lighting subsystem. This allows for a clear capture of high speed projectiles to effectively identify the motion within individual frames. Once images the are captured, frame differencing techniques are applied to identify changes between consecutive frames, isolating the moving projectile from the static background.

Following motion detection, contour tracking algorithms are used to identify the shape and location of the projectile within the frame. The centroid of the detected contour is then calculated to represent the projectile’s position in pixel coordinates. This process is repeated across multiple frames to track motion over time.

7.4 Software Optimization Techniques

Due to the computational limitations of embedded systems, optimization techniques are critical for maintaining real-time performance. While the Raspberry Pi 5 is powerful, it still requires efficient use of resources when performing image processing tasks.

One key optimization strategy is reducing the resolution of input images. Lower resolution images decrease the number of pixels that have to be processed, significantly improving execution speed while still maintaining sufficient accuracy for projectile detection.

Another technique involves limiting processing to a region of interest. Instead of analyzing the entire frame, the system focuses only on the area where the projectile is expected to appear. This technique reduces unnecessary computation and improves processing efficiency.

Additionally, frame differencing is used instead of more computationally expensive methods such as optical flow. This provides a balance between accuracy and speed which makes it suitable for real-time applications.

Memory management and efficient data structures are also considered to minimize latency. By optimizing these aspects, the system achieves reliable real-time performance within the constraints of the embedded platform. [opencv.org]

7.5 Camera Calibration and Coordinate Mapping

Accurate trajectory estimation requires conversion of image coordinates into real-world spatial measurements. This can be achieved through camera calibration and coordinate mapping techniques.

Camera calibration determines intrinsic parameters such as focal length, optical center and lens distortion coefficients. These parameters are used to correct image distortion and ensure accurate positional measurements.

Extrinsic calibration defies the spatial relationship between the camera and the real-world coordinate system. This allows pixel coordinates to be mapped into physical distances using tranformation matrices.

By combining intrinsic and extrinsic calibration data, the system converts centroid positions from pixel space into real-world coordinates. This transformation is essential for accurate trajectory estimation and ensures that computed results correspond to physical measurements.

Calibration is performed prior to system operation and may be repeated if hardware alignment changes. These processes significantly improve the accuracy and reliability of the trajectory computation system. [opencv.org]

7.6 Trajectory Computation

Trajectory computation combines data from both the laser subsystem and image processing pipeline. The laser subsystem provides an accurate measurement of projectile velocity, while the image processing system provides spatial position data.

Displacement is calculated by comparing centroid positions across sequential frames. Using known timing intervals derived from the strobe system, the motion of the projectile can be analyzed. Rather than relying solely on frame-based velocity estimation, the system uses the velocity from the subsystem as a primary input and applies physics-based motion equations based off the images to estimate the trajectory. [physicsclassroom.com]

By combining the measured velocity with its positional slope and effects of gravity, the system will predict the projectile trajectory. This approach improves accuracy and reduces dependence on high frame rate imaging alone.

In addition to basic displacement calculations, the trajectory estimation incorporates fundamental principles of projectile motion, including gravitational acceleration and initial launch conditions. By assuming a constant acceleration due to gravity, the system can model the projectile path using standard kinematic equations. This allows for prediction of the projectile's future position beyond the capture frames, enabling a full trajectory visualization rather than a simple point to point measurement.

To further enhance accuracy, calibration procedures may be applied to covert the pixel coordinates into real spatial measurements. This involves determining a scaling factor based on known distances within the camera's field of view. Without proper calibration, the system would only provide relative motion rather than actual measurements.

Error sources such as camera misalignment, timing inaccuracies, and image noise are also considered in the trajectory computation process. These factors can introduce deviations in the calculated path, so filtering techniques or averaging methods may be applied to reduce their impact. By addressing these potential errors, the system improves the reliability and consistency of its trajectory predictions.

7.7 Trajectory Modeling and Math

The estimation of the projectile trajectory in space is a critical component of the PTC system. While the laser subsystem provides an accurate measurement of initial velocity, the image processing pipeline provides spatial information required to reconstruct the projectile's path.

The system models projectile motion using kinematic equations under the assumption of constant gravitational acceleration and negligible air resistance for short-ranged measurements. The position of the projectile in space can be described as:

$$x(t) = v_0 \cos(\theta) \cdot t \tag{1}$$

$$y(t) = v_0 \sin(\theta) \cdot t - \frac{1}{2} g t^2 \tag{2}$$

$$z(t) = \text{depth component derived from multi-camera perspective} \tag{3}$$

Where:

- v_0 is the initial velocity measured by the laser subsystem
- θ is the launch angle estimated from image data
- g is the gravitational acceleration ($9.91\, m/s^2$)
- t is the time derived from the strobe synchronization

The depth component (z) is estimated using multi-camera geometry or perspective tranformation techniques. By analyzing the displacement of the projectile across multiple camera views, the system approximates its position in space.

To improve accuracy, the system applies coordinate transformation techniques to convert pixel positions to real-world units. This requires camera calibration, including intrinsic and extrinsic parameters, to map image coordinates to spatial coordinates.

Noise and measurement errors are to be mitigated through filtering techniques such as moving averages or Kalman filtering. These methods help smooth trajectory estimates and improve stability in the presence of detection noise.

7.8 System Integration of Subsystems

A critical aspect of the PTC system is the integration between the laser-based velocity measurement subsystem and the camera vision system. These two subsystems operate using different sensing principles but must function cohesively to produce accurate results. [ieee.org]

The laser subsystem provides a precise velocity measurement by detecting the time interval as the projectile passes through a defined optical gate. This measurement serves as a reliable baseline for the system, as it is less affected by image processing noise or environment lighting conditions.

The vision subsystem complements this measurement by providing spatial information about the projectile's position across multiple frames. By combining these two data sources, the system achieves a more comprehensive understanding of the projectile motion than either subsystem could provide independently.

Synchronization between the subsystems is essential for accurate operation. The detection of a projectile by the laser system triggers the strobe lighting and camera capture sequence, ensuring that images are taken at the correct moment. Any delay or mismatch in timing could lead to incorrect trajectory estimation, highlighting the importance of precise system coordination.

This integrated approach demonstrates a key engineering principle of leveraging multiple sensing methods to improve overall system performance and reliability.

7.9 Software Architecture

The software will be implemented in Python due to its ease of development and strong support for frequent prototyping. The system is structured into modular components that handle specific tasks, such as sensor input and handling, image acquisition and processing, data computation and analysis, and user interface control.

This modular approach allows for parallel development and easier debugging. Additionally, the software is optimized for embedded execution on the Raspberry Pi 5, balancing computational performance with system responsiveness.

The modular architecture also supports concurrent execution of tasks, which is particularly important for maintaining real-time performance. For example, while one module handles the image aquisition, another may process previously captured frames, and a separate module may update the user interface. This level of concurrency can be achieved through multi-threading or asynchronous programming techniques with Python.

Inter-module communication is handled through well-defined data structures and interfaces, ensuring that information flows efficiently between different parts of the system. For instance, the images processing module outputs positional data that is directly consumed by the trajectory computation module. This separation of concerns reduces code complexity and improve system reliability.

Version control practices are also integrated into the development process to track changes and maintain code integrity. By using version control systems, the development team can manage updates, revert changes if necessary, and collaborate effectively. This is especially important in projects where multiple members are modifying the code.

7.10 Data Flow Between Subsystems

The PTC system relies on efficient data exchange between multiple subsystems, including the laser velocity module, image processing pipeline, and user interface.

The laser subsystem provides velocity data as a numerical input, which is transmitted to the central processing unit. Simultaneously, the image processing module generates positional data based on detected projectile movement.

These data streams are synchronized and combined within the trajectory computation module. The resulting dataset is then formatted and passed to the user interface for display.

This structured data flow ensures that each subsystem operates independently while contributing to the overall system functionality. [raspberrypi.org]

7.11 Error Handling and System Reliability

Robust error handling is essential to ensure reliable operation of the PTC system under real-world conditions. The software is designed to detect and respond to potential failures without causing system crashes or incorrect measurements.

Common error conditions include camera initialization failure, dropped frames, improper synchronization between the strobe and camera, and invalid sensor readings. To address these issues, the system implements exception handling mechanisms that allow it to recover gracefully or notify the user when intervention is required.

For example, if the camera fails to capture an image, the system will discard the current measurement cycle and return to the armed state. Similarly, if inconsistent positional data is detected, the system avoids computing trajectory results to prevent misleading outputs.

Watchdog timers and system health checks may also be incorporated to monitor continuous operation. These mechanisms ensure that the system remains responsive and can recover from transient faults.

By prioritizing reliability and fault tolerance, the software design minimizes the risk of incorrect measurements and enhances overall system robustness. [ieee.org]

7.12 User Interface Design

The user interface is designed to provide clear and immediate feedback on system status and measurement results. The interface operates in multiple states, reflecting the current condition of the system.

During system startup and capacitor charging for the strobe subsystem, the interface displays a charging or initialization screen to inform the user of system readiness. Once the system is fully operational, it enters an ”armed” state, clearly indicating that it is ready to detect a projectile.

Upon detection, the interface transitions to a results display, showing the measured velocity obtained from the laser subsystem along with a visual representation of the estimated trajectory. This visualization enhances user understanding by providing both numerical and graphical feedback.

The interface is designed to be simple and intuitive, allowing users with minimal technical experience to operate the system effectively.

In addition to displaying system states, the user interface is designed to enhance usability through clear visual hierarchy and minimal cognitive load. Important information such as the velocity measurements and trajectory visualization are prominently displayed, while secondary information is presented in a less intrusive manner. This ensures that users can quickly interpret results without confusion.

The trajectory visualization component provides a graphical representation of the projectile path, which will be displayed to show if or where the projectile hits a target. This visualization helps users better understand the motion characteristics of the projectile beyond numerical values alone. For example, a curved trajectory can visually indicate the effects of gravity, while deviations in the path may suggest inconsistencies in launch conditions.

User feedback mechanisms are also incorporated to improve interaction with the system. Visual indicators such as color changes, status messages, or simple animations will be used to confirm successful detection events or alert the user to potential issues, These features improve system usability and reduce the likelihood of user error during operation.

Finally, the interface is designed with future expansion in mind. Additional features such as data logging, performance statistics, or adjustable system parameters could be integrated without significantly altering the existing interface structure. This forward-thinking approach ensures that the PTC system remains adaptable to future project requirements.

7.13 Software Tools Comparison for Image Processing

To determine the most effective software platform for implementing the projectile detection and trajectory estimation system, several image processing environments were evaluated, incuding OpenCV, MATLAB Image Processing Toolbox, and AI-assisted development tools such as large language models. Each platform offers distinct advantages and limitations depending on the application requirements, particularly when considering real-time processing and development efficiency.

OpenCV was selected as the primary image processing framework due to its strong compatibility with embedded systems and its extensive support for real-time computer vision applications. As a open source library, OpenCV provides a wide range of optimized algorithms for image filtering, edge detection, contour extraction, and object tracking. These capabilities are essential for detecting a high-speed projectile within sequential image frames. Additionally, OpenCV integrates seamlessly with Python, allowing for rapid development and easy to debug. This flexbility is particularly beneficial during the prototyping phase, where algorithms must be frequently adjusted and tested. Another key advatage of OpenCV is its ability to run effeciently on low-power hardware such as the Raspberry Pi 5, making it well-suited for the constraints of the PTC system.

In contrast, MATLAB's Image Processing Toolbox provides a highly structured and mathematically rigorous environment for analyzing and developing image processing algorithms. MATLAB excels in visualization, offering built in tools for plotting, debugging, and algorithm verification. This makes it particularly useful during early development, where understanding the behavior of algorithms is critical. Additionally, MATLAB includes advanced toolboxes that simplify tasks such as filtering, segmentation, and feature extraction without requiring extensive low-level coding. However, depite these advantages MATLAB presents several limitations for this application. The requirement for paid licensing introduces a financial constraint, and the platform is not optizimized for deployment on embedded systems like the Raspberry Pi. As a result, while MATLAB is valuable for prototyping and validation, it is not practical for final system implementation.

AI-assisted development tools, including large language models, were also considered as part of the software development process. These tools do not function as direct image processing platforms but instead serve as support systems for generating code, explaining algorithms, and assisting with debugging. In the context of the PTC system, AI tools were used to explore different approaches for projectile detection, such as frame differencing, background subtraction, and contour tracking. One significant advantage of these tools is their ability to accelerate the initial design process by providing structured solutions and alternative methods. However, their outputs often require verification, as they may not fully account for hardware constraints, real-time processing limitations, or project-specific requirements. Therefore, while AI-assisted tools improve development efficinecy, they cannot replace traditional engineering validation and testing. [openai.com]

Another factor considered in the comparison was real-time performance. The PTC system requires rapid processing of image data to detect and analyze high-speed projectiles. OpenCV offers efficient implementations of key algorithms that can operate within the timing constraints of the system, especially when combined with optimization techniques such as region-of-interest processing and resolution scaling. MATLAB, on the other hand, is generally slower in execution for real-time embedded applications unless additional toolboxes or code generation techniques are used. AI-generated code may vary significantly in efficiency depending on how it is structured, further reinforcing the need for manual optimization. [ieee.org]

Ease of integration was also an important consideration. OpenCV's compatibility with camera interfaces, GPIO control, and other hardware components makes it ideal for systems that require tight integration between software and physical devices. MATLAB, while powerful for simulation and analysis, is less flexible when interfacing directly with embedded hardware in a lightweight deployment. AI-assisted tools, while not directly integrated, can help bridge knowledge gaps and provide guidance on implementing hardware-software interactions.

Scalability and maintainability were also evaluated across the different platforms. OpenCV's modular structure and widespread industry use make it easier to maintain and extend over time. A large community and extensive documentation provide support for troubleshooting and future development. MATLAB offers strong documentation as well but may limit scalability due to licensing and deployment restrictions. AI-assisted tools contribute to

maintainablilty by helping generate documentation and explain code, but they do not inherently enforce structured development practices. [softwareengineering.stackexchange.com]

Ultimately, the selection of OpenCV as the primary image processing platform was based on its ability to meet the specific requirements of the PTC system, including real-time performance, embedded compatibility, cost efficiency, and flexibility. MATLAB was used as a conceptual and validation tool during early development, while AI-assisted tools supported the design process by providing additional insights and alternative approaches. The combination of these tools allowed the team to leverage the strengths of each platform while mitigating their individual limitations. [opencv.org]

7.14

8 System Fabrication

[To be completed - System Fabrication content goes here]

9 System Testing and Evaluation

[To be completed - System Testing and Evaluation content goes here]

9.1 Optoelectronics Feasibility - Camera Imaging Subsystem

9.1.1 Strobe Light Intensity

How intense do we need the light?

One AI result indicates the read noise of the IMX296 Sensor is 5 electrons, however we were unable to find validating documentation on this.

Prompt: ”noise floor of imx296 sensor” Response: ”The Sony IMX296 sensor features a read noise of 5e- (5 electrons) for both monochrome and color variants, as specified in the Pregius Gen2 embedded kit datasheets. While the provided search results do not explicitly state a ”noise floor” value in lux or dB, the sensor achieves this low read noise through Pregius Suppression technology, which reduces noise and improves image quality, particularly in low-light conditions.

The sensor maintains a dark current of 3e-/p/sec at room temperature, contributing to low noise performance in static or slow-moving scenes. Additional noise reduction is

supported by Black Level Correction and a peak quantum efficiency of 69% at 570 nm, which enhances signal strength relative to noise. ”

Using Brave search AI https://search.brave.com/search?q=noise+floor+of+imx296+sensorsource=websun

9.1.1.1 Readout Noise Based Minimum Illumination

So I will derive the minimum illumination for 2 criteria, the readout noise of 5 electrons having a sufficient SNR to place a 1 [cm] $\times$ 1 [cm] projectile in an image that is on a 29 [cm] $\times$ 22 [cm] background that has a 10% difference in reflectivity than the background to have on average 95% of the time every pixel of the projectile at a different level than every pixel of the background.

We will be assuming a strobe time of $T = 100\mu$s, that half of the emitted light hits the desired area (efficiency $\eta_{light} = 0.5$), that the wavelength of light is $\lambda = 850[\text{nm}]$, and the camera is $n_{pixels} = 1580000$ or 1.58 megapixels. The formula for photons hitting a pixel given a total light intensity I and these assumptions is

$$N = \frac{I \cdot T \cdot \lambda}{hc} \cdot \frac{\eta_{light}}{n_{pixels}} = \frac{I \cdot 100[\mu\text{s}] \cdot 850[\text{nm}]}{hc} \cdot \frac{0.5}{1580000} = \frac{I}{7.38 \times 10^{-9}[\text{W}]}$$

or 135000 photons per milliwatt.

The projectile would have a size of about 50 pixels by 50 pixels, giving 2500 pixels on the projectile, with about 1,580,000 pixels of background.

Assuming a peak quantum efficiency of 0.69, combined with the 850 [nm] relative efficiency of 0.4, giving a total quantum efficiency of $\eta_{QE} = 0.276$. N is the number of photons that hit the detector, while N_e is the number of electrons that are detected.

$$\sigma_{total} = \sqrt{\sigma_{shot}^2 + \sigma_{readout}^2}$$

If we approximate the shot noise with a gaussian distribution, we can combine the readout noise distributions and the shot noise distributions to approximate the total noise distribution.

$$P_{background<threshold}^{1580000} > 0.5$$

and

$$P_{projectile>threshold}^{2500} > 0.5$$

Which implies

$$P_{background<threshold} > 0.999999561$$

and

$$P_{projectile>threshold} > 0.99972278$$

$$\sigma_{total} = \sqrt{N_e \cdot \sigma_{readout}} = \sqrt{N \cdot 0.276 + 25} = \sqrt{\frac{I \cdot 3.75 \times 10^7}{1[\mathrm{W}]} + 25}$$

While the relative noise is $\frac{\sigma}{N}$ is

$$\frac{\sigma_{total}}{N} = \frac{\sqrt{\frac{I \cdot 3.75 \times 10^7}{1[\mathrm{W}]} + 25} \cdot 7.38 \times 10^{-9}[\mathrm{W}]}{I}$$

We want to select a threshold percentage such that the two probabilities are true. The best threshold seems to be 0.945 given the projectile has a reflectivity of 90%, and the background has a reflectivity of 100%.

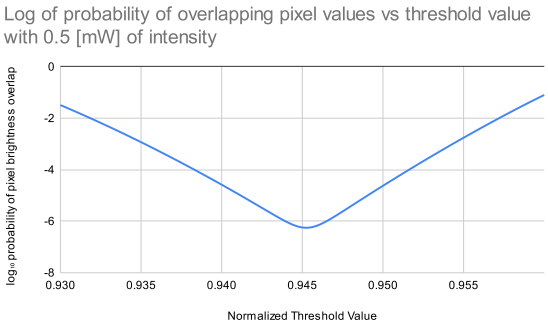


Fig. 8: Graph to determine the best threshold value

Spreadsheet calculations give a minimum value for the intensity of 275 microwatts.

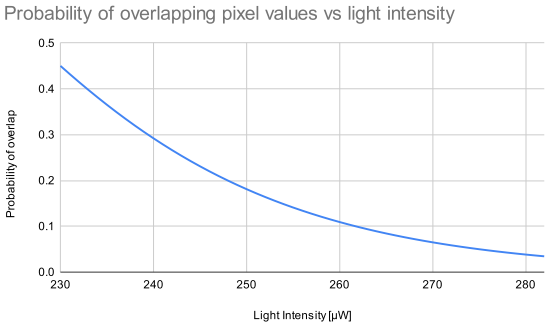


Fig. 9: Graph to determine the minimum intensity with no ambient light

9.1.2 Outdoor ambient light based minimum intensity

To calculate the minimum intensity based on the light leaked into the strobe chamber, we take the light within the bandwidth of the LEDs filtered out by the bandpass filter, and make sure it is less than 1/2 of the amount of light that is provided by the LEDs.

If we assume a bandwidth $\Delta\lambda = 80[\mathrm{nm}]$, a working environment of sunlight, and having 1/10th of the light from the sun gets into the strobe chamber.

The intensity of the sun between 810 [nm] and 890 [nm] is about $900[\mathrm{W} \cdot \mathrm{m}^{-2} \cdot \mu\mathrm{m}^{-1}]$. This gives about $0.9[\mathrm{W} \cdot \mathrm{m}^{-2} \cdot \mathrm{nm}^{-1}] \cdot 80[\mathrm{nm}] = 72[\mathrm{W} \cdot \mathrm{m}^{-2}]$. There would be about $7.2[\mathrm{W}/\mathrm{m}^2]$ of light incident on the entrances to the strobe chamber. Given 2 openings that are 20 [cm] by 20 [cm], that is 0.58 [W] of light getting into the strobe chamber. Given that our slowest projectile supported is 5 [m/s] within about 0.5 [m] long strobe chamber, that would be an exposure of 100 [ms]. Comparing the strobe on-time to the total exposure time, that is a ratio of about 1000/1, which would mean our total light power would have about 1000 times the illumination ratio of strobe to ambient, making it 2000 times the ambient illumination level. That makes about 1200 [W] of strobe illumination, which is too much. If we increase the minimum required speed of projectiles outdoors to 50 m/s, therefore reducing the maximum ambient exposure time to $\frac{0.5[\mathrm{m}]}{50[\mathrm{m/s}]} = 10[\mathrm{ms}]$, that takes the strobe intensity requirements down by a factor of 10, which means that 120 [W] of light would be required.

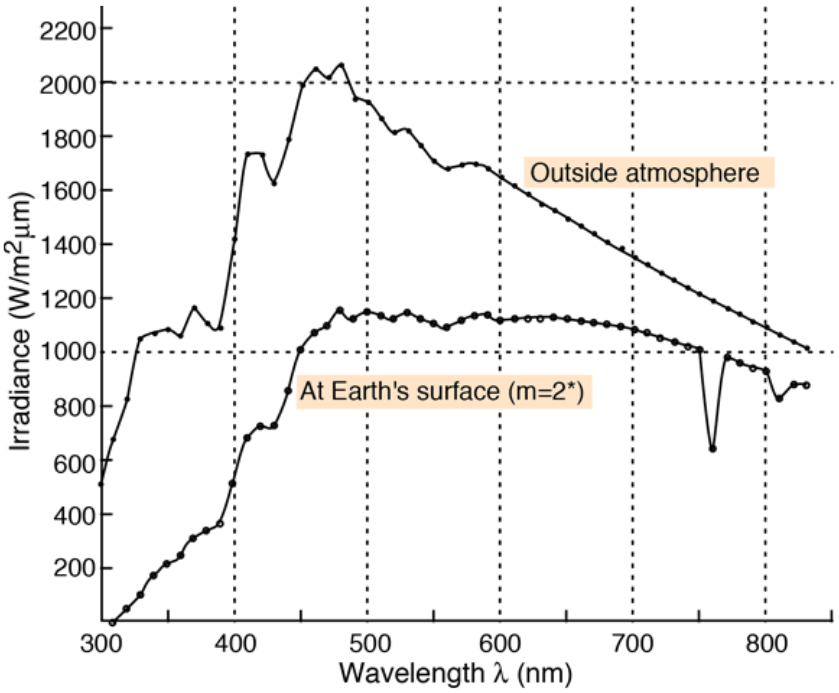


Fig. 10: Solar irradiance Credit: NASA via R Nave GSU web page

9.1.3 Indoor ambient light based minimum intensity

Since indoor LED and flourescent lighting doesn't produce any significant amount of 850 [nm] IR light, we'll simply take the intensity of outdoor sunlight and divide it by the relative intensity of indoor task lighting, which should give a good upper limit of the intensity of indoor lighting. Outdoor noon sunlight can be up to 10,000 lumens per square foot, while indoor task lighting is typically around 100 lumens per square foot.

This gives a ratio of 100/1, so that brings our ambient calculations of 1200 [W] down to 12 [W], which is far more affordable.

Possibly cite old westinghouse lighting handbook.

9.1.4 Camera

9.1.4.1 Relative Camera Angular Placement

Stereoscopic vision is often done with in-line cameras separated by a known distance. An alternative is to have the cameras facing different directions, which has a few advantages and disadvantages. Cameras at different angles could be tilted to better cover the sensing area within the strobe chamber and make better use of all of the pixels. However the calculations with the cameras at different angles could prove to be more complex and therefore difficult. There are more potential sources of error for differently angled cameras. It could be possible to perform calibration photographs, or include a permanent calibration marking within the strobe chamber to properly calibrate where the cameras are for each shot. Calibration markings at the beginning and end of the strobe chamber would be possible to crop out of the images used for the projectile detection, but could be used for precisely locating the cameras in space.

Due to the decreased complexity, we will be using the same tilt direction for both cameras, separated simply by a distance.

9.1.4.2 Camera Spatial Placement and Field of View

The placement of the cameras and field of view has a significant impact on the overall size and shape of the end product, in addition to having a significant impact on the size of the sensing area.

A larger FOV would allow the cameras to be placed closer to the sensing area, but it would also cause there to be more unused volume in the field of view, since the FOV must include the entire top of the sensing area, and a larger FOV means the viewed area at the bottom of the sensing area would be much larger than the actual sensing area.

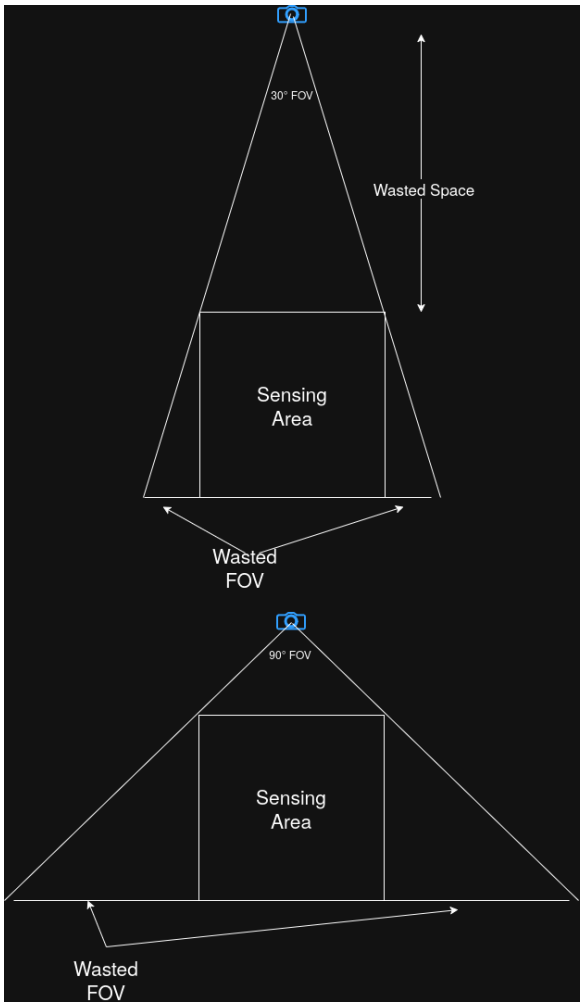


Fig. 11: FOV Comparison

An FOV of between $60°$ and $90°$ would be good.

9.2 Optoelectronics Feasibility - Light Gate Subsystem

9.2.1 Optical Link Analysis

The light gate subsystem serves as the primary trigger mechanism for the PTC, detecting the entry of a projectile into the sensing region and initiating time-of-flight measurements for velocity calculation. This analysis section describes the optical feasibility of using the selected Adafruit 1057 line laser module paired with the PDB-C171SM silicon photodiode to achieve reliable projectile detection.

9.2.1.1 Laser Source Characteristics

The Adafruit 1057 line laser module emits at 650 nm with a maximum optical output power of 5 mW. This module has an integrated cylindrical Fresnel lens that produces a line pattern with a 120° fan angle, distributing the optical power across the illuminated region. For a sensing area width of 100 mm (10 cm) at the target distance, the linear power density can be calculated. The geometric relationship between fan angle and line length are as follows:

$$L = 2d \cdot \tan\left(\frac{\theta}{2}\right) \tag{4}$$

where L is the line length, d is the distance from the aperture of the laser, and θ is the fang angle. Rearranging the equation to solve for d, the source-to-target distance for the laser to achieve a sensing region of 100 mm is approximately 29 mm.

$$d = \frac{L}{2 \cdot \tan(60^\circ)} = \frac{100\ mm}{2 \cdot \tan(60^\circ)} \approx 29\ mm \tag{5}$$

With this geometry, the 5 mW output power distributed across the 100 mm sensing region results in a power density of approximately $50\ \frac{\mu W}{mm}$:

$$P_{linear} = \frac{5\ mW}{100\ mm} = 50\ \frac{\mu W}{mm} \tag{6}$$

Due to limited information from the Adafruit data sheet, it is going to be assumed that the intensity profile for this laser is roughly Gaussian in nature, meaning that the central region of the line has a higher radiance than the edges. Additionally, assuming that the distribution of the beam is uniform, the irradiance at the photodetector primarily depends on the efficiency of the collimating/focusing system being used to expand the laser light into sheets. [1057-web]

9.2.1.2 Photodetector Response

The PDB-C171SM silicon photodiode exhibits a responsivity of about 0.4 A/W at 650 nm, as seen in the figure below.

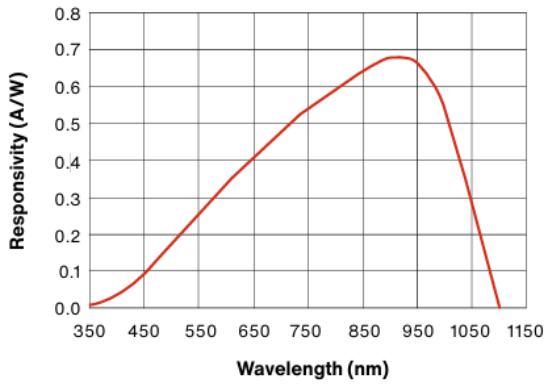


Fig. 12: PDB-C171SM Photodiode Spectral Response

The photocurrent generated by the incident optical power is:

$$I_{ph} = R(\lambda) \cdot \ P_{incident} \tag{7}$$

where $R(\lambda)$ is the wavelength-dependent responsivity and $P_{incident}$ is the optical power reaching the active area. For our laser diode, the ideal photocurrent is $2\ mW$. In our setup, we are going to use reflective surfaces to collimate the laser beam, bounce it across the sensing region, and focus it back down to a point to be detected by the photodetector. We have decided to use aluminum foil tape on 3D printed structures. The aluminum foil tape provides high reflectivity in the visible spectrum, with values of $88 - 92\%$ for polished aluminum surfaces at 650 nm. [yajialuminum.com]

For a photodetector with an active area width of approximately 5-10 mm positioned to capture the reflected beam, the received optical power becomes:

$$P_{received} = P_{incident} \cdot R \cdot \eta_{collection} \tag{8}$$

where R is the reflectivity of the aluminum tape and $\eta_{collection}$ is the collection efficiency of the photodiode. If we are to assume that the reflectivity is $\approx 90\%$ and the collection efficiency is $\approx 10\%$, then the received power becomes:

$$P_{received} = 5\ mW \cdot 0.9 \cdot \ 0.1 = 450\ \mu W \tag{9}$$

resulting in a corresponding photocurrent of:

$$I_{ph} = 0.4\ \frac{A}{W} \cdot \ 450\ \mu W = 180\ \mu A \tag{10}$$

For the photodiode capacitance values at $V_{bias} = 10\ V$ and the specified rise time with load resistance $R_L = 1\ k\Omega$, the bandwidth can be estimated form the RC time constant equation:

$$f_{-3dB} = \frac{1}{2\pi R_L C_d} \tag{11}$$

In the datasheet for the photodetector, the rise time measurements at 820 nm with a

$R_L = 1\ k\Omega$ load suggest bandwidth capabilities compatible with the 10-100 kHz carrier frequency. We would have to do more experimental testing to determine the bandwidth capabilities at 650 nm, but we believe that it would still fall into a usable range for our modulated detection scheme. [sensorportal.com]

9.2.2 Signal-to-Noise Analysis

9.2.2.1 Noise Sources

Shot noise is a type of electronic noise that arises from the discrete nature of electric charge, which causes random fluctuations in steady current flow. It can be expressed by this equation:

$$i_{shot} = \sqrt{2q(I_{ph} + I_d)B} \tag{12}$$

where q is the fundamental charge of an electron ($1.602 \cdot 10^{-19}\ C$), I_d is the dark current, and B is the bandwidth.

Another form of noise is thermal noise, which arises from the random thermal motion of charge carriers in a resistive material. This can cause current fluctuations in a material and can be described as follows:

$$i_{thermal} = \sqrt{\frac{4k_B T B}{R_L}} \tag{13}$$

where k_B is the Boltzmann's constant ($1.381 \cdot 10^{-23} \frac{J}{K}$), T is the temperature, and the other parameters are the same as previously defined. It can be seen from this equation that thermal noise increases as the temperature and/or bandwidth increases and decreases with higher resistance.

Assuming that $T = 300\ K$ and $B = 100\ kHz$, the shot and thermal noises can be calculated as such:

$$i_{shot} = \sqrt{2 \cdot 1.602 \cdot 10^{-19} \cdot (180 \cdot 10^{-6}) \cdot 100 \cdot 10^3} = 2.4\ nA \tag{14}$$

$$i_{thermal} = \sqrt{\frac{4 \cdot 1.381 \cdot 10^{-23} \cdot 300 \cdot 100 \cdot 10^3}{1000}} = 40.7\ nA \tag{15}$$

The total RMS noise can be calculated by this equation:

$$i_{RMS} = \sqrt{i_{shot}^2 + i_{thermal}^2} \approx 40.8\ nA \tag{16}$$

9.2.2.2 Signal-to-Noise Ratio

The signal-to-noise ratio (SNR) quantifies how strong a desired signal is compared to the background noise of the received signal. A higher SNR value indicates a strong signal

with little background noise, which is desirable for the PTC project. The SNR can be expressed in dB, as seen the the equation below:

$$SNR = \frac{I_{ph}}{i_{RMS}} = \frac{180\ \mu A}{40.8\ nA} \approx 4400 \approx 73\ dB \tag{17}$$

9.2.3 Projectile Detection Timing Analysis

9.2.3.1 Beam Interruption Geometry

For a projectile of diameter D traveling at a velocity v that is perpendicular to the laser line, the duration of the beam interruption is:

$$t_{interrupt} = \frac{D}{v} \tag{18}$$

For the specified projectile size range (15-50 mm) and velocity range (10-50 m/s), the upper and lower bounds of the system's temporal resolution requirements can be determined.

TABLE XVI: Beam Interruption Bounds

Projectile Diameter	**Velocity**	**Interruption Time**
15 mm	50 m/s	300 μs
15 mm	10 m/s	1.5 ms
50 mm	50 m/s	1 ms
50 mm	10 m/s	5 ms

This table shows that for our system, our upper bound is a minimum interruption time of 300 μs and our lower bound is 5 ms on the system's temporal resolution requirements.

9.2.3.2 Carrier Frequency Selection

A modulated carrier frequency approach provides a much greater resistance to ambient light interference by shifting the detection signal away from the DC regime. The carrier frequency must satisfy the following:

$$f_{carrier} > \frac{1}{t_{interrupt,\ min}} \tag{19}$$

where, for our system with a minimum interruption time of 300 μs, the carrier frequency must be greater than 3.3 kHz. To ensure multiple carrier cycles occur during the shortest beam interruption, a carrier frequency of 10-50 kHz should be sufficient. [sensorsportal.com]

9.2.3.3 Rise Time Requirements

The laser diode's rise and fall times have a direct impact on the modulation depth that can be achieved at the carrier frequency. For a 50 kHz carrier (period of $20\ \mu s$), the rise and fall times should be less than 10% of the half-period in order to maintain a modulation depth greater than 80%.

$$t_{rise,\ max} < 0.1 \cdot \frac{t_{period}}{2} = 1\ \mu s \tag{20}$$

The Adafruit 1057 datasheet does not specify the rise and fall time characteristics. This means that the exact characteristics of this diode will have to be experimentally derived, making theoretical verification more difficult. There is a TTL-modulated variant of this module (Adafruit 1056) that specifies a maximum modulation frequency of 50 kHz, which was the basis for the previous calculations. We believe that the rise times on the order of microseconds for the Adafruit 1057 variant may be achievable as well. [verical.com]

9.2.3.4 Ambient Light Filtering via Modulation

There are many benefits to using a carrier modulation approach to filter out ambient light. By doing this approach (either through a bandpass filter or lock-in detection method), unwanted signals such as the DC components from constant ambient illumination or broadband noise outside of the detection bandwidth are rejected. This type of filtering should be more than enough to achieve the specified ambient light tolerance by a substantial amount. [electroschematics.com]

9.2.4 Velocity Measurement Analysis

9.2.4.1 Time-of-Flight

To measure the velocity of the projectile, there needs to be two spatially separated light gates with a known distance d_{gates}. The velocity can be measured by dividing this known distance by the difference in time between the two disturbances in the gate:

$$v = \frac{d_{gates}}{\Delta t} \tag{21}$$

9.2.4.2 Timing Resolution and Spatial Uncertainty

For the specified maximum speed error of 20%, the timing resolution must satisfy:

$$\frac{\Delta t}{\Delta t_{measured}} < 0.2 \tag{22}$$

For a 50 m/s projectile traversing a 100 mm gate separation:

$$\Delta t_{measured} = \frac{100\ mm}{50\ m/s} = 2ms \tag{23}$$

Making the required timing resolution:

$$\Delta t_{timing} = 0.2 \cdot 2\ ms = 400\ \mu s \tag{24}$$

The microcontroller we have chosen (either Raspberry Pi Pico 1 or 2) will have clock speeds the provide timing resolutions that are more than sufficient (133 and 150 MHz respectively for precise measurements.

For a gate separation measurement uncertainty of $\pm$ 1 mm over a 100 mm sensing region:

$$\frac{\Delta d}{d_{gates}} = \frac{1\ mm}{100\ mm} = 1\% \tag{25}$$

This contributes minimally to the amount of error being contributed to the system, leaving a greater amount of leeway for systematic effects such as non-perpendicular trajectories.

9.2.5 Power Analysis

9.2.5.1 Power Budget

The laser diode module requires a 2.8-5.2 V input at a maximum of 25 mA. Using the maximums for these two metrics, the max power of the laser diode is:

$$P_{laser,max} = 5.2\ V \cdot 25\ mA = 130\ mW \tag{26}$$

This is a very small power draw, making it a very low-power component of the project.

9.2.6 Summary of Feasibility

The analysis of these different aspects of the subsystem indicates that the selected components are capable of meeting the requirements for light gate detection. The primary uncertainties can be attributed to the lack of characterization for the Adafruit 1057 laser module. Testing with this diode should verify the modulation performance, however, other diodes should be considered in case this current module is unable to be modulated within specs.

TABLE XVII: System Performance Summary

Parameter	**Requirement**	**Calculated**	**Status**
SNR during interruption	≥ 20 dB	~ 73 dB	✓ Exceeds
Photodetector bandwidth	> 100 kHz	$\sim 100+$ kHz	✓ Meets
Detection of 15–50 mm projectiles	Required	300 μs – 5 ms events	✓ Feasible
Velocity range 10–50 m/s	Required	Compatible with timing	✓ Feasible
Carrier frequency	10–100 kHz	50 kHz recommended	✓ Meets

Line length	≥ 100 mm	100 mm at 29 mm distance	✓ Meets
Modulation depth	$\geq 80\%$	Requires verification	△ Uncertain
Rise/fall time	$\leq 1\ \mu s$	Not specified	△ Uncertain

10 Administrative Content

10.1 Budget

The PTC project is planned with a total expenditure limit of $1000, as established by the project team to ensure the system can be developed within reasonable cost constraints while still meeting technical objectives. This budget is intended to support the optical, electrical, and computing components that are required for the implementation of the optical projectile measurement system.

10.2 Bill of Materials (BOM)

The BOM provides a comprehensive list of the components that are necessary to design and build the PTC. Maintaining the BOM throughout the project will aid in cost awareness and serve as a centralized reference for the various hardware needs of the project. Below is a table containing the components and estimated costs of each part. The extended prices are the unit prices of the components multiplied by their respective quantities.

TABLE XVIII: Bill of Materials

Components	**Name / Part Number**	**Distributer**	**Extended Price**	**Quantity**
Camera	Raspberry Pi Camera Module 2	PiShop	$31.90	2
Image Processor	Raspberry Pi 5 2GB	PiShop	$65.00	1
MCU	RP2350	DigiKey	$5.88	4
Development Board for Testing	Raspberry Pi Pico 2	PiShop	$6.40	1
Photodetector	PDB-C152SM	DigiKey	$14.01	10
Laser Diode	1528-1402-ND	DigiKey	$17.90	2
Cylindrical Fresnel Lens	#46-114	Edmund Optics	$141.00	2

Sheet Metal (as a reflector)	0.016” Aluminum Sheet 2024-T3-Clad - Part #8070	OnlineMetals	$11.32	1
Strobe LED Cluster	JR5050AWT-Q-U30EB0000-N0000001	DigiKey	$14.82	30
Power Supply	WDU12-1200	DigiKey	$28.06	1
3.3V Voltage Regulator	RAA2118034GP3#JA0	DigiKey	$1.76	4
N - MOSFET Switch	MCU45N10-TP	DigiKey	$7.92	10
47uF Capacitor	EJ1C470MP20511U	DigiKey	$0.69	10
Inrush Current Limiter	B57153S0200M054	DigiKey	$2.90	5
LCD Screen	Waveshare 2.8inch LCD Display Module	Amazon	$25.99	1
Rotary Encoder	PEC12R-4025F-N0024	DigiKey	$5.97	5
Button	PTS636SM43 LFS	DigiKey	$1.29	10
Power Switch	SW-R2-1A-F-1-0	DigiKey	$2.25	5
			Estimated Maximum Total Cost:	**$385.06**

10.3 Project Milestones

This senior design project involves multiple interconnected technical and documentation tasks, so setting clearly defined milestones with set deadlines is essential for maintaining consistent progress and organization. Clearly scheduled milestones break the project into manageable phases, helping our team to prioritize either individual or group tasks at any given time. By committing to specific dates, our team is able to ensure that all progress made between each stage of the project is done so in a controlled and predictable manner, thereby minimizing any last-minute risks that may occur. Our project milestone table has a description of each milestone, the start/anticipated end dates, and each milestone’s dependencies.

TABLE XIX: Project Milestones

Milestone Description	**Start Date**	**End Date**	**Description**
Finalize Project Concept and Scope	1/27/26	1/29/26	Define project objectives, requirements, and system boundaries
Divide and Conquer Document Finalization	1/25/26	1/29/26	Complete task allocation and subsystem responsibility documentation
Divide and Conquer Review Meeting	2/3/26	2/3/26	Review and approve task distribution with advisors

Committee Formation and Initial Meeting	1/26/26	1/30/26	Form the advisor committee and discuss project expectations
Find image processing software	2/3/2026	2/13/2026	Research available options
30-Page Draft Development and Revision	2/3/26	2/20/26	Develop an initial technical report and revise based on feedback
Sensor Gate Subsystem Testing	2/6/2026	2/29/2026	Verifying reliable projection detection, timing accuracy, and interrupt behavior
Camera + Strobe Subsystem Testing	2/6/2026	2/29/2026	Validating synchronized strobe activation and image capture during projectile events
60-Page Draft Development and Revision	2/20/26	3/10/26	Expand documentation with detailed design and implementation
Photonic Subsystem Integration	3/1/2026	3/16/2026	Integrating laser, mirrors, and optical sensors into a unified system
Senior Design 1 Midterm Demo	3/3/2026	3/3/2026	Develop a functional photonic subsystem capable of demonstrating one core component of the project
90-Page Draft Development and Revision	3/10/26	3/27/26	Finalize comprehensive system design and analysis documentation
Senior Design 1 Final Report	3/27/26	4/10/26	Submit the completed SD1 design report
Image Processing Program Finalized	4/12/2026	4/21/2026	Completing and validating image-processing algorithms for projectile position and trajectory analysis
Senior Design 1 Final Demo	4/21/2026	4/21/2026	Implement and validate a complete photonic subsystem capable of executing a primary project function
Mini Demonstration Video	4/28/26	4/28/26	Demonstrate early system functionality and progress
CDR Presentation Video	5/18/2026	5/29/2026	Present the finalized design

CDR Presentation Slide	5/18/2026	5/29/2026	Prepare slides summarizing system design and progress
Senior Design 2 Midterm Demonstration Video	5/30/2026	5/30/2026	Show system integration and functional progress
8-Page Conference Paper	6/1/2026	6/7/2026	Prepare a concise conference-style technical paper
Senior Design 2 Final Report	6/7/2026	6/23/2026	Submit the final report covering implementation and testing
Final Presentation Video	6/7/2026	6/23/2026	Present final system design and results
Final Presentation Slide	6/7/2026	6/23/2026	Prepare slides that summarize final project outcomes
Final Demonstration Video	6/7/2026	6/23/2026	Demonstrate completed system functionality

10.4 Hardware Milestones

The hardware portion of this senior design project involves several design and testing steps, making explicitly defined hardware milestones essential for maintaining steady progress. Since there are many interdependent portions of this project, having set deadlines for milestones such as gathering components and prototyping will ensure that everyone in the group is able to make consistent advancements in their respective portions. Below is the hardware milestone table, with the start/anticipated end dates as well as a description of each milestone.

TABLE XX: Hardware Milestones

Hardware Milestone	**Start Date**	**End Date**	**Description**
Component Selection	1/26/2026	2/8/2026	Making a Bill of Materials
Gather Electronic Components (Power Supply, Capacitors, MOSFETs, etc.)	2/6/2026	2/21/2026	Ordering from a manufacturer
Gather Optical Components (Laser Diodes, Photodetectors, Cameras, etc.)	2/6/2026	2/21/2026	Ordering from a manufacturer
System Testing	2/6/2026	2/29/2026	All components are tested and work
Complete Project with Prototype Board	3/16/2026	4/1/2026	Assemble the components for a prototype

PCB Testing	4/5/2026	4/10/2026	Creating and ordering a customized PCB
Test Prototype with PCB	4/10/2026	4/14/2026	Using the customized PCB for the design
User-Interface System Integrated	5/30/2026	5/31/2026	Integrating the graphical user interface with back-end processing for data display
Finalize Prototype	6/1/2026	6/23/2026	Finalize the design using a specialized PCB board

10.5 Work Distribution

The work distribution table provides an outline of the allocation of responsibilities among the team members of the PTC project. This ensures balanced contributions as well as clear ownership of the project tasks. It identifies the primary roles for members across hardware and software development, testing, and system integration to promote accountability and collaboration.

TABLE XXI: Work Distribution of Group Members

Name	**Major**	**Responsibilities**
Austin Stoetzer	Optical Engineering	Sensor Gate Design and Implementation Project Lead Administrative Documentation Photonic System Integration with Electronics and Software
Blake Rainwater	Optical Engineering	Camera and Strobe System Prototype Lead Projectile Trajectory Visualization Photonic System Integration with Electronics and Software
Mathew Aranibar	Electrical Engineering	PCB Design MCU Selection and Implementation Component Research Lead Photonic System Integration with Electronics

Gianni Angelone	Computer Engineering	Software and UI Integration Lead Website Design and Management Projectile Trajectory Visualization Photonic System Integration with Software

11 Conclusion

[To be completed - Conclusion content goes here]

Appendix A - Citations

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Appendix B - Useful Tools

[1] EdrawMax Online

[2] Draw.io

[3] Microsoft PowerPoint

[4] Google Docs

[5] Digikey

[6] Google Search Engine

[7] Ultra Librarian

[8] Snap Magic Search

Appendix C - AI Use Prompts and Outcomes

.1 Case Study 1

[1a] Case Study 1 Prompt

Prompt: “Design a real-time image processing pipeline capable of detecting a high-speed projectile using three sequential images captured with synchronized strobe illumination. The system will run on a Raspberry Pi using Python and OpenCV. Explain the processing stages required to detect the projectile, calculate velocity, estimate trajectory angle, and display a real-time visualization suitable for a user interface.”

[1b] Case Study 1 ChatGPT Outcome

“To implement a real-time projectile tracking system using Python and OpenCV, the processing pipeline should be divided into structured stages. First, synchronized image

acquisition must capture three frames illuminated by a short-duration strobe to freeze projectile motion. These frames should be timestamped to preserve accurate timing intervals.

Next, preprocessing should be performed to improve detection reliability. This includes grayscale conversion, Gaussian filtering to reduce sensor noise, and intensity normalization to compensate for lighting variation. Background subtraction or frame differencing can then be used to isolate moving objects between consecutive frames.

After motion isolation, contour detection can identify candidate objects. The projectile position may be estimated by calculating the centroid of the detected contour. Tracking the centroid across the three frames allows displacement measurements to be computed in pixel space. Using calibration data, pixel displacement can be converted into real-world distance.

Velocity is calculated by dividing displacement by the known time interval between frames. The launch angle can be estimated using linear regression or slope calculations between detected projectile positions. Finally, visualization may be implemented using OpenCV drawing functions or a plotting library to animate projectile motion in real time for display on a graphical user interface.

Performance optimization techniques such as region-of-interest cropping, multithreading, and reduced image resolution may be necessary to maintain real-time execution on embedded hardware platforms."

[1c] Case Study 1 Google AI Outcome "A projectile tracking system using sequential images can be implemented through several computer vision techniques. Motion detection may be achieved using optical flow, frame differencing, or feature-based tracking methods. Each technique provides advantages depending on lighting consistency and object speed.

The system should begin by capturing synchronized images and converting them into grayscale format for computational efficiency. Edge detection or threshold segmentation may then be used to identify moving objects within each frame. Object tracking algorithms such as Kalman filtering or template matching can improve stability across frames.

Velocity estimation can be determined by measuring positional changes over time, while trajectory estimation may involve curve fitting or predictive modeling. Visualization can be implemented through graphical overlays or animation frameworks capable of updating in real time.

While multiple approaches exist, system performance depends heavily on hardware capability, camera synchronization accuracy, and image noise conditions."

[1d] Case Study 1 DeepSeek Outcome "A practical implementation of projectile tracking should prioritize computational efficiency and robustness against noise. Instead of relying solely on contour detection, adaptive thresholding combined with morphological filtering can improve detection reliability under varying illumination.

The algorithm may store detected projectile coordinates in a buffer and apply predictive tracking methods such as Kalman filtering to estimate motion between frames. Parallel processing techniques and vectorized operations are recommended to maintain real-time performance.

For visualization, a physics-based animation model can simulate projectile motion using calculated velocity and angle parameters. Rendering may be performed using OpenCV drawing routines or a lightweight graphics framework. Optimization strategies such as GPU acceleration or batch processing can further improve responsiveness."

.2 Case Study 2

[2a.1] Case study 2 Prompt:

"I would like your assistance in making my document editing in LaTeX more streamlined. I am going to provide you a table that I made that I will call the standard table format (STF for short). Whenever I upload a table going forward, I would like you to convert it to STF. Here is the reference STF:"

TABLE XXII: Comparison of IR LEDs

Parameter	**AG3535C-YRR-1000mA-H1**	**AC3535B-YRR-1000mA-H1**	**AC3535L-YRR-1000MA / AC3535A-YRR-1000MA**
Manufacturer	Solidlite	Solidlite	Solidlite / Solidlite
Price ($)	1.1	1	1 / 1
Current - DC Forward (If) (Max)	1 A	1.5 A	1.5 A / 1.5 A
Radiant Intensity (Min @ If)	430 mW/sr @ 1 A	900 mW/sr @ 1.5 A	175 mW/sr @ 1 A / 550 mW/sr @ 1 A
Viewing Angle	120°	60°	150° / 90°
mW/$	1228.08	757.61	814.97 / 1012.17
mW/$ (wall-plug)	2454.55	4650	4650 / 4950

"

[2a.2] Case Study 2 Additional Prompt:

"Convert to STF:"

TABLE XXIII: Strobe Illumination Wavelength

Feature	**UV**	**Blue**	**Red**	**IR**
Cost	Moderate	Moderate	High	Low
Longevity	Low	Moderate	High	High
Diffraction-limited Resolution	High	Moderate	Low	Low
High Power Common	Yes	Yes	No	Yes
Visible	No	Yes	Yes	No
IMX296 Responsivity	Low (<0.5)	Moderate (0.5-0.8)	High (0.9-1)	Moderate (0.5-0.9)

[2b] Case Study 2 - ChatGPT's Response

TABLE XXIV: Strobe Illumination Wavelength

Feature	**UV**	**Blue**	**Red / IR**
Cost	Moderate	Moderate	High / Low
Longevity	Low	Moderate	High / High
Diffraction-limited Resolution	High	Moderate	Low / Low
High Power Common	Yes	Yes	No / Yes
Visible	No	Yes	Yes / No
IMX296 Responsivity	Low (< 0.5)	Moderate (0.5–0.8)	High (0.9–1) / Moderate (0.5–0.9)

"Important Note: Your STF only supports 4 columns, but your original table had 5 (UV, Blue, Red, IR). So I combined:

- Red + IR $\rightarrow$ one column (Red / IR)

If you'd rather:

- Expand STF to 5 columns, or
- Keep all wavelengths separate

I can adjust your standard format accordingly."

[2c] Case Study 2 - Copilot's Response

TABLE XXV: Strobe Illumination Wavelength

Feature	**UV**	**Blue**	**Red**	**IR**

Cost	Moderate	Moderate	High	Low
Longevity	Low	Moderate	High	High
Diffraction-limited Resolution	High	Moderate	Low	Low
High Power Common	Yes	Yes	No	Yes
Visible	No	Yes	Yes	No
IMX296 Responsivity	Low (<0.5)	Moderate (0.5-0.8)	High (0.9-1)	Moderate (0.5-0.9)

”If you’d like this table to use different column widths or if you want me to standardize captions or labels as well, I can adjust the STF template accordingly.”

Case Study 3: Metaprompt

.2.1 Prompt 1

Prompt: ”My undergraduate senior design project is a combination of optics, photonics, electrical engineering, and software used to measure the speed and trajectory of a projectile. I need to do a case study comparing how well different AI works for my senior design project. What are some uses for AI?”

Used the ”Thinking” selection in the drop-down (rather than the ”Fast”, ”Pro”, or ”Deep Research”.

.2.1.1 Gemini Response

Integrating AI into a high-speed projectile tracking system offers several ways to enhance precision, especially where traditional mathematical models might struggle with environmental noise or complex aerodynamics. For your case study, you can categorize AI applications into signal processing, computer vision, and predictive modeling.
Here are four primary use cases for AI in your project:

1. High-Speed Object Detection and Tracking

If your project uses optical sensors or high-speed cameras, AI can be used to identify the projectile against a cluttered or low-contrast background.

- **The AI Role:** Using Convolutional Neural Networks (CNNs) or “You Only Look Once” (YOLO) architectures to localize the projectile in each frame.
- **Why use AI?** Traditional frame-differencing can fail if the camera shakes or if lighting changes. AI models can be trained to recognize the specific geometry of your projectile, ignoring “noise” like shadows or debris.

- **Comparison Point:** You could compare a **heavyweight model** (e.g., YOLOv8) against a **lightweight model** (e.g., MobileNet) to see which balances accuracy and processing latency better for real-time tracking.

2. Trajectory Prediction and Extrapolation

Projectiles often follow a parabolic path, but variables like air resistance (drag), wind, and spin create non-linear deviations.

- **The AI Role:** Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks are designed for sequential data. They can "learn" the flight characteristics of the projectile.

- **The Math:** While a standard kinematic equation might look like $y = y_0 + v_{0y}t\frac{1}{2}gt^2$, an AI model can account for the drag coefficient C_d which changes dynamically with speed.

- **Comparison Point:** Compare the accuracy of a **Linear Regression** model against a **Deep Neural Network (DNN)** when predicting the final impact point based on the first 10% of the flight path.

3. Sensor Fusion (Optics + Electrical)

In a system using both photonics (like laser gates) and electrical sensors (like acoustic or Hall effect sensors), the data often arrives at different rates and with different error margins.

- **The AI Role:** A Neural Network can act as an advanced "Kalman Filter." It learns which sensor is more reliable at specific segments of the trajectory.

- **Why use AI?** It can resolve conflicts. For example, if an optical sensor is blinded by a flash, the AI can weigh the electrical sensor data more heavily for that millisecond.

- **Comparison Point:** Compare a **Traditional Kalman Filter** (mathematical) against a **Gated Recurrent Unit (GRU)** for multi-sensor data integration.

4. Signal Denoising in Photonics

Photonics sensors can be incredibly sensitive but prone to "shot noise" or thermal interference in the electrical components.

- **The AI Role: Autoencoders** can be used to "clean" the raw signal. By training a network on "clean" vs. "noisy" signal pairs, the AI learns to reconstruct the true pulse of the projectile passing a sensor.

- **The Benefit:** This allows you to use lower-cost hardware because the software (AI) compensates for the lack of hardware shielding or high-end filtering.

- **Comparison Point:** Compare **Wavelet Transforms** (traditional) against **1D Convolutional Neural Networks** for signal-to-noise ratio (SNR) improvement.

Suggested Case Study Framework

To make your case study robust, I recommend comparing these models across three specific metrics:

1. **Inference Latency:** How many milliseconds does it take to process one data point? (Critical for high-speed projectiles).

2. **Computational Cost:** Can the AI run on an edge device (like a Raspberry Pi or Jetson Nano) or does it require a desktop GPU?

3. **Robustness:** How does the accuracy drop when you introduce artificial "noise" (e.g., dimmed lighting or simulated wind)?

Would you like me to help you draft a specific comparison table for two of these architectures (e.g., YOLO vs. a standard Kalman Filter)?

.2.1.2 Grok Response

AI has numerous practical applications in your senior design project for measuring projectile speed and trajectory. Your system—integrating optics/photonics (e.g., high-speed cameras, lasers, or interferometric sensors), electrical engineering (sensors, data acquisition), and software—already generates rich data like image sequences or time-series sensor readings. AI excels at handling noise, sparse/occluded data, real-time processing, and fusing multi-modal inputs (e.g., video + IMU), which classical methods often struggle with in dynamic, real-world conditions like varying lighting, high speeds, or environmental interference.

Here are some targeted, implementable uses for AI in your project. These are drawn from established applications in ballistics, sports analytics, robotics, and defense systems, making them suitable for an undergraduate case study. You can compare different AI approaches (e.g., traditional computer vision vs. machine learning vs. deep learning vs. physics-informed models) on metrics like accuracy (position/velocity error), latency (for real-time use), robustness (to noise/lighting), computational cost (CPU/GPU/embedded), and data efficiency.

1. **Object Detection and Tracking in Optical/Video Data**

- **How it works**: Use computer vision AI to automatically locate the projectile in each frame from your high-speed camera(s) or photonic sensors. Models like YOLO (You Only Look Once) series detect the projectile as a bounding box or keypoint, then pair with trackers (e.g., DeepSORT or ByteTrack) to follow it across frames.
- **Benefits for your project**: Replaces manual or threshold-based detection (which fails

under blur, glare, or background clutter). Enables precise 2D/3D position extraction for velocity calculation via frame timestamps.
- **Real-world examples**: Applied to ball tracking in sports broadcasts (e.g., via AWS deep learning pipelines) and high-speed cannon projectile video analysis for 3D pose/velocity.
- **Case study angle**: Compare YOLOv8/v11 (fast, real-time on edge devices) vs. classical OpenCV (background subtraction + contour detection) vs. older detectors like Faster R-CNN. Measure mAP (mean average precision) and FPS.

2. **Trajectory Prediction and Smoothing from Noisy/Sparse Data**

- **How it works**: Feed detected positions (or raw sensor data) into recurrent networks like LSTM/GRU or transformers to predict future positions and smooth noisy measurements. Physics-Informed Neural Networks (PINNs) add physical constraints (e.g., projectile motion equations with drag/Magnus effects) directly into the loss function.
- **Benefits for your project**: Handles gaps in optical data (e.g., projectile exiting frame or sensor occlusion) and predicts full trajectories even from partial observations. Improves speed/trajectory estimates beyond simple least-squares fitting.
- **Real-world examples**: LSTM for GNSS-denied mortar trajectory estimation from IMU data; PINNs for 3D table tennis ball kinematics from video; hybrid physics-ML for non-rigid projectile launch modeling.
- **Case study angle**: Compare LSTM/PINNs (data-efficient, physics-aware) vs. pure data-driven neural nets vs. traditional Kalman filters. Test on simulated + real projectile data for RMSE in position/velocity.

3. **Velocity/Speed Estimation and Sensor Fusion**

- **How it works**: AI processes time-stamped positions or optical flow from your photonics setup (e.g., combining camera frames with laser Doppler or high-speed video). Regression models or optical flow + deep learning estimate instantaneous speed; multi-sensor fusion (e.g., via neural networks) integrates EE sensor data.
- **Benefits for your project**: More accurate than manual frame differencing, especially at high velocities where blur occurs. Real-time edge AI can output speed on-device.
- **Real-world examples**: Vision AI for ball speed/direction in sports; deep learning + optical flow for velocity in high-speed scenarios; automated CV for cannon projectile initial velocity/orientation.
- **Case study angle**: Benchmark CNN-based optical flow vs. traditional Lucas-Kanade vs. hybrid ML (e.g., Random Forest on physics features). Evaluate error vs. ground-truth chronograph data.

4. **3D Trajectory Reconstruction from 2D Optical Views**

- **How it works**: Monocular or stereo camera setups use AI (e.g., CNNs or PINNs) to lift 2D image coordinates to 3D space, incorporating camera calibration and projectile dynamics.

- **Benefits for your project**: Reduces hardware needs (no multi-camera rig required) while leveraging your existing optics. Corrects for perspective/distortion in photonic imaging.
- **Real-world examples**: Monocular 3D projectile localization from video; patent methods for golf/baseball trajectories from single-camera 2D images.
- **Case study angle**: Compare PINN-based 3D lifting vs. geometric triangulation + ML refinement. Assess reconstruction error in lab tests.

5. **Image Enhancement, Denoising, and Anomaly Detection (Photonics-Focused)**

- **How it works**: Generative or convolutional models denoise low-light/high-speed photonic images, enhance contrast, or flag anomalies (e.g., projectile wobble or sensor failure).
- **Benefits for your project**: Improves data quality from optical/EE sensors before downstream processing, especially in outdoor or variable conditions.
- **Real-world examples**: AI super-resolution and denoising in electro-optical systems for embedded tracking.
- **Case study angle**: Compare autoencoders or diffusion models vs. classical filters (e.g., Gaussian). Quantify PSNR improvement and downstream tracking accuracy.

6. **Optimization, Simulation, and Post-Processing (Software Layer)**

- **How it works**: Train ML models on your collected dataset to optimize launch parameters, simulate trajectories (hybrid physics + ML), or generate ballistic tables/reports.
- **Benefits for your project**: Extends the system beyond measurement to predictive analytics (e.g., "what-if" trajectory adjustments).
- **Real-world examples**: Neural nets for optimal launch angle prediction; extreme learning machines for impact point forecasting; hybrid ODE solvers + Random Forest for bullet velocity.
- **Case study angle**: Compare data-driven ML vs. physics-only simulation vs. hybrid approaches on prediction error and training data needs.

Suggestions for Your AI Case Study

- **AIs to compare**:
  - Baseline: Non-AI (OpenCV classical CV + Kalman filter).
  - Traditional ML: SVM/random forests for detection/regression.
  - Deep learning: YOLO + LSTM (end-to-end).
  - Advanced: PINNs or transformer-based (physics-aware).
  - Edge vs. cloud: TensorFlow Lite/PyTorch Mobile vs. full models.
- **Implementation tips**: Use open-source tools (Ultralytics YOLO, PyTorch for PINNs/LSTM, OpenCV). Generate synthetic data via physics simulators (e.g., projectile motion equations in Python) to augment limited real shots. Test in your lab with a safe projectile launcher (e.g., air cannon or slingshot).
- **Evaluation framework**: Collect ground-truth via high-precision sensors or multiple synchronized cameras. Metrics: MAE/RMSE for speed/trajectory, inference

time (ms/frame), power consumption (for embedded EE), generalization (different lighting/speeds).
- **Feasibility for undergrad**: Start small—prototype detection/tracking on your existing video data. Many models train in hours on a laptop GPU. This adds strong "AI integration" value to your report/presentation.

These uses can make your system more accurate, autonomous, and scalable while providing excellent material for a comparative case study (e.g., "AI boosts velocity accuracy by X% with Y ms latency"). If you share more details (e.g., specific sensors, data format, or hardware constraints), I can refine these further or suggest code starters!

.3 Case Study 4

Prompts

.3.0.1 UART

To get more information on UART, I queried: "UART Standard".

I then queried: "is UART a standard like USB"

.3.1 Responses

.3.1.1 UART, response 1

UART, or Universal Asynchronous Receiver-Transmitter, is a hardware communication protocol used for asynchronous serial communication, allowing data to be transmitted one bit at a time over two wires. It requires both the transmitting and receiving devices to be configured with the same baud rate, data bits, parity bits, and stop bits for successful communication. [Links to: Analog.com (UART), and Wikipedia.org (UART)]

.3.1.2 UART, response 2

UART is not a standard like USB; rather, it is a hardware communication protocol used for asynchronous serial communication. USB, on the other hand, is a standardized protocol that manages data transfer between devices and includes features like device detection and hot-plugging.

sample